

NEET **CHEMISTRY** MODULE 1

- Some Basic Concepts of Chemistry
- Structure of Atom
- Redox Reactions

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Chapter 1

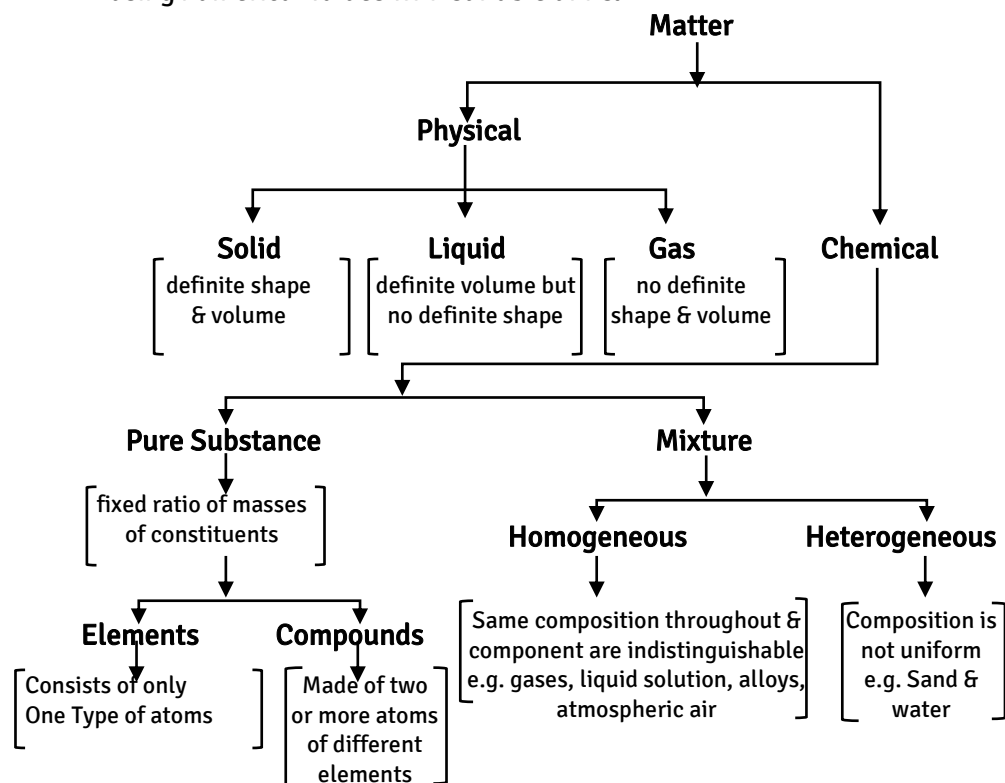
SOME BASIC CONCEPTS OF CHEMISTRY

Chapter Summary

- Introduction
- SI Units
- Mole Concept
- Some Related Definition
- Relation Between Molecular Weight & Vapour Density
- Percentage Composition, Empirical Formula & Molecular Formula
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- Limiting Reagent
- Percentage Yield & Percentage Purity
- Equivalent Weight
- Methods For Calculation of Atomic Weights & Molecular Weights
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Introduction

Chemistry deals with the composition, structure and properties of matter. These aspects can be best described and understood in terms of basic constituents of matter: **atoms** and **molecules**. That is why chemistry is called the science of atoms and molecules. Can we see, weigh and perceive these entities? Is it possible to count the number of atoms and molecules in a given mass of matter and have a quantitative relationship between the mass and number of these particles (atoms and molecules)? We will like to answer some of these questions in this Unit. We would further describe how physical properties of matter can be quantitatively described using numerical values with suitable units.



Classification of Universe

(A) Matter

(B) Energy

(A) Matter: Anything that occupies and has mass is called matter.

Matter is further classified into two categories:

(I) Physical classification

(II) Chemical classification

Physical Classification

It is based on physical state under ordinary conditions of temperature and pressure, on the basis of which matter can be classified into the following three states:

(a) Solid

(b) Liquid

(c) Gas

(a) Solid: A substance is said to be solid if it possesses a definite volume and a definite shape.

Example: sugar, iron, gold, wood etc.

(b) Liquid: A substance is said to be liquid if it possesses a definite volume but not definite shape. They take up the shape of the vessel in which they are put.

Example: water, milk, oil, mercury, alcohol etc.

(c) Gas: A substance is said to be gas if it neither possesses a definite volume nor a definite shape. This is because they fill up the whole vessel in which they are put.

Example: Hydrogen(H_2), Oxygen(O_2), Carbon Dioxide (CO_2) etc.

Chemical Classification

It may be classified into two types:

(a) Pure Substance

(b) Mixture

(a) Pure Substance: A material containing only one type of substance. Pure Substance cannot be separated into simpler substance by physical method.

Example: Element = Na, Mg, Ca etc.

Compound = HCl, H_2O , CO_2 , HNO_3 etc.

Pure substance is classified into two types:

(i) Element

(ii) Compound

(i) Element: Any pure substance containing only one kind of atom.

It is classified into 3 types (depending on physical and chemical properties)

(i) Metal

→

Zn, Cu, Hg, Ag, Sn, Pb etc.

(ii) Non-metal

→

N_2 , O_2 , Cl_2 , Br_2 , F_2 , P_4 , S_8 etc.

(iii) Metalloids

→

B, Si, As, Te etc.

(ii) Compound: It is defined as a pure substance containing more than one kind of elements or atoms which are combined together in a fixed proportion by weight and which can be decomposed into simpler substance by suitable chemical methods. The properties of a compound are completely different from those of its constituent elements.

Example: HCl, H_2O , H_2SO_4 , $HClO_4$, HNO_3 etc.

(b) Mixture: A material which contain more than one type of substance and which are mixed in any ratio by weight is called as mixture. The property of the mixture is the same as / based on the property of its components. The mixture can be separated by simple physical method.

Mixture is classified into two types:

(i) **Homogeneous mixture:** Any mixture, in which all the components are present uniformly is called homogeneous mixture. Components of mixture are present in single phase.

Example: Water + Salt, Water + Sugar, Water + Alcohol.

(ii) **Heterogenous mixture:** The mixture, in which all the components are present non-uniformly

Example: Water + Sand, Water + Oil, blood, petrol etc.

Exercise 1.1

- Which is an example of matter according to physical state at room temperature and pressure.
(1) solid (2) liquid (3) gas (4) all of these
- Which of the following are the types of the compound.
(1) Organic compound (2) Inorganic compound
(3) Both (1) and (2) (4) None of these
- Which of the following is an example of a homogeneous mixture.
(1) Water + Alcohol (2) Water + Sand (3) Water + Oil (4) None of these
- Which of the following is a solution.
(1) Heterogeneous mixture (2) Homogeneous mixture
(3) Both (1) and (2) (4) None of these
- Which of the following is a compound
(1) graphite (2) producer gas (3) cement (4) marble

S.I. Units (International System of Units)

Different types of units of measurements have been in use in different parts of the world e.g. kilograms, pounds etc. for mass; miles, furlongs, yards etc. for distance.

To have a common system of units throughout the world. French Academy of Science, in 1791, introduced a new system of measurements called metric system in which the different units of a physical quantity are related to each other as multiples of powers of 10, e.g. $1 \text{ km} = 10^3 \text{ m}$, $1 \text{ cm} = 10^{-2} \text{ m}$ etc. This system of units was found to be so convenient that scientists all over the world adopted this system for scientific data.

(A) Seven Basic Units:

The seven basic physical quantities in the International System of Units, their symbols, the names of their units (called the base units) and the symbols of these units are given in Table.

Table: Seven Basic Physical Quantities and their S.I. Units

Physical Quantity	Symbol	S.I. Unit	Symbol
Length	l	metre	m
Mass	m	kilogram	kg
Time	t	second	s
Electric current	I	ampere	A
Temperature	T	kelvin	K
Luminous intensity	I_v	candela	cd
Amount of the substance	n	mole	mol

(B) Prefixes Used with Units:

The S.I. system recommends the multiples such as 10^3 , 10^6 , 10^9 etc. and fraction such as 10^{-3} , 10^{-6} , 10^{-9} etc., i.e. the powers are the multiples of 3. These are indicated by special prefixes. These along with some other fractions or multiples in common use, along with their prefixes are given below in Table and illustrated for length (m)

Table: Seven Basic Physical Quantities and their S.I. Units

Prefix	Symbol	Multiplication Factor	Example
deci	d	10^{-1}	1 decimetre (dm) = 10^{-1} m
centi	c	10^{-2}	1 centimetre (cm) = 10^{-2} m
milli	m	10^{-3}	1 millimetre (mm) = 10^{-3} m
micro	μ	10^{-6}	1 micrometre (μ m) = 10^{-6} m
nano	n	10^{-9}	1 nanometre (nm) = 10^{-9} m
pico	p	10^{-12}	1 picometre (pm) = 10^{-12} m
femto	f	10^{-15}	1 femtometre (fm) = 10^{-15} m
atto	a	10^{-18}	1 attometre (am) = 10^{-18} m
deca	da	10^1	1 decametre (dam) = 10^1 m
hecto	h	10^2	1 hectometre (hm) = 10^2 m
kilo	k	10^3	1 kilometre (km) = 10^3 m
mega	M	10^6	1 megametre (Mm) = 10^6 m
giga	G	10^9	1 gigametre (Gm) = 10^9 m
tera	T	10^{12}	1 terametre (Tm) = 10^{12} m
peta	P	10^{15}	1 petametre (Pm) = 10^{15} m
exa	E	10^{18}	1 exametre (Em) = 10^{18} m

As volume is very often expressed in litres, it is important to note that the equivalence in S.I. units for volume is as under:

$$1 \text{ litre (1L)} = 1 \text{ dm}^3 = 1000 \text{ cm}^3$$

$$\text{and } 1 \text{ millilitre (1mL)} = 1 \text{ cm}^3 = 1 \text{ cc}$$

(C) Some Important Unit Conversions:**1. Length:** 1 mile = 1760 yards

$$1 \text{ yard} = 3 \text{ feet}$$

$$1 \text{ foot} = 12 \text{ inches}$$

$$1 \text{ inch} = 2.54 \text{ cm}$$

$$1 \text{ \AA} = 10^{-10} \text{ m} \text{ or } 10^{-8} \text{ cm}$$

2. Mass:

$$1 \text{ Ton} = 1000 \text{ kg}$$

$$1 \text{ Quintal} = 100 \text{ kg}$$

$$1 \text{ kg} = 2.205 \text{ Pounds (lb)}$$

$$1 \text{ kg} = 1000 \text{ g}$$

$$1 \text{ gram} = 1000 \text{ milli gram}$$

$$1 \text{ amu} = 1.67 \times 10^{-24} \text{ g}$$

3. Volume:

$$1 \text{ L} = 1 \text{ dm}^3 = 10^{-3} \text{ m}^3 = 10^3 \text{ cm}^3 = 10^3 \text{ mL} = 10^3 \text{ cc}$$

$$1 \text{ mL} = 1 \text{ cm}^3 = 10^{-6} \text{ m}^3 = 1 \text{ cc}$$

4. Energy :

$$1 \text{ calorie} = 4.184 \text{ joules} \approx 4.2 \text{ joules}$$

$$1 \text{ joule} = 10^7 \text{ ergs}$$

$$1 \text{ litre atmosphere (L-atm)} = 101.3 \text{ joule}$$

$$1 \text{ electron volt (eV)} = 1.602 \times 10^{-19} \text{ joule}$$

5. **Pressure:** 1 atmosphere (atm) = 760 torr
 = 760 mm of Hg
 = 76 cm of Hg
 = 1.01325×10^5 pascal (Pa)
 = 1.01325×10^5 N/m²

6. **Temperature:** $^{\circ}\text{C} + 273.15 = \text{K}$; $\frac{5}{9} (^{\circ}\text{F} - 32) = ^{\circ}\text{C}$

Points to Ponder

- The unit named after a scientist is started with a small letter and not with a capital letter e.g. unit of force is written as newton and not as Newton. Likewise, unit of heat and work is written as joule and not as Joule.
- Symbols of the units do not have a plural ending like 's'. For example, we have 10 cm and not 10 cms.
- Words and symbols should not be mixed e.g. we should write either joules per mole or J mol⁻¹ and not joules mol⁻¹.
- Prefixes are used with the basic units e.g. kilometre means 1000 m (because meter is the basic unit). Exception. Though kilogram is the basic unit of mass, yet prefixes are used with gram because in kilogram, kilo is already a prefix.
- A unit written with a prefix and a power is a power for the complete unit e.g. cm³ means (centimetre)³ and not centi (meter)³.

Exercise 1.2

- Convert 2 litre atmosphere into erg.
 (1) 101.3×10^7 erg (2) 98.7×10^7 erg (3) 202.6×10^7 erg (4) 213.7×10^7 erg
- Convert 2 atm into cm of Hg.
 (1) 152 cm of Hg (2) 168 cm of Hg (3) 84 cm of Hg (4) 76 cm of Hg
- Convert 20 dm³ into mL.
 (1) 2×10^2 mL (2) 2×10^3 mL (3) 2×10^4 mL (4) 2×10^{-1} mL
- Convert 59 F into $^{\circ}\text{C}$.
 (1) 8°C (2) 11°C (3) 15°C (4) 20°C

Mole Concept

In SI Units we represent mole by the symbol 'mol'. It is defined as follows:

- (i) **A mole is the amount of a substance that contains as many entities (atoms, molecules or other particles) as there are atoms in exactly 12g of the carbon - 12 isotope.**

It may be emphasised that the mole of a substance always contains the same number of entities, no matter what the substance may be. In order to determine this number precisely, the mass of a carbon-12 atom was determined by a mass spectrometer and found to be equal to 1.992648×10^{-23} g. Knowing that 1 mole of carbon weighs 12g, the number of atoms in it is equal to:

$$\frac{12\text{g} / \text{mol C}^{12}}{1.992648 \times 10^{-23}\text{g} / \text{C}^{12}\text{atom}} = 6.0221367 \times 10^{23} \text{ atoms / mol}$$

- (ii) In a simple way, we can say that mole has 6.0221367×10^{23} entities (atom, molecules or ions etc.) The number of entities in 1 mol is so important that it is given a separate name and symbol, known as 'Avogadro constant' denoted by N_A .

Here entities may represent atoms, ions, molecules or other subatomic entities. Chemists count the number of atoms and molecules by weighing. In a reaction we require these particles (atoms, molecules and ions) in a definite ratio. We make use of this relationship between numbers and masses of the particles for determining the stoichiometry of reactions.

Formula to get moles are following:

$$(i) \quad \text{Number of moles}(n) = \frac{\text{Weight(g)}}{\text{molar mass}}$$

Where molar mass = gram atomic mass or gram molecular mass or gram ionic mass

$$(ii) \quad \text{Number of moles}(n) = \frac{V_{(L)}}{22.4} \quad (\text{Where } V = \text{Volume of gas in L at NTP or STP})$$

$$(iii) \quad \text{Number of moles}(n) = \frac{N}{N_A} \quad (\text{Where } N = \text{Number of particles})$$

$$\text{mole atoms} = \frac{\text{number of atoms}}{N_A} \quad \text{and mole molecules} = \frac{\text{number of molecules}}{N_A}$$

Ex.1: Calculate the number of molecules of sulphur dioxide in 0.064 g of the gas.

- (A) 6.023×10^{23} (B) 6.023×10^{20} (C) 6.023×10^{18} (D) 6.023×10^{16}

Sol. (B)

Gram molecular weight of sulphur dioxide (SO_2) = 64gm

Given mass = 0.064 gm

Gram molecular weight of any gas contains avogadro number of molecules = 6.023×10^{23}

$\therefore$ 0.064 g of sulphur dioxide contain molecules = 6.023×10^{20}

Ex.2: Which of the following contains the least number of molecules -

- (A) 16g of CO_2 (B) 8g of O_2 (C) 4g of N_2 (D) 2g of H_2

Sol. (C)

$$(A) \text{ No. of moles of } \text{CO}_2 = \frac{\text{Weight}}{\text{Molecular weight}} = \frac{16}{44} = 0.36$$

$$(B) \text{ Number of moles of } \text{O}_2 = \frac{8}{32} = 0.25$$

$$(C) \text{ Number of moles of } \text{N}_2 = \frac{4}{28} = 0.14$$

$$(D) \text{ Number of moles of } \text{H}_2 = \frac{2}{2} = 1$$

Ex.3: Calculate the volume at STP occupied by 240gm of SO_2 .

- (A) 84 Litre (B) 48 Litre (C) 54 Litre (D) 64 Litre

Sol. (A)

Molecular weight of $\text{SO}_2 = 32 + 2 \times 16 = 64$

64 gm of SO_2 occupies 22.4 litre at STP

$$240 \text{ gm of } \text{SO}_2 \text{ occupies} = \frac{22.4}{64} \times 240 = 84 \text{ litre at STP}$$

Some Related Definitions

Atomic Mass (Relative Atomic Mass):

It is defined as the number which indicates how many times the mass of one atom of an element is heavier in comparison to $\frac{1}{12}$ th part of the mass of one atom of C-12.

Atomic mass unit (a.m.u.): The quantity $\frac{1}{12}$ th mass of an atom of C^{12} is known as atomic mass unit.

Since mass of 1 atom of C - 12 = 1.9924×10^{-23} g

$$\therefore \frac{1}{12} \text{ part of the mass of 1 atom} = \frac{1.9924 \times 10^{-23} \text{ g}}{12} = 1.67 \times 10^{-24} \text{ g} = 1 \text{ a.m.u.} = \frac{1}{6.023 \times 10^{23}}$$

It may be noted that the atomic masses as obtained above are the relative atomic masses and not the actual masses of the atoms. These masses on the atomic mass scale are expressed in terms of atomic mass units (abbreviated as amu). Today, 'amu' has been replaced by 'u' which is known as unified mass.

One atomic mass unit (amu) is equal to $\frac{1}{12}$ th of the mass of an atom of carbon - 12 isotope.

Thus, the atomic mass of hydrogen is 1.008 amu while that of oxygen is 15.9994 amu (or taken as 16 amu).

Gram Atomic Mass (or Mass of 1 Gram Atom):

When numerical value of atomic mass of an element is expressed in grams then the value becomes gram atomic mass.

$$\begin{aligned} \text{gram atomic mass.} &= \text{mass of 1 gram atom} = \text{mass of 1 mole atom} \\ &= \text{mass of } N_A \text{ atoms} = \text{mass of } 6.023 \times 10^{23} \text{ atoms.} \end{aligned}$$

Ex.: gram atomic mass of oxygen = mass of **1 g atom** of oxygen = mass of **1 mol atom** of oxygen.

$$= \text{mass of } N_A \text{ atoms of oxygen.} = \left(\frac{16}{N_A} \text{ g} \right) \times N_A = 16 \text{ g}$$

Molecular Mass (Relative Molecular Mass):

The number which indicates how many times the mass of one molecule of a substance is heavier in comparison to $\frac{1}{12}$ th part of the mass of an atom of C-12

Gram Molecular Mass (Mass of 1 Gram Molecule):

When numerical value of molecular mass of the substance is expressed in grams then the value becomes gram molecular mass.

$$\begin{aligned} \text{gram molecular mass} &= \text{mass of 1 gram molecule} = \text{mass of 1 mole molecule} \\ &= \text{mass of } N_A \text{ molecules} = \text{mass of } 6.023 \times 10^{23} \text{ molecules} \end{aligned}$$

Ex.: gram molecular mass of H_2SO_4 = mass of **1 gram molecule** of H_2SO_4

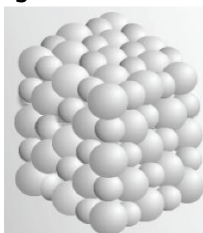
$$= \text{mass of 1 mole molecule of } H_2SO_4$$

$$= \text{mass of } N_A \text{ molecules of } H_2SO_4$$

$$= \left(\frac{98}{N_A} \text{ g} \right) \times N_A = 98 \text{ g}$$

Formula Mass:

Some substances, such as sodium chloride, do not contain discrete molecules as their constituent units. In such compounds, positive (sodium ion) and negative (chloride ion) entities are arranged in a three-dimensional structure, as shown in Fig.



It may be noted that in sodium chloride, one Na^+ ion is surrounded by six Cl^- ion and vice-versa. The formula, such as NaCl , is used to calculate the formula mass instead of molecular mass as in the solid-state sodium chloride does not exist as a single entity.

Thus, the formula mass of sodium chloride is atomic mass of sodium + atomic mass of chlorine
 $= 23.0 \text{ u} + 35.5 \text{ u} = 58.5 \text{ u}$

Actual Mass:

The mass of one atom or one molecule of substance expressed in gram is called as actual mass.

- Ex.: (i) Actual mass of $\text{O}_2 = 32 \text{ amu} = 32 \times 1.67 \times 10^{-24} \text{ g} \rightarrow \text{Actual mass}$
 (ii) Actual mass of $\text{H}_2\text{O} = (2 + 16) \text{ amu} = 18 \times 1.67 \times 10^{-24} \text{ g} = 2.99 \times 10^{-23} \text{ g}$

Atomicity:

Total number of atoms in a **molecule** of elementary substance is called as atomicity.

Ex.:

Molecule	Atomicity
H_2	2
O_2	2
O_3	3
NH_3	4

Ex.4: Atomic weight of helium is 4. Calculate the number of atoms in 1g of helium.

- (A) 6×10^{23} (B) 1.51×10^{23} (C) 2.5×10^{23} (D) 3×10^{23}

Sol. (B)

4g of Helium contains 6.023×10^{23} atoms

1g of Helium contains $\frac{6.023 \times 10^{23}}{4} = 1.506 \times 10^{23}$ atoms

Ex.5: What is the mass of 1 molecule of CO

- (A) $1.21 \times 10^{-24} \text{ g}$ (B) $2.32 \times 10^{-23} \text{ g}$ (C) $1.15 \times 10^{-23} \text{ g}$ (D) $4.65 \times 10^{-23} \text{ g}$

Sol. (D)

Gram molecular weight of CO = $12 + 16 = 28 \text{ g}$

6.023×10^{23} molecules of CO weighs 28gm

1 molecule of CO weighs = $\frac{28}{6.02 \times 10^{23}} = 4.65 \times 10^{-23} \text{ g}$

Ex.6: Calculate the number of atoms in each of the following -

- (A) 52 mole of He (B) 52 amu of He (C) 52 g of He

Sol. (A) 1mole of He contain 6.02×10^{23} atoms

$$\therefore 52 \text{ mole of He contain} = 52 \times 6.02 \times 10^{23} = 31.3 \times 10^{24} \text{ atoms}$$

(B) Atomic weight of He = 4amu

$$\therefore 52 \text{ amu of He contain} = \frac{52}{4} = 13 \text{ atoms of He}$$

(C) Number of moles of He in 52g = $\frac{52}{4} = 13$ moles

$$\therefore \text{no. of atoms in 52g of He i.e. 13 moles} = 13 \times 6.02 \times 10^{23} \text{ atoms} = 78.26 \times 10^{23} \text{ atoms}$$

Relation Between Molecular Weight and Vapour Density

Vapour density (V.D): Vapour density of a gas is the ratio of densities of gas & hydrogen at the same temperature & pressure.

$$\text{Vapour Density (V.D.)} = \frac{\text{Density of gas}}{\text{Density of hydrogen}} = \frac{d_{\text{gas}}}{d_{\text{H}_2}} \left\{ d = \frac{m(\text{mass})(g)}{V(\text{volume})(\text{mL})} \right\}$$

$$\text{V.D.} = \frac{(m_{\text{gas}}) \text{ for certain V litre volume}}{(m_{\text{H}_2}) \text{ for certain V litre volume}}$$

If N molecules are present in the given volume of a gas and hydrogen under similar condition of temperature and pressure.

$$\text{V.D.} = \frac{(m_{\text{gas}}) \text{ of N molecules}}{(m_{\text{H}_2}) \text{ of N molecules}} = \frac{(m_{\text{gas}}) \text{ of 1 molecule}}{(m_{\text{H}_2}) \text{ of 1 molecule}} = \frac{\text{Molecular mass of gas}}{2}$$

$$\therefore \boxed{\text{Molecular mass of gas } (M_w) = 2 \times \text{V.D.}}$$

Relation Between Molar Mass & Volume:

$$\text{At STP. } M_w = 2 \times \text{V.D} = 2 \times \frac{d_{\text{gas}}}{d_{\text{H}_2}} = 2 \times \frac{(m_{\text{gas}}) \text{ for certain V litre volume}}{(m_{\text{H}_2}) \text{ for certain V litre volume}}$$

$$\text{or } M_w = 2 \times \frac{\text{mass of 1 litre gas}}{\text{mass of 1 litre H}_2}$$

$$d_{\text{H}_2} = 0.000089 \frac{\text{g}}{\text{mL}} = \frac{m}{V} = \frac{m}{1000\text{mL}}$$

$$V = 1 \text{ litre} = 1000 \text{ mL}$$

$$\text{Then } m_{\text{H}_2} = 0.089$$

$$\text{or } M_w = 2 \times \frac{\text{mass of 1 litre gas}}{0.089\text{g}}$$

$$M_w(\text{g}) = 22.4 \times \text{mass of 1 litre gas}$$

$$\boxed{M_w(\text{g}) = \text{Mass of 22.4 litre gas}} \text{ or } \boxed{M_w(\text{g}) = 22.4 \text{ litre (at STP)}}$$

Gram Molecular Volume (GMV):

At NTP, the volume of 1 mole of gaseous substance is 22.4 litre is called as gram molecular volume.

At NTP, $d_{H_2} = 0.000089 \text{ g/mL} = \text{mass/volume} = \text{mass}/1000 \text{ mL}$

If volume = 1 litre = 1000 mL then mass = 0.089 g

$\therefore 0.089 \text{ g } H_2 \text{ occupies} = 1 \text{ litre at STP.}$

$\therefore 1 \text{ g } H_2 \text{ occupies} = \frac{1 \text{ litre}}{0.089} = \text{at STP.}$

$\therefore 2 \text{ g or } 1 \text{ mol } H_2 \text{ occupies} = \frac{1 \text{ litre}}{0.089} \times 2 = 22.4 \text{ litre at STP}$

1 mole of any gaseous substance occupy 22.4 litre of volume at NTP or STP

$$\boxed{1 \text{ mol} = 22.4 \text{ litre (at STP)}}$$

Ex.7: At NTP the density of a gas is 0.00445 g/mL then find out its V.D. and molecular mass.

Sol. $V.D. = \frac{\text{Density of gas}}{\text{Density of } H_2} = \frac{0.004450}{0.000089} = 50$

Molecular mass = $2 \times V.D. = 2 \times 50 = 100$

Ex.8: Weight of 1 L gas is 2 g then find out its V.D. and molecular mass.

Sol. Density of gas = $\frac{\text{Mass}}{\text{Volume}} = \frac{2}{1000} = 0.002 \text{ g/mL}$

$V.D. = \frac{\text{Density of gas}}{\text{Density of } H_2} = \frac{0.002000}{0.000089} = 22.4$

Molecular mass = $2 \times V.D. = 44.8$

Exercise 1.3

- | | |
|--|---|
| <p>1. Mass of 1 atom of hydrogen is –
 (1) $1.66 \times 10^{-24} \text{ g}$ (2) 10^{-22} g
 (3) 10^{-23} g (4) 10^{-25} g</p> <p>2. 2 moles of H_2 at NTP occupy a volume of
 (1) 11.2 litre (2) 44.8 litre
 (3) 2 litre (4) 22.4 litre</p> <p>3. The number of molecules in 4.25 gm of NH_3 is
 (1) 1.505×10^{23} (2) 3.01×10^{23}
 (3) 6.02×10^{23} (4) None of these</p> <p>4. How many atoms are contained in a mole of $Ca(OH)_2$?
 (1) $30 \times 6.02 \times 10^{23} \text{ atoms / mol}$
 (2) $6 \times 6.02 \times 10^{23} \text{ atoms / mol}$
 (3) $6.02 \times 10^{23} \text{ atoms / mol}$
 (4) $5 \times 6.02 \times 10^{23} \text{ atoms / mol}$</p> | <p>5. Number of oxygen atoms in 8 gm of ozone is-
 (1) 6.02×10^{23} (2) $\frac{6.02 \times 10^{23}}{2}$
 (3) $\frac{6.02 \times 10^{23}}{3}$ (4) $\frac{6.02 \times 10^{23}}{6}$</p> <p>6. Total number of atoms represented by the compound $CuSO_4 \cdot 5H_2O$ is:
 (1) 27 (2) 21
 (3) 5 (4) 8</p> <p>7. Total number of atoms present in 64 gm of SO_2 is –
 (1) $2 \times 6.02 \times 10^{23}$ (2) 6.02×10^{23}
 (3) $4 \times 6.02 \times 10^{23}$ (4) $3 \times 6.02 \times 10^{23}$</p> |
|--|---|

8. 5.6 litre of oxygen at STP contains
 (1) 6.02×10^{23} atoms (2) 3.01×10^{23} atoms
 (3) 1.505×10^{23} atoms (4) 0.7525×10^{23} atoms
9. The volume of a gas in discharge tube is 1.12×10^{-7} ml. at STP. Then the number of molecules of gas in the tube is
 (1) 3.01×10^4 (2) 3.01×10^{15}
 (3) 3.01×10^{12} (4) 3.01×10^{16}
10. Which of the following contains the least number of molecules
 (1) 4.4 gm CO_2 (2) 3.4 gm NH_3
 (3) 1.6 gm CH_4 (4) 3.2 gm SO_2
11. Which of the following contains largest number of atoms?
 (1) 4 gm. of H_2 (2) 16 gm. of O_2
 (3) 28 gm. of N_2 (4) 18 gm. of H_2O
12. Which of the following contains greatest number of oxygen atoms?
 (1) 1 g of O
 (2) 1 g of O_2
 (3) 1 g of O_3
 (4) All have the same number of atoms
13. One mole of P_4 molecule contains –
 (1) 1 molecule
 (2) 4 molecules
 (3) $1/4 \times 6.022 \times 10^{23}$ atoms
 (4) 24.088×10^{23} atoms
14. Number of moles of water in 488 gm of $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ are – (Ba = 137)
 (1) 2 moles (2) 4 moles
 (3) 3 moles (4) 5 moles
15. A person adds 1.71 gram of sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in order to sweeten his tea. The number of carbon atoms added are (mol. mass of sugar = 342)
 (1) 3.6×10^{22} (2) 7.2×10^{21}
 (3) 0.05 (4) 6.6×10^{22}
16. The number of sodium atoms in 2 moles of sodium ferrocyanide $\text{Na}_4[\text{Fe}(\text{CN})_6]$, is-
 (1) 2 (2) 6.023×10^{23}
 (3) $8 \times 6.02 \times 10^{23}$ (4) $4 \times 6.02 \times 10^{23}$
17. The number of moles of carbon dioxide which contain 8 gm of oxygen is –
 (1) 0.5 mole (2) 0.20 moles
 (3) 0.40 moles (4) 0.25 moles
18. The number of oxygen atoms present in 14.6 g of magnesium bicarbonate $[\text{Mg}(\text{HCO}_3)_2]$ is
 (1) $6 N_A$ (2) $0.6 N_A$
 (3) N_A (4) $0.5 N_A$
19. Which of the following will contain same number of atoms as 20 g of calcium?
 (1) 24 g Magnesium (2) 12 g Carbon
 (3) 8 g Oxygen Gas (4) 16 g Oxygen Atom
20. The number of atoms present in 0.5 g- atoms of nitrogen is same as the atoms in –
 (1) 12 g of C (2) 32 g of S
 (3) 8 g of oxygen (4) 24 g of Mg
21. Calculate g-atom of element in 40 kg, if mass of one atom of an element is 6.644×10^{-23} g:
 (1) 10^2 g-atom
 (2) 10 g-atom
 (3) 10^3 g-atom
 (4) None of these
22. Which of the following has maximum mass?
 (1) 0.1 gm atom of carbon
 (2) at STP, 1120 cc volume of carbon dioxide
 (3) 6.02×10^{22} molecule of hydrogen
 (4) 0.1 mole of ammonia
23. What is correct for 10 g of CaCO_3 ?
 (1) It contains 1 g -atom of carbon
 (2) It contains 0.3 g -atoms of oxygen
 (3) It contains 12 g of calcium
 (4) All of these

24. The maximum volume at NTP is occupied by-
 (1) 12.8 gm of SO_2
 (2) 6.02×10^{22} molecules of CH_4
 (3) 0.5 mol of NO_2
 (4) 1 gm -molecule of CO_2
25. 4.48 litres of methane at N.T.P. correspond to-
 (1) 1.2×10^{22} molecules of methane
 (2) 0.5 mole of methane
 (3) 3.2 gm of methane
 (4) 0.1 mole of methane
26. Calculate the mass in gm of 2 g atom of Mg.
 (1) 12 gm (2) 24 gm
 (3) 6 gm (4) 48 gm
27. The actual weight of a molecule of water is
 (1) 18 gm
 (2) 2.99×10^{-23} gm
 (3) both (1) and (2) are correct
 (4) None of these
28. Number of neutrons present in 1.7 gm of ammonia is -
 (1) N_A (2) $N_A/10 \times 4$
 (3) $(N_A/10) \times 7$ (4) $N_A \times 10 \times 7$
29. The total number of protons, electrons and neutrons in 12 gm of ${}_6\text{C}^{12}$ is -
 (1) 1.084×10^{25}
 (2) 6.022×10^{23}
 (3) 6.022×10^{22}
 (4) 18
30. 16 gm of SO_x occupies 5.6 litres at STP. Assuming ideal gas nature, the value of x is -
 (1) 1 (2) 2
 (3) 3 (4) None of these
31. 5.6 litre of a gas at N.T.P. weights equal to 8 gm. Vapour density of the gas is-
 (1) 32 (2) 16
 (3) 8 (4) 40
32. 4.4 gm of an unknown gas occupies 2.24 litres of volume at STP. The gas may be
 (1) N_2O (2) CO
 (3) CO_2 (4) 1 & 3 both
33. If V ml of the vapours of substance at NTP weight W gm. Then molecular wt. of substance is
 (1) $(W/V) \times 22400$
 (2) $V/W \times 22.4$
 (3) $(W - V) \times 22400$
 (4) $\frac{W \times 1}{V \times 22400}$
34. Given that one mole of N_2 at NTP occupies 22.4 litre the density of N_2 is
 (1) 1.25 g/litre
 (2) 0.80 g/litre
 (3) 2.5 g/litre
 (4) 1.60 g/litre
35. The total number of electrons present in 18 mL water (density 1 g/mL) is -
 (1) 6.023×10^{23}
 (2) 6.023×10^{24}
 (3) 6.023×10^{25}
 (4) 6.023×10^{21}
36. 22.4 litre of water vapour at NTP, when condensed to water occupies an approximate volume of -
 (1) 18 litre (2) 1 litre
 (3) 1 ml (4) 18 ml
37. Mol. Wt. = vapor density $\times 2$, is valid for:
 (1) Metals (2) Non-metals
 (3) Solid (4) Gases
38. Vapour density of gas is 11.2 volume occupied by 2.4 gm of this at STP will be
 (1) 11.2 lit (2) 2.24 lit
 (3) 22.4 lit (4) 2.4 lit

Percentage Composition, Empirical Formula & Molecular Formula

Percentage Composition (% By Mass):

(In a molecule or compound) Mass % of an element = $\frac{\text{Number of atom (Atomicity)} \times \text{atomic mass}}{\text{molecular mass}} \times 100$

If number of atoms = 1: Molecular mass = minimum molecular mass

Ex.9: Find out percentage composition of each element present in glucose?

Sol. % of C = $\frac{12 \times 6}{180} \times 100 = 40\%$
 % of H = $\frac{12 \times 1}{180} \times 100 = 6.67\%$
 % of O = $\frac{16 \times 6}{180} \times 100 = 53.33\%$

Empirical Formula:

The empirical formula of a compound expresses the simplest whole number ratio of atoms of various elements present in 1 molecule of the compound.

Molecular Formula:

The molecular formula of a compound represents the **actual number** of atoms present in 1 molecule of the compound i.e. it shows the real formula of its 1 molecule.

Ex.:	Molecular Formula	→	H ₂ O ₂	CH ₄	C ₂ H ₆	C ₂ H ₄ O ₂
			2 : 2	1 : 4	2 : 6	2 : 4 : 2
			1 : 1	1 : 4	1 : 3	1 : 2 : 1
	Empirical Formula	→	HO	CH ₄	CH ₃	CH ₂ O

Relationship between Empirical & Molecular Formula:

Molecular Formula = n × Empirical Formula

[Where n = natural no. (1, 2, 3,)]

or $n = \frac{\text{Molecular Formula}}{\text{Empirical Formula}}$

or $n = \frac{\text{Molecular Formula mass}}{\text{Empirical Formula mass}}$

Determination of Empirical Formula:

Following steps are involved to determine the empirical formula of the compounds -

- (1) First of all, find the % by weight of each element present in 1 molecule of the compound.
- (2) The % by weight of each element is divided by its atomic weight. It gives atomic ratio of elements present in the compounds.
- (3) Atomic ratio of each element is divided by the minimum value of atomic ratio as to get simplest ratio of atoms.
- (4) If the value of simplest atomic ratio is fractional then raise the value to the nearest whole number or multiply with suitable coefficient to convert it into nearest whole number.
- (5) Write the Empirical formula as we get the simplest ratio of atoms.

Ex.10: In a compound x is 75.8% and y is 24.2% by weight present. If atomic weight of x and y are 24 and 16 respectively. Then calculate the empirical formula of the compound.

Elements	%	Atomic weight	$\frac{\%}{\text{Atomic weight}}$	Simplest Ratio	Ratio
x	75.8%	24	$\frac{75.8}{24} = 3.1$	$\frac{3.1}{1.5} = 2.06$	2
y	24.2%	16	$\frac{24.2}{16} = 1.5$	$\frac{15}{15} = 1$	1

Then empirical formula is x_2y

Ex.11: Phosgene, a poisonous gas used during World war-I, contains 12.1% C, 16.2% O and 71.7% Cl by mass. What is the empirical formula of phosgene.

Sol.	Element	%	Mole ratio	Simplest mole ratio
	C	12.1	$\frac{12.1}{12} = 1.01$	$\frac{1.01}{1.01} = 1$
	O	16.2	$\frac{16.2}{16} = 1.01$	$\frac{1.01}{1.01} = 1$
	Cl	71.7	$\frac{71.7}{35.5} = 2.02$	$\frac{2.02}{1.01} = 2$

Then empirical formula = COCl_2

Ex.12: 5.325g sample of methyl benzoate, a compound used in the manufacture of perfumes is found to contain 3.758 g of carbon, 0.316g hydrogen and 1.251g of oxygen. What is empirical formula, of compound. If mol. weight of methyl benzoate is 136.0, calculate its molecular formula.

Sol.

Element	%	Mole ratio	Simplest whole ratio
C	$\frac{3.758 \times 100}{5.325} = 70.57$	$\frac{70.57}{12} = 5.88$	$\frac{5.88}{1.47} = 4$
H	$\frac{0.316 \times 100}{5.325} = 5.93$	$\frac{5.93}{1} = 5.93$	$\frac{5.93}{1.47} = 4$
O	$\frac{1.251 \times 100}{5.325} = 23.50$	$\frac{23.50}{16} = 1.47$	$\frac{1.47}{1.47} = 1$

Empirical = $\text{C}_4\text{H}_4\text{O}$

$$n = \frac{\text{Mol. wt}}{\text{Empirical formula wt.}} = \frac{136}{68} = 2$$

$\Rightarrow$ Molecular formula = $\text{C}_8\text{H}_8\text{O}_2$

Exercise 1.4

- The percentage of nitrogen in urea is about:
(1) 38.4 (2) 46.6
(3) 59.1 (4) 61.3
- Insulin contains 3.4% sulphur by mass. What will be the minimum molecular weight of insulin
(1) 94.117 (2) 1884
(3) 941 (4) 976
- Caffeine has a molecular weight of 194. It contains 28.9% by mass of nitrogen. Number of atoms of nitrogen in one molecule of it –
(1) 2 (2) 3 (3) 4 (4) 5
- The haemoglobin of most mammals contains approximately 0.33% of iron by mass. The molecular mass of haemoglobin is 67200. The number of iron atoms in each molecule of haemoglobin is –
(1) 3 (2) 4 (3) 2 (4) 6
- A giant molecule contains 0.25% of a metal whose atomic weight is 59. Its molecule contains one atom of that metal. Its minimum molecular weight is
(1) 5900 (2) 23600
(3) 11800 (4) $\frac{100 \times 59}{0.4}$
- The formula which represents the simple ratio of atoms in a compound is called:
(1) molecular formula
(2) structural formula
(3) empirical formula
(4) rational formula
- Which of the following compounds has same empirical formula as that of glucose: –
(1) CH_3CHO (2) CH_3COOH
(3) CH_3OH (4) C_2H_6
- A compound is found to contain 80% of carbon and 20% of hydrogen, then the molecular formula of the compound is –
(1) C_6H_6 (2) $\text{C}_2\text{H}_5\text{OH}$
(3) C_2H_6 (4) C_2H_4
- A compound contains 69.5% oxygen and 30.5% nitrogen and its molecule weight is 92. The molecular formula of the compound will be –
(1) N_2O (2) NO_2
(3) N_2O_4 (4) N_2O_5
- An element (A) (at wt. = 75) and another element (B) (at. wt. = 25) combine to form a compound. The compound contains 75% (A) by weight. The formula of the compound will be –
(1) A_2B (2) A_3B (3) AB_3 (4) AB
- On analysis, a certain compound was found to contain 254 gm of iodine (at. mass 127) and 80 gm oxygen (at. mass 16). What is the formula of the compound?
(1) IO (2) I_2O (3) I_5O_3 (4) I_2O_5
- A hydrocarbon contains 75% of carbon. Then its molecular formula is
(1) CH_4 (2) C_2H_4
(3) C_2H_6 (4) C_2H_2
- An oxide of sulphur contains 50% of sulphur in it. Its empirical formula is –
(1) SO_2 (2) SO_3
(3) SO (4) S_2O
- A compound of X and Y has an equal mass of them. If their atomic weights are 30 and 20 respectively. The molecular formula of that compound (mol. wt. is 120) could be –
(1) X_2Y_2 (2) X_3Y_3
(3) X_2Y_3 (4) X_3Y_2
- The number of atoms of Cr and O are 4.8×10^{10} and 9.6×10^{10} respectively. Its empirical formula is –
(1) Cr_2O_3 (2) CrO_2
(3) Cr_2O_4 (4) CrO_5
- An oxide of a metal (M) contains 40% by mass of oxygen. Metal (M) has atomic mass of 24. The empirical formula of the oxide is –
(1) M_2O (2) MO
(3) M_2O_3 (4) M_3O_4

17. A hydrocarbon contains 80% C. The vapour density of compound is 30. Empirical formula of compound is: -

- (1) CH_3 (2) C_2H_6
(3) C_4H_{12} (4) C_4H_8

18. The hydrated salt $\text{Na}_2\text{SO}_4 \cdot n\text{H}_2\text{O}$, undergoes 55% loss in mass on heating and becomes anhydrous. The value of n will be:

- (1) 5 (2) 3
(3) 7 (4) 10

19. Element X and Y can combine to form two different compounds. If 1.60 g of X reacts with exactly 1.60 g of Y, the compounds produced has the formula XY_2 . However, under different conditions, 2.40 g of X will react with 1.60 g of Y to form a second compound, whose empirical formula is:

- (1) X_3Y_4 (2) XY
(3) XY_3 (4) X_2Y

Chemical Equation (Stoichiometric Calculation)

Representation of the chemical change in terms of symbol and formula of the reactants & products is called a chemical equation.

Information conveyed by a chemical equation:

- (1) Qualitatively, a chemical equation tells us the names of the various reactants
- (2) Quantitatively, it expresses
 - (a) The relative no. of molecules of reactants and products
 - (b) The relative no. of moles of reactant and products
 - (c) The relative masses of reactants and products
 - (d) The relative volumes of gaseous reactants and products

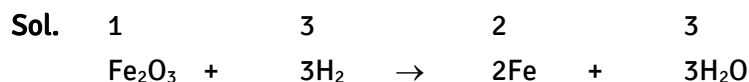
Limitations of chemical equations:

- (1) The physical state of the reactants and products
- (2) The energy changes during chemical reaction
- (3) The conditions of P, T etc at which reaction occurs.
- (4) The rate of chemical reaction

Single Reactant Based:

(A) Type-I (Involving Mass-Mass Relationship):

Ex.13: How much iron can be theoretically obtained in the reduction of 1 kg of Fe_2O_3 by H_2 .



$$n = \frac{\text{weight}}{M_w} = \frac{1000}{160} \text{ mol}$$

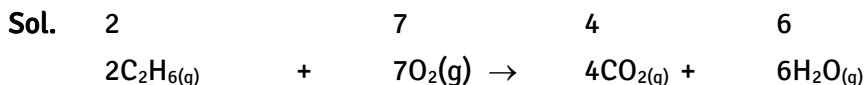
The equation shows that 2 mol of iron are obtained from 1 mol of ferric oxide.

$$\text{Hence, the obtained no. of moles of Fe} = \frac{2 \times 1000}{160} = 12.5 \text{ mol} = \frac{\text{weight}}{\text{Atomic weight}} = \frac{\text{weight}}{56}$$

$$\text{Weight of iron obtained} = 12.5 \times 56 \text{ g} = 700 \text{ g}$$

(B) Type-II (Mass - Volume Relationship):

Ex.14: For complete combustion of 3g ethane the required volume of O_2 & produced volume of CO_2 at STP will be.



$$n = \frac{\text{weight}}{M_w} = \frac{3}{30} = \frac{1}{10} = 0.1 \text{ mol}$$

(a) Required moles of $O_2 = \frac{7}{2} \times 0.1 = 0.35 \text{ mol}$

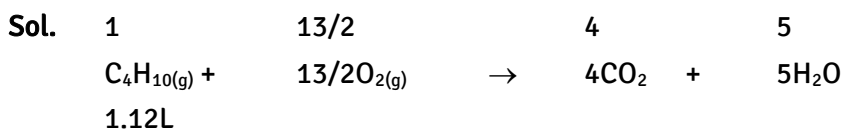
volume of O_2 at STP = $0.35 \times 22.4 = 7.84 \text{ L}$

(b) Produced moles of $CO_2 = \frac{4}{2} \times 0.1 = 0.2 \text{ mol}$

volume of CO_2 at STP = $0.2 \times 22.4 = 4.48 \text{ L}$

(C) Type-III (Volume-Volume Relationship):

Ex.15: For complete combustion of 1.12 L of butane (C_4H_{10}), the produced volume of $H_2O_{(g)}$ & $CO_{2(g)}$ at STP will be.



Volume of $H_2O_{(g)}$ at STP = $5 \times 1.12 = 5.6 \text{ L}$

Volume of $CO_{2(g)}$ at STP = $4 \times 1.12 = 4.48 \text{ L}$

Limiting Reagent

Limiting Reagent (L.R.): The reactant which is completely consumed in a reaction is called as limiting reagent.



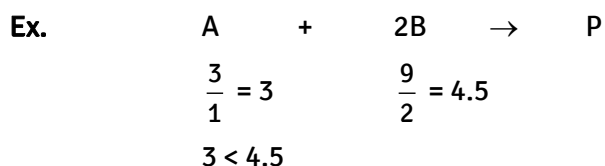
Given

	3 mol	9mol		
	3 – 3	9 – 6	3 mol	6 mol
	= 0 mol	= 3 mol		

L.R. = A

$$x = \frac{\text{given value (may moles, volume, or molecules)}}{\text{Stoichiometry Co-efficient}}$$

Reactants having least value of x are limiting reagents.

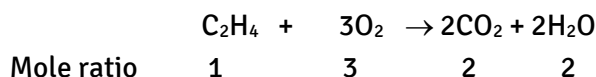


So A is L.R.

Ex.16: Calculate the mass of oxygen required to burn 14g C_2H_4 completely-

- (A) 84 g (B) 48 g (C) 28 g (D) 82 g

Sol. (B)



$$\text{Moles of } C_2H_4 \text{ to be burnt} = \frac{14}{28} = \frac{1}{2} \text{ mole.}$$

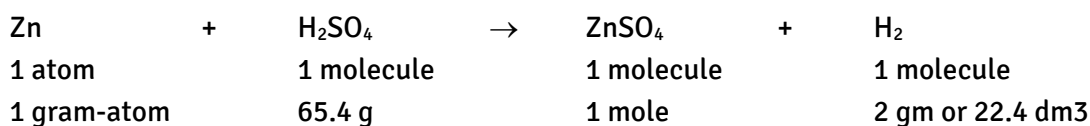
$\therefore$ 1 mole C_2H_4 requires 3 mol O_2 for combustion

$$\therefore \frac{1}{2} \text{ mole } C_2H_4 \text{ requires } 3 \times \frac{1}{2} \text{ mole } O_2 = \frac{3}{2} \text{ mol } O_2. \text{ \& Mass of } O_2 = \frac{3}{2} \times 32 = 48 \text{ gm}$$

Ex.17: Calculate the weight and volume of H_2 at STP that will be displaced by 1 gram of Zn when it is completely dissolved in dilute sulphuric acid.

- (A) 0.3425 dm³ (B) 1.65 dm³ (C) 5.16 dm³ (D) 4.34 dm³

Sol. (A)



$\therefore$ 65.4 g of Zn displaces 2 g of Hydrogen

$$\therefore 1.0 \text{ g of Zn displaces } \frac{2}{65.4} \times 1 = 0.0306 \text{ g of } H_2$$

$\therefore$ 65.4 g of Zn displaces 22.4 dm³ of H_2 at S.T.P.

$$\therefore 1.0 \text{ g of Zn displaces } \frac{22.4}{65.4} \times 1.0 = 0.3425 \text{ dm}^3$$

Ex.18: Find out the mass of O_2 obtained from 90 kg of water?

- (A) 70 kg (B) 80 kg (C) 100 kg (D) 50 kg

Sol. (B) $2H_2O \longrightarrow 2H_2 + O_2$

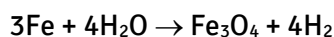
2 mole H_2O forms 1 mole of O_2

$2 \times 18 \text{ g of } H_2O$ forms 32 g of O_2 from above reaction.

$$1 \text{ g of } H_2O \text{ will form } \frac{32}{2 \times 18} \text{ g of } O_2$$

$$\text{So, from } 90 \times 10^3 \text{ g of } H_2O = \frac{32 \times 90 \times 10^3}{2 \times 18} \text{ g of } O_2 = 80 \text{ kg of } O_2$$

Ex.19: Calculate the weight of Fe which will be converted into its oxide (Fe_3O_4) on action of 18g of steam?



- (A) 21 gm (B) 42 gm (C) 63 gm (D) 84 gm

Sol. (B) From the compound given the following can be deduced

Molar ratio 3 mol Fe : 4 mol H_2O

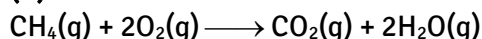
Mass ratio $3 \times 56 \text{ g Fe} : 4 \times 18 \text{ g } H_2O$

$$\text{Using stoichiometry } m_{Fe} = \frac{3 \times 56 \times 18}{4 \times 18} = 42 \text{ gram}$$

Ex.20: Calculate the amount of water (g) liberated on burning of 16g of CH_4 .

- (A) 36 g (B) 72 g (C) 48 g (D) 84 g

Sol. (A)



(i) 16 g of Methane is equal to one mole.

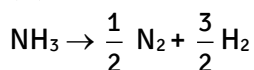
(ii) From above equation, a mol of $\text{CH}_4(\text{g})$ gives 2 moles of $\text{H}_2\text{O}(\text{g})$.

So, mass of $\text{H}_2\text{O} = 2 \times 18 = 36\text{g}$

Ex.21: When 170 g NH_3 ($M = 17$) decomposes how many grams of N_2 & H_2 is produced.

- (A) 30 g, 140 g (B) 140 g, 30 g (C) 70 g, 15 g (D) 15 g, 70 g

Sol. (B)



$$\frac{\text{moles of NH}_3}{1} = \frac{\text{moles of N}_2}{1/2} = \frac{\text{moles of H}_2}{3/2}$$

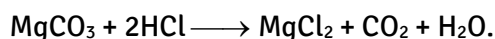
$$\text{So, moles of N}_2 = \frac{1}{2} \times \frac{170}{17} = 5 \text{ mol}$$

$$\text{So, wt. of N}_2 = 5 \times 28 = 140 \text{ g.}$$

$$\text{Similarly, moles of H}_2 = \frac{3}{2} \times \frac{170}{17} = 15 \text{ mol}$$

$$\text{So, wt. of H}_2 = 15 \times 2 = 30 \text{ g.}$$

Ex.22: 4 mole of MgCO_3 is reacted with 6 moles of HCl solution. Find the volume of CO_2 gas produced at STP, the reaction is



- (A) 33.6 litre (B) 3.36 litre (C) 6.72 litre (D) 67.2 litre

Sol. (D)

Here HCl is limiting reagent. So, moles of CO_2 formed = 3.

$$\text{So, vol. at STP} = 3 \times 22.4 = 67.2 \text{ lit.}$$

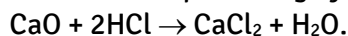
Percentage Yield & Percentage Purity

Percentage Yield:

- (i) The amount of product formed by a chemical reaction is less than the amount predicted by theoretical calculations.
 (ii) The ratio of the amount of product formed to the amount predicted when multiplied by 100 gives the percentage yield

$$\text{Percentage Yield} = \frac{\text{Actual Yield}}{\text{Theoretical Yield}} \times 100$$

Ex.23: Calculate the percentage yield for the reaction,



When 1.85 gm CaCl_2 is formed by reaction of 1.12 gm of CaO with excess of HCl .

Sol. $\text{CaO} + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{H}_2\text{O}$

56 gm CaO will produce 111 gm CaCl_2

$$1.12 \text{ gram of CaO will produce} \rightarrow \frac{111}{56} \times 1.12 = 2.22 \text{ gm}$$

Thus, Theoretical yield = 2.22 gm

Actual yield = 1.85 gm

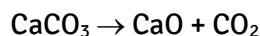
$$\% \text{ yield} = \frac{1.85}{2.22} \times 100 = 83.33\%$$

Percentage Purity:

$$\text{Percentage purity} = \left[\frac{\text{Amount of pure substance}}{\text{Actual amount of sample taken}} \right] \times 100$$

Ex.24: Find the quantity of (CaO) in kg which is produced when 200 kg of lime stone (90% pure CaCO_3) is heated?

Sol. Mass of pure $\text{CaCO}_3 = \frac{200 \times 90}{100} = 180 \text{ kg}$



100 kg 56 kg

180 x

$$\frac{100}{180} = \frac{56}{x} \Rightarrow x = 100.8 \text{ kg}$$

Exercise 1.5

1. The reaction, $2C + 2O_2 \rightarrow 2CO_2$ is carried out by taking 24 g carbon and 96 g O_2 . Which one is limiting reagent?
 (1) C (2) O_2
 (3) CO_2 (4) None of these
 2. According to following reaction;
 $N_2 + 3H_2 \rightarrow 2NH_3$
 If 6 moles of H_2 were used, then find the moles of NH_3 produced.
 (1) 4 mol (2) 2 mol
 (3) 5 mol (4) 6 mol
 3. In a reaction; $A + B_2 \rightarrow AB_2$
 If 300 atoms of A reacted with 200 molecules of B_2 then Identify the limiting reagent.
 (1) B_2
 (2) A
 (3) AB_2
 (4) Both are limiting reagent
 4. For the reaction: $A + 2B \rightarrow C$, 5 mole of A and 8 mole of B will produce –
 (1) 5 mole of C (2) 4 mole of C
 (3) 8 mole of C (4) 13 mole of C
 5. 4 gm of hydrogen are ignited with 4 gm of oxygen. The weight of water formed is-
 (1) 0.5 gm (2) 3.5 gm
 (3) 4.5 gm (4) 2.5 gm
 6. 9 gm of Al will react, with
 $2Al + \frac{3}{2}O_2 \rightarrow Al_2O_3$
 (1) 6 gm O_2 (2) 8 gm O_2
 (3) 9 gm O_2 (4) 4 gm O_2
 7. 12 lit. of H_2 and 11.2 lit. of Cl_2 are mixed and exploded. The composition by volume of mixture is-
 (1) 24 lit. of HCl (g)
 (2) 0.8 lit. Cl_2 and 20.8 lit. HCl (g)
 (3) 0.8 lit. H_2 and 22.4 lit. HCl (g)
 (4) 22.4 lit. HCl (g)
 8. The equation:
 $2Al(S) + \frac{3}{2}O_2(g) \rightarrow Al_2O_3(S)$ shows that
 (1) 2 mole of Al reacts with $\frac{3}{2}$ mole of O_2 to produce $\frac{7}{2}$ mole of Al_2O_3
 (2) 2 gm of Al reacts with $\frac{3}{2}$ g of O_2 to produce one mole of Al_2O_3
 (3) 2 gm mole of Al reacts with $\frac{3}{2}$ lite of O_2 to produce 1 mole of Al_2O_3
 (4) 2 mole of Al reacts with $\frac{3}{2}$ mole of O_2 to produce 1 mole of Al_2O_3

9. What quantity of limestone (CaCO_3) on heating will give 56 kg of CaO ?
 (1) 1000 kg (2) 10 kg
 (3) 50 kg (4) 100 kg
10. A mixture containing 100 gm H_2 and 100 gm O_2 is ignited so that water is formed according to the reaction, $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$; How much water will be formed?
 (1) 112.5 gm (2) 50 gm
 (3) 25 gm (4) 200 gm
11. The moles of O_2 required for reacting with 6.8 g of ammonia
 $4\text{NH}_3 + 5\text{O}_2 \rightarrow 4\text{NO} + 6\text{H}_2\text{O}$ is-
 (1) 5 (2) 2.5 (3) 1 (4) 0.5
12. In a gaseous reaction of the type
 $a\text{A} + b\text{B} \rightarrow c\text{C} + d\text{D}$,
 which statement is wrong?
 (1) a litre of A combines with b litre of B to give C and D
 (2) a mole of A combines with b moles B of to give C and D
 (3) a gm of A combines with b gm of B to give C and D
 (4) a molecule of A combines with b molecules of B to give C and D
13. How many moles of ethane are required to produce 44 g of $\text{CO}_2(\text{g})$, after the combustion of ethane?
 (1) 2 (2) 1
 (3) 0.5 (4) 0.25
14. What volume of oxygen gas (O_2) measured at 0°C and 1 atm, is needed to burn completely 1L, of propane gas (C_3H_8) measured under the same conditions?
 (1) 5 L (2) 10 L
 (3) 7 L (4) 6 L
15. 0.5 mole of H_2SO_4 is mixed with 0.2 mole of $\text{Ca}(\text{OH})_2$. The maximum number of moles of CaSO_4 formed is
 (1) 0.2 (2) 0.5
 (3) 0.4 (4) 1.5
16. If 1.6 gm of SO_2 and 1.5×10^{22} molecules of H_2S are mixed and allowed to remain in contact in a closed vessel until the reaction proceeds to completion.
 $2\text{H}_2\text{S} + \text{SO}_2 \rightarrow 3\text{S} + 2\text{H}_2\text{O}$
 Which of the following statement is true?
 (1) Only S' and H_2O ' remains in the reaction vessel
 (2) ' H_2S ' will remain in excess
 (3) ' SO_2 ' will remain in excess
 (4) None of these
17. 26 CC of CO_2 are passed over red-hot coke. The volume of CO evolved is -
 (1) 15 CC (2) 10 CC
 (3) 32 CC (4) 52 CC
18. The number of litres of air required to burn 8 litres of C_2H_2 is approximately
 (1) 40 (2) 60
 (3) 80 (4) 100
19. The volume of gas C_2H_2 at NTP produced by 100g of CaC_2 with water is: -
 $\text{CaC}_2 + 2\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{C}_2\text{H}_2$
 (1) 70L (2) 35L
 (3) 17.5L (4) 22.4L
20. 170g of sample of Al_2O_3 produce 18g Al. According to following equation
 $\text{Al}_2\text{O}_3 \rightarrow 2\text{Al} + \frac{3}{2}\text{O}_2$
 Determine the percentage purity of Al_2O_3 in the original sample
 (1) 2% (2) 10%
 (3) 30% (4) 20%

21. 100 mL of PH_3 when decomposed produces phosphorus and hydrogen. The change in volume is:
 (1) 50 mL increase (2) 500 mL decrease
 (3) 900 mL decrease (4) None of these
22. What is the mass of oxygen that is required for the complete combustion of 2.8 kg of ethylene?
 (1) 2.8 kg (2) 6.4 kg
 (3) 96.0 kg (4) 9.6 kg
23. 1.35 g of pure Ca metal was quantitatively converted into 1.88 g of pure CaO, what is atomic mass of Ca?
 (1) 40.75 (2) 50
 (3) 60 (4) 70
24. How many moles of potassium chlorate (KClO_3) need to be heated to produce 22.4 L oxygen at NTP
 $\text{KClO}_3 \rightarrow \text{KCl} + \frac{3}{2}\text{O}_2$
 (1) $\frac{1}{3}$ mol (2) $\frac{3}{2}$ mol
 (3) $\frac{2}{3}$ mol (4) 1 mol
25. How many grams of CaO will be produced on the decomposition of 40g of 20% pure CaCO_3
 (1) 4.48g (2) 2.24g
 (3) 1.12g (4) 0.6g
26. How many grams of 40% pure FeO can produce 5.6g Fe according to given equation. (At. wt. of Fe = 56)
 $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
 (1) 7.2g (2) 18g
 (3) 9g (4) 27g
27. MgCO_3 sample is 10% pure. Volume of CO_2 collected at STP when 84g of MgCO_3 sample is decomposed
 (1) 2.24 L (2) 22.4 L
 (3) 11.2 L (4) 1.12 L
28. Calculate the percentage yield for the reaction
 $\text{MgCO}_3 \rightarrow \text{MgO} + \text{CO}_2$
 When 2g of MgO is produced by the decomposition of 21g MgCO_3
 (1) 40% (2) 60%
 (3) 20% (4) 50%

Equivalent Weight

The equivalent weight of a substance is the number of parts by weight of the substance that combine displace directly or indirectly 1.008 parts by weight of hydrogen or 8 parts by weight of oxygen or 35.5 parts by weight of chlorine or 108 parts by weight of Ag.

(a) Calculation of Equivalent Weight:

(i)
$$\text{Equivalent weight} = \frac{\text{Atomic weight}}{\text{Valency factor}}$$

(ii)
$$\text{Equivalent weight of ions} = \frac{\text{formula weight of ion}}{\text{Valency}}$$

(iii)
$$\text{Equivalent weight of ionic compound} = \text{equivalent weight of cation} + \text{equivalent weight of anion}$$

Ex.
$$\text{Equivalent weight of } \text{H}_2\text{SO}_4 = \text{Equivalent weight of } \text{H}^+ + \text{Equivalent weight of Anion } (\text{SO}_4^{-2})$$

$$= 1 + 48 = 49$$

(iv)
$$\text{Equivalent weight of acid / base} = \frac{\text{Molecular weight}}{\text{Basicity / Acidity}}$$

(v)
$$\text{Equivalent weight of salt} = \frac{\text{Molecular weight}}{\text{Total charge on cation or anion}}$$



Total charge on cation or anion is 2 molecular weights of Na_2SO_4 is $= (2 \times 23 + 32 + 16 \times 4) = 142$

$$\text{Equivalent weight of } \text{Na}_2\text{SO}_4 = \frac{142}{2} = 71$$

(vi) Equivalent weight of an oxidizing or reducing agent

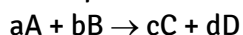
$$= \frac{\text{Molecular weight of the substance}}{\text{Number of electrons gain/lost by one molecule}}$$

(b) **Concept of Gram Equivalent or Law of Chemical Equivalence:**

$$\text{Number of gram equivalent} = \frac{W_{(\text{ram})}}{E} = n \times \text{valence factor; where}$$

$$\left(\text{Normality} = \frac{\text{number of gram equivalent of solute}}{\text{volume of solution in (L)}} \right)$$

According to it in a reaction equal gram equivalent of reactant reacts to give same number of gram equivalent of products.



Number of gram equivalent of A = Number of gram equivalent of B = Number of gram equivalent of C = Number of gram equivalent of D

(c) **Methods for Determination of the Equivalent Weights:**

(i) **Hydrogen displacement method:** This method is used for those elements which can evolve hydrogen from acids, i.e., active metals.

$$\text{equivalent weight of metal} = \frac{\text{weight of metal}}{\text{weight of } \text{H}_2 \text{ gas (displaced)}} \times 1.008$$

(ii) **Oxide formation method:** A known mass of the element is changed into oxide directly or indirectly. The mass of oxide is noted

$\Rightarrow$ weight of oxygen = weight of oxide weight of element

$$\text{equivalent weight of element} = \frac{\text{Weight of element}}{\text{weight of oxygen}} \times 8$$

(iii) **Chloride formation method:** A known mass of the element is changed into chloride directly or indirectly. The mass of the chloride is determined.

$$\text{equivalent weight of element} = \frac{\text{weight of element}}{\text{weight of chlorine}} \times 35.5$$

(iv) **Metal to metal displacement method:** More active metal can displace less active metal from its salt's solution. The mass of the displaced metal bears the same ratio as their equivalent weights.

$$\frac{m_1}{m_2} = \frac{E_1}{E_2}$$

(v) **Double decomposition method:** this method is based on the following points –

(a) The mass of the compound reacted and the mass of product formed are in the ratio of their equivalent masses.

(b) The equivalent mass of the compound (electrovalent) is the sum of equivalent masses of its radicals.

(c) The equivalent mass of a radical is equal to the formula mass of the radical divided by its charge.

$$\frac{\text{Mass of AB}}{\text{Mass of AD}} = \frac{\text{Equivalent mass of AB}}{\text{Equivalent mass of AD}} = \frac{\text{Equivalent mass of A} + \text{Equivalent mass of B}}{\text{Equivalent mass of A} + \text{Equivalent mass of D}}$$

Methods for Calculation of Atomic Weights and Molecular Weights

(a) Methods for Determination of Atomic Weight:

(i) Atomic weight = equivalent weight $\times$ valency

(ii) **Dulong and Petit's law:** This law is applicable only for solids (except Be, B, Si, C)

$$\text{Atomic mass} \times \text{specific heat (in cal / gram} \times ^\circ\text{C)} \approx 6.4 \text{ or atomic mass (approximate)} = \frac{6.4}{\text{specific heat}}$$

(iii) **Law of isomorphism:** Isomorphous substances form crystals which have same shape and size and can grow in the saturated solution of each other.

Examples of isomorphous compounds:

(1) H_2SO_4 and K_2CrO_4

(2) $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ and $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$

(3) KClO_4 and KMnO_4

(4) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$ and $\text{K}_2\text{SO}_4 \cdot \text{Cr}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

Conclusions:

Masses of two elements that combine with same mass of other elements in their respective compounds are in the ratio of their atomic masses.

$$\frac{\text{Mass of one element (A) that combine with a certain mass of other element}}{\text{Mass of other element (B) that combine with the same mass of other element}} = \frac{\text{Atomic mass of A}}{\text{Atomic mass of B}}$$

(iv) Volatile chloride method:

Required condition - chloride of element should be vapour.

Required data - (i) Vapour density of chloride.

(ii) Equivalent weight of element.

Let the valency of the element be x . The formula of its chloride will be MCl_x

Molecular weight = Atomic weight of $M + 35.5x$

Atomic weight = Equivalent weight $\times$ valency or $A = E \times x$

$$\therefore \text{Molecular weight} = Ex + 35.5x \text{ or } 2 \times \text{V.D.} = x(E + 35.5) \text{ or } x = \frac{2 \times \text{V.D.}}{E + 35.5}$$

(v) **Specific heat method:** If $\frac{C_p}{C_v} = \gamma$ is given, then

Case I. If $\gamma = 5/3 = 1.66$ Atomicity will be one

Case II. If $\gamma = 7/5 = 1.4$ Atomicity will be two

Case III. If $\gamma = 4/3 = 1.33$ Atomicity will be three

(b) Methods for determination of molecular weight:

(i) Molecular weight = $2 \times \text{V.D.}$

(ii) Victor Mayer's Method

Laws of Chemical Combination

(a) Law of Mass Conservation (Law of Indestructibility of Matter):

"It was given by Lavoisier and tested by Landolt"

According to this law, the mass can neither be created nor be destroyed in a balanced chemical reaction or physical reaction. But one form is changed into another form is called as law of mass conservation.

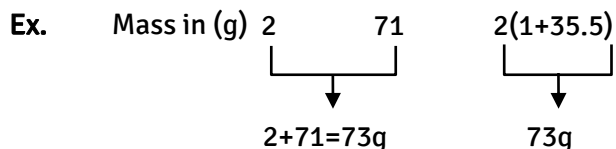
If the reactant is completely converted in products, then the sum of the mass of reactants is equal to the sum of the mass of products.

Total mass of reactants = Total mass of product

Total mass of reactants = Total mass of products

If reactants are not completely consumed then the relationship will be:

Total mass of reactants = Total mass of product + Mass of unreacted reactants



(b) Law of Definite Proportion / Law of Constant Composition:

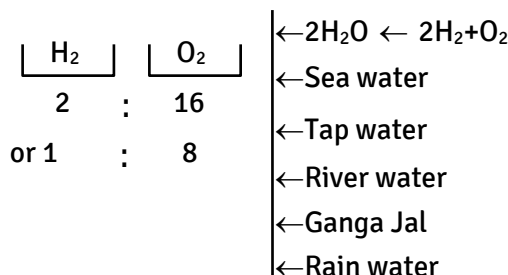
"It was given by Proust."

According to this law, a compound can be obtained from different sources. But the ratio of each component (by weight) remains same. i.e. it does not depend on the method of its preparation or the source from which it has been obtained.

For example:

molecule of ammonia always has the formula NH_3 . That is one molecule of ammonia always contains, one atom of nitrogen and three atoms of hydrogen or 17.0 g of NH_3 always contains 14.0 g of nitrogen and 3 g of hydrogen.

Ex.: Water can be obtained from different sources but the ratio of weight of H and O remains same.



(c) Law of Multiple Proportion:

"It was given by John Dalton"

According to law of Multiple proportion if two elements combine to form more than one compound then the different mass of one element which combine with a fixed mass of other element bear a simple ratio to one another. The following examples illustrate this law.

- (i) Nitrogen and Oxygen combine to form five oxides, which are Nitrous oxide (N_2O), nitric oxide (NO), nitrogen trioxide (N_2O_3), nitrogen tetroxide (N_2O_4) and nitrogen pentoxide (N_2O_5). Weights of oxygen which combine with the fixed weight of nitrogen in these oxides are calculated as Oxide Ratio of weights of nitrogen and oxygen in each compound

N_2O	NO	N_2O_3	N_2O_4	N_2O_5
28 : 16	14 : 16	28 : 48	28 : 64	28 : 80

Number of parts by weight of oxygen which combine with 14 parts by weight of nitrogen from the above are 8, 16, 24, 32 and 40 respectively. Their ratio is 1 : 2 : 3 : 4 : 5, which is a simple ratio. Hence, the law is illustrated.

- (ii) Sulphur combines with oxygen to form two oxides SO_2 and SO_3 the weights of oxygen which combine with a fixed weight of sulphur, i.e. 32 parts by weight of sulphur in two oxides are in the ratio of 32 : 48 or 2 : 3 which is a simple ratio. Hence the law of multiple proportions is illustrated.

(d) **Law of Gaseous Volume:**

"It was given by **Gay Lussac's**"

According to this law, in the gaseous reaction, the reactants are always combined in a simple ratio by volume and form products, which is **simple ratio by volume** at same temperature and pressure.

Ex.:

One volume of hydrogen combines with one volume of chlorine to produce 2 volumes of hydrogen chloride.

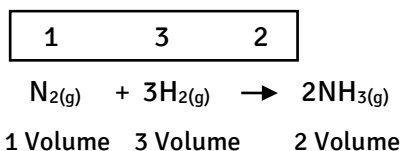
Simple ratio = 1 : 1 : 2.



Ex.:

One volume of nitrogen combines with 3 volumes of hydrogen to form 2 volumes of ammonia.

Simple ratio = 1 : 3 : 2

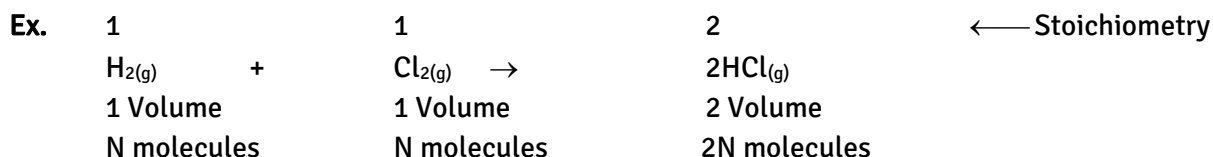


Special Note:

This law is used only for gaseous reaction. It relates volume to mole or molecules. But not relate with mass.

(e) **Avogadro's law:**

"Equal volume of all gases contain equal number of molecules at same temperature and pressure.



Exercise 1.6

- A, E, M and n are the atomic weight, equivalent weight, molecular weight and valency of an element. The correct relation is:
 (1) $A = E \times n$ (2) $A = M/E$
 (3) $A = M/n$ (4) $M = A \times n$
- If equivalent weight of S in SO_2 is 8 then equivalent weight of S in SO_3 is
 (1) $\frac{8 \times 2}{3}$ (2) $\frac{8 \times 3}{3}$
 (3) $8 \times 2 \times 3$ (4) $\frac{2 \times 3}{8}$
- Equivalent weight of bivalent metal is 32.7. Molecular weight of its chloride is –
 (1) 68.2 (2) 103.7
 (3) 136.4 (4) 166.3
- Molecular weight of tribasic acid is W. Its equivalent weight will be
 (1) $\frac{W}{2}$ (2) $\frac{W}{3}$
 (3) W (4) 3 W
- In a compound A_xB_y ,
 (1) Mole of A = mole of B = mole of A_xB_y
 (2) eq. of A = eq. of B = eq. of A_xB_y
 (3) yx mole of A = yx mole of B = $(x + y) \times$ mole of A_xB_y
 (4) $y \times$ mole of A = $y \times$ mole of B
- 0.84 gm. of metal hydride contains 0.04 gm of hydrogen. The equivalent wt. of metal is
 (1) 80 (2) 40
 (3) 20 (4) 60
- 2.8 g of iron displaces 3.2 g of copper from a solution of copper sulphate solution. If the equivalent mass of iron is 28, then equivalent mass of copper will be:
 (1) 16 (2) 32
 (3) 48 (4) 64
- 1.0 gm of a metal combines with 8.89 gm of Bromine. Equivalent weight of metal is nearly: (at. wt. of Br = 80)
 (1) 8 (2) 9 (3) 10 (4) 7
- 0.45 gm of acid (molecular wt. = 90) was exactly neutralised by 20 ml. of 0.5 N NaOH. Basicity of the acid is –
 (1) 1 (2) 2 (3) 3 (4) 4
- If m_1 gm of a metal A displaces m_2 g of another metal B from its salt solution and if their equivalent weights are E_2 and E_1 respectively then the equivalent weight of A can be expressed by
 (1) $\frac{m_1}{m_2} \times E_2$ (2) $\frac{m_2}{m_1} \times E_2$
 (3) $\frac{m_1}{m_2} \times E_1$ (4) $\frac{m_2}{m_1} \times E_1$
- 0.84 gm of a metal carbonate reacts exactly with 40 ml of $\frac{N}{2} \text{H}_2\text{SO}_4$. The equivalent weight of the metal carbonate is –
 (1) 84 (2) 64 (3) 42 (4) 32
- A metal oxide is reduced by heating it in a stream of hydrogen. It is found that after complete reduction 3.15 gm of the oxide have yielded 1.05 gm of the metal. We may conclude that
 (1) Atomic weight of the metal is 4
 (2) Equivalent weight of the metal is 8
 (3) Equivalent weight of the metal is 4
 (4) Atomic weight of the metal is 8
- If 1.2 g of a metal displaces 1.12 litre of hydrogen at NTP, equivalent mass of the metal would be –
 (1) 1.2×11.2 (2) 12
 (3) 24 (4) $1.2 + 11.2$

14. Equivalent weight of a divalent metal is 24. The volume of hydrogen liberated at STP by 12 gm of the same metal when added to excess of an acid solution is –
 (1) 2.8 litres (2) 5.6 litres
 (3) 11.2 litres (4) 22.4 litres
15. 2 g of a base whose eq. wt. is 40 reacts with 3 g of an acid. The eq. wt. of the acid is
 (1) 40 (2) 60 (3) 10 (4) 80
16. 1.6 gm of Ca and 2.60 gm of Zn when treated with an acid in excess separately, produced the same amount of hydrogen. If the equivalent weight of Zn is 32.6, what is the equivalent weight of Ca: –
 (1) 10 (2) 20 (3) 40 (4) 5
17. When an element forms an oxide in which oxygen is 20% of the oxide by mass, the equivalent mass of the element will be –
 (1) 32 (2) 40 (3) 60 (4) 128
18. 0.126 g of acid requires 20 ml of 0.1 N NaOH for complete neutralisation. Eq. wt. of the acid will be:
 (1) 45 (2) 53 (3) 40 (4) 63
19. A_1 g of an element gives A_2 g of its oxide. The equivalent mass of the element is –
 (1) $\frac{A_2 - A_1}{A_1} \times 8$ (2) $\frac{A_2 - A_1}{A_2} \times 8$
 (3) $\frac{A_1}{A_2 - A_1} \times 8$ (4) $(A_2 - A_1) \times 8$
20. The specific heat of an element is 0.214 cal/gm°C. The approximate atomic weight is –
 (1) 0.6 (2) 12 (3) 30 (4) 65
21. Vapour density of metal chloride is 77. Equivalent weight of metal is 3, then its atomic weight will be –
 (1) 3 (2) 6 (3) 9 (4) 12
22. 74.5 g of a metallic chloride contains 35.5 g of chlorine. The equivalent mass of the metal is –
 (1) 19.5 (2) 35.5
 (3) 39.0 (4) 78.0
23. The chloride of a metal contains 71% chlorine by weight and the vapour density of it is 50. The atomic weight of the metal will be –
 (1) 29 (2) 58
 (3) 35.5 (4) 71
24. The carbonate of a metal is isomorphous with MgCO_3 and contains 6.091% of carbon. Atomic weight of the metal is nearly –
 (1) 48 (2) 68.5
 (3) 137 (4) 120
25. 71 gm of chlorine combines with a metal giving 111 gm of its chloride. The chloride is isomorphous with $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$. The atomic mass of the metal is –
 (1) 20 (2) 30
 (3) 40 (4) 69
26. A metal M forms a sulphate which is isomorphous with $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$. If 0.6538 gm of metals M displaced 2.16 gm of silver from silver nitrate solution, then the atomic weight of the metal M is
 (1) 32.61 (2) 56.82
 (3) 65.38 (4) 74.58
27. Specific heat of a solid element is 0.1 Cal/gm°C and its equivalent weight is 31.8 its exact atomic weight is:
 (1) 31.8 (2) 63.6
 (3) 318 (4) 95.4
28. The equivalent weight of an element is 4. Its chloride has a vapour density 59.25. The valency of the element is –
 (1) 4 (2) 3 (3) 2 (4) 1

29. Vapour density of a gas is 16. The ratio of specific heat at constant pressure to specific heat at constant volume is 1.4, then its atomic weight is –
(1) 8 (2) 16
(3) 24 (4) 32
30. The atomic weight of a metal (M) is 27 and its equivalent weight is 9. The formula of its chloride will be –
(1) MCl (2) MCl_2
(3) M_3Cl (4) MCl_3
31. A container of volume V, contains 0.28 gm of N_2 gas. If same volume of an unknown gas under similar condition of temperature and pressure weighs, 0.44 gm, the molecular mass of the gas is
(1) 22 (2) 44 (3) 66 (4) 88
32. 20 g of an acid furnishes 0.5 moles of H_3O^+ ions in its aqueous solution. The value of g equivalent of the acid.
(1) 40 g (2) 20 g
(3) 10 g (4) 100 g
33. 1.5 g of a divalent metal displaced 4 g of copper (atomic mass = 63.8) from a solution of copper sulphate. The atomic mass of the metal is:
(1) 12 (2) 24 (3) 48 (4) 6
34. Which one of the following pairs of compounds illustrate the law of multiple proportions?
(1) H_2O , Na_2O (2) MgO , Na_2O
(3) Na_2O , BaO (4) SnCl_2 , SnCl_4
35. In the reaction $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$, ratio by volume of N_2 , H_2 and NH_3 is 1 : 3 : 2. This illustrates law of –
(1) Definite proportion
(2) Multiple proportion
(3) Reciprocal proportion
(4) Gaseous volumes
36. Different proportions of oxygen in the various oxides of nitrogen prove the law of-
(1) Equivalent proportion
(2) Multiple proportion
(3) Constant proportion
(4) Conservation of matter
37. Two flask A & B of equal capacity of volume contain NH_3 and SO_2 gas respectively under similar conditions which flask has more number of moles –
(1) A
(2) B
(3) Both have same moles
(4) None
38. If law of conservation of mass was to hold true, then 20.8 gm. of BaCl_2 on reaction with 9.8 gm. of H_2SO_4 will produce 7.3 gm. of HCl and BaSO_4 equal to –
(1) 11.65 gm.
(2) 23.3 gm.
(3) 25.5 gm.
(4) 30.6 gm.
39. After a chemical reaction, the total mass of reactants and products
(1) is always increased
(2) is always decreased
(3) is not changed
(4) is always less or more
40. Four ten litre flasks are separately filled with the gases hydrogen, helium, oxygen and nitrogen at same room temp. and pressure. The ratio of total number of atoms of these gases present in the different flasks would be –
(1) 1 : 1 : 1 : 1
(2) 1 : 2 : 2 : 3
(3) 2 : 1 : 2 : 2
(4) 2 : 1 : 3 : 2

Definition of Solution:

When two or more chemically non-reacting substances are mixed together forming homogeneous mixture the mixture is called solution.

When the solution is composed of only two chemical substances, it is termed as binary solution, similarly, it is called ternary and quaternary if it is composed by three and four components respectively.

For binary solution: Solution = solute + solvent

- Generally, the component present in lesser amount than other component in solution, is called solute.
- Generally, the component present in greater amount than all other component, is called solvent.
- Physical state of solvent and solution is same

Solution	Solute+ Solvent	
	(B)	(A)
moles	n	N
mass	$W_{(g)}$	$W_{(g)}$
molar mass	m	M
mole fraction	X_B	X_A

Ex.: In a syrup (liquid solution) containing 60g sugar (a solid) and 40g water (a liquid) same aggregation as solution water is termed as the solvent.

Ex.: In a solution of alcohol and water, having 10 mL alcohol and 20 mL water, water is solvent and alcohol will be solute.

- On the basis of amount of solute, solutions can be classified in two ways
 - (a) Dilute solution: A solution in which relatively a small amount of solute is dissolved in large amount of solvent is called a dilute solution.
 - (b) Concentrated solution: A solution in which relatively a large amount of solute is dissolved in large amount of solvent is called a dilute solution.

Concentration Terms**Molarity (M):**

- (i) Molarity is defined as the number of moles of the solute per liter of solution. Molarity expressed in mol/litre or mol/dm³.
- (ii) Mathematically,

$$M = \frac{\text{No. of moles of solute}(n)}{\text{Vol. of solution in litres}} = \frac{\text{wt. of solute(gm)}}{\text{gm mol. wt. of solute}} \times \frac{1000}{\text{vol. of solution(ml)}}$$
- (iii) Temperature dependent

Molality (m):

- (i) Molality is defined as the number of moles of the solute per 1000 g of the solvent.
- (ii) Mathematically, molality can be calculated by the following formulae,

$$m = \frac{\text{Number of moles of solute}}{\text{weight of solvent in kg}} = \frac{\text{Number of moles of solute}}{\text{weight of solvent in gm}} \times 1000$$

$$m = \frac{\text{wt. of solute}}{\text{mol. wt. of solute}} \times \frac{1000}{\text{Weight of solvent in gm}}$$
- (iii) Unit of molality is mol/kg.
- (iv) Temperature independent

In terms of percentage:

$$(i) \text{ Weight by weight percentage (\%w/W)} = \frac{\text{Wt. of solute(g)}}{\text{Wt. of solution(g)}} \times 100$$

e.g. 10% Na₂CO₃ solution w/W means 10g of Na₂CO₃ is dissolved in 100 g of the solution.

$$(ii) \text{ Weight by volume percent (\%w/V)} = \frac{\text{Wt. of solute(g)}}{\text{Volume of solution(cm}^3\text{)}} \times 100$$

e.g., 10% Na₂CO₃ (w/V) means 10 g Na₂CO₃ is dissolve in 100 cm³ of solution.

$$(iii) \text{ Volume by volume percent (\%v/V)} = \frac{\text{Volume of solute(cm}^3\text{)}}{\text{Volume of solution(cm}^3\text{)}} \times 100$$

e.g., 10 cm³ of ethanol dissolved in 100 cm³ of solution means 10% ethanol (v/V)

Parts per million (ppm):

$$\text{ppm} = \frac{\text{Mass of solute}}{\text{Total mass of solution}} \times 10^6$$

Mole fraction (X):

- (i) It is defined as ratio of number of moles of a component to total moles of all components (solvent and solute) present in solution.
- (ii) It is denoted by the letter X.
- (iii) Unitless
- (iv) Temperature Independent

Ex: if weight of A & B are W_A & W_B and molar mass are M_A & M_B then calculate mole fraction of A and B.

$$\text{Number of moles of component A is given by, } n_A = \frac{W_A}{M_A}$$

$$\text{Number of moles of component B is given by, } n_B = \frac{W_B}{M_B}$$

$$\text{Total number of moles of A and B} = n_A + n_B$$

$$\text{Moles fraction of A, } X_A = \frac{n_A}{n_A + n_B}$$

$$\text{Moles fraction of B, } X_B = \frac{n_B}{n_A + n_B}$$

In a solution for all component the sum of mole fractions is always unity.

$$X_A + X_B = \frac{n_A}{n_A + n_B} + \frac{n_B}{n_A + n_B} = 1$$

Exercise 1.7

1. Molality is expressed in units of
(1) mol kg^{-1} (2) mol L^{-1}
(3) $\text{mol L}^{-1} \text{s}^{-1}$ (4) $\text{mol g}^{-1} \text{s}^{-1}$
2. Which of the following is correct?
(1) The sum of mole fractions of all the components in a solution is always unity
(2) Mole fraction depends upon temperature
(3) Mole fraction is always negative
(4) Mole fraction is independent of content of solute in solution
3. Which of the following methods of expressing concentration varies with temperature?
(1) Molality (2) Weight percent
(3) Molarity (4) Mole fraction
4. What is the molarity of NaOH solution if 250 ml of it contains 1 mg of NaOH?
(1) 10^{-1} M (2) 10^{-2} M
(3) 10^{-4} M (4) 10^{-3} M
5. How many g of NaOH will be needed to prepare 250 mL of 0.1 M solution?
(1) 1 g (2) 10 g
(3) 4 g (4) 6 g
6. 8 g NaOH is dissolved in one litre of solution, its molarity is:
(1) 0.8 M (2) 0.4 M
(3) 0.2 M (4) 0.1 M
7. If 18 g of glucose is present in 1000 g of solvent, the solution is said to be:
(1) 1 molar (2) 0.1 molar
(3) 0.5 molar (4) 0.1 molal
8. A molal solution is one that contains one mole of a solute in
(1) 1000 g of the solvent
(2) One litre of the solvent
(3) One litre of the solution
(4) 22.4 litres of the solution
9. Number of HCl molecules present in 10 ml of 0.1 M solution is:
(1) 6.022×10^{23}
(2) 6.023×10^{22}
(3) 6.022×10^{21}
(4) 6.022×10^{20}
10. 1000g aqueous solution of CaCO_3 contains 10g of CaCO_3 . Concentration of the solution is
(1) 10 ppm (2) 100 ppm
(3) 1000 ppm (4) 10000 ppm
11. The mole fraction of oxygen in a mixture of 7g of nitrogen and 8g of oxygen is:
(1) 8/15 (2) 0.5
(3) 0.25 (4) 1
12. The molality of 15% (w/vol.) solution of H_2SO_4 of density 1.1 g/cm^3 is approximately:
(1) 1.2 (2) 1.4
(3) 1.8 (4) 1.6

Exercise 2

- What is the weight of 3.01×10^{23} molecules of ammonia to
 (1) 17 g (2) 8.5 g
 (3) 34 g (4) None of these
- How many molecules are present in one mL of water vapours at STP to?
 (1) 1.69×10^{19} (2) 2.69×10^{-19}
 (3) 1.69×10^{-19} (4) 2.69×10^{19}
- The amount of sulphur required to produce 100 moles of H_2SO_4 is –
 (1) 3.2×10^3 g (2) 32.65 g
 (3) 32 g (4) 3.2 g
- How many molecules are in 5.23 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) to?
 (1) 1.65×10^{22} (2) 1.75×10^{22}
 (3) 1.75×10^{21} (4) None of these
- The weight of 1 mole of a gas of density 0.1784 g/l at NTP is
 (1) 0.1784 g (2) 1 g
 (3) 4 g (4) Can not be Calculate
- A sample of ammonium phosphate $(\text{NH}_4)_3\text{PO}_4$ contains 3.18 mol of H atoms. The number of mol of O atoms in the sample is:
 (1) 0.265 (2) 0.795
 (3) 1.06 (4) 3.18
- How many moles of magnesium phosphate, $\text{Mg}_3(\text{PO}_4)_2$ will contain 0.25 mole of oxygen atoms?
 (1) 2.5×10^{-2} (2) 0.02
 (3) 3.125×10^{-2} (4) 1.25×10^{-2}
- If 3.01×10^{20} molecules are removed from 98 mg of H_2SO_4 , then the number of moles of H_2SO_4 left are –
 (1) 0.1×10^{-3} (2) 0.5×10^{-3}
 (3) 1.66×10^{-3} (4) 9.95×10^{-2}
- Number of molecules in 100 ml of each of O_2 , NH_3 and CO_2 at STP are
 (1) In the order $\text{CO}_2 < \text{O}_2 < \text{NH}_3$
 (2) In the order $\text{NH}_3 < \text{O}_2 < \text{CO}_2$
 (3) The same
 (4) $\text{NH}_3 < \text{CO}_2 < \text{O}_2$
- The number of water molecules in 1 liter of water is
 (1) 18 (2) 18×1000
 (3) N_A (4) 55.55 N_A
- The numbers of moles of BaCO_3 which contain 1.5 moles of oxygen atoms is
 (1) 0.5 (2) 1
 (3) 3 (4) 6.02×10^{23}
- 2g of oxygen contains number of atoms equal to that in
 (1) 0.5g of hydrogen (2) 4g of sulphur
 (3) 7g of nitrogen (4) 2.3g of sodium
- The number of gram atoms of oxygen present in 0.3-gram mole of $(\text{COOH})_2 \cdot 2\text{H}_2\text{O}$ is
 (1) 0.6 (2) 1.8
 (3) 1.2 (4) 3.6
- The number of atoms present in 0.5 g atom of nitrogen is same as the atoms in –
 (1) 12g of C (2) 32g of S
 (3) 8g of oxygen (4) 24g of Mg
- Rearrange the following (I to IV) in the order of increasing masses and choose the correct answer from (1), (2), (3) and (4) (Atomic mass: N=14, O=16, Cu=63).
 I. 1 molecule of oxygen
 II. 1 atom of nitrogen
 III. 1×10^{-10} g molecular weight of oxygen
 IV. 1×10^{-10} g atomic weight of copper
 (1) II<I<III<IV (2) IV<III<II<I
 (3) II<III<I<IV (4) III<IV<I<II

16. The number of molecules in 8.96 L of a gas at 0°C and 1 atmospheric pressure is approximately
 (1) 6.02×10^{23} (2) 12.04×10^{23}
 (3) 18.06×10^{23} (4) 24.08×10^{22}
17. The atomic weights of two elements 'A' and 'B' are 40 and 80, respectively. If x g of 'A' contains y atoms, how many atoms are present in 2x g of 'B'?
 (1) $\frac{y}{2}$ (2) $\frac{y}{4}$ (3) y (4) 2y
18. Elements A and B form two compounds B_2A_3 and B_2A . 0.05 moles of B_2A_3 weight 9.0 gm and 0.10 mole of B_2A weight 10 gm atomic weight of A and B are-
 (1) 20 and 30 (2) 30 and 40
 (3) 40 and 30 (4) 30 and 20
19. Calculate the gram quantity of Na_2CO_3 which has same no. of atoms as the no. of protons present in 10 g CaCO_3 to.
 (1) 20 g (2) 88.33 g
 (3) 44 g (4) 60 g
20. Cortisone is a molecular substance containing 21 atoms of carbon per molecule. The mass percentage of carbon in cortisone is 69.98%. Its molar mass is:
 (1) 176.5 (2) 252.2
 (3) 287.6 (4) 360.1
21. An organic compound has found 71.7% chlorine, 4.04% Hydrogen and remaining carbon. If molecular weight compound is 99 then molecular formula will be
 (1) CHCl_3 (2) $\text{C}_2\text{H}_4\text{Cl}_2$
 (3) $\text{C}_2\text{H}_2\text{Cl}_2$ (4) CH_3Cl
22. If a compound, on analysis was found to contain C = 18.5 %, H = 1.55 %, Cl = 55.04 % and O = 24.81 %, then its empirical formula is.
 (1) $\text{C}_2\text{H}_2\text{OCl}$ (2) CH_2ClO
 (3) CHClO (4) ClCH_2O
23. The empirical formula of an organic compound containing carbon and hydrogen is CH_2 . The mass of one litre of this organic gas is exactly equal to that of one litre of N_2 at same temperature and pressure. Therefore, the molecular formula of the organic gas is –
 (1) C_2H_4 (2) C_3H_6
 (3) C_6H_{12} (4) C_4H_8
24. If 0.5 mol of BaCl_2 is mixed with 0.1 mole of Na_3PO_4 , the maximum number of mole of $\text{Ba}_3(\text{PO}_4)_2$ that can be formed is:
 $3\text{BaCl}_2 + 2\text{Na}_3\text{PO}_4 \rightarrow \text{Ba}_3(\text{PO}_4)_2 + 6\text{NaCl}$
 (1) 0.7 (2) 0.05 (3) 0.30 (4) 0.10
25. The mass of Mg_3N_2 produced if 48 gm of Mg metal is reacted with 34 gm NH_3 gas is
 $3\text{Mg} + 2\text{NH}_3 \rightarrow \text{Mg}_3\text{N}_2 + 3\text{H}_2$
 (1) $\frac{200}{3}$ gm (2) $\frac{100}{3}$ gm
 (3) $\frac{400}{3}$ gm (4) $\frac{150}{3}$ gm
26. Equal weight of 'X' (At. wt. = 36) and 'Y' (At. wt. = 24) are reacted to form the compound X_2Y_3 . Then
 (1) X is the limiting reagent
 (2) Y is the limiting reagent
 (3) No reactant is left over and mass of X_2Y_3 formed is double the mass of 'X' taken
 (4) none of these
27. If 1 gm of HCl and 1 gm of MnO_2 heated together the maximum weight of Cl_2 gas evolved will be
 $[\text{MnO}_2 + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}]$
 [Atomic mass of Mn = 55]
 (1) 2 gm (2) 0.975 gm
 (3) 0.486 gm (4) 0.972 gm
28. 0.01 mole of iodoform (CHI_3) reacts with Ag to produce a gas whose volume at NTP is
 $2\text{CHI}_3 + 6\text{Ag} \rightarrow \text{C}_2\text{H}_2 + 6\text{AgI(s)}$
 (1) 224 ml (2) 112 ml
 (3) 336 ml (4) None of these

29. A sample of hard water is found to contain 40 mg of Ca^{+2} ion per litre.
 $\text{Ca}^{+2} + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 + 2\text{Na}^+$
 The amount of washing soda (Na_2CO_3) required to soften five litres of the sample would be –
 (1) 1.06 g (2) 5.3 g
 (3) 53 mg (4) 530 mg
30. How much oxygen is required for complete combustion of 560 g of ethene (C_2H_4)?
 (1) 6.4 kg (2) 1.92 kg
 (3) 2.8 kg (4) 9.6 kg
31. 0.5 mole of Mg combine with O_2 to form oxide MgO. The mass of O_2 that has combined is:
 (1) 8g (2) 32g (3) 24g (4) 94g
32. 2.76g of silver carbonate on being strongly heated yields a residue weighing (At. wt. Ag = 108u)
 $\text{Ag}_2\text{CO}_3 \rightarrow 2\text{Ag} + \text{CO}_2 + \frac{1}{2}\text{O}_2$
 (1) 2.16g (2) 2.48g
 (3) 2.32g (4) 2.64g
33. A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g. of CO_2 . The empirical formula of the hydrocarbon is:
 (1) C_2H_4 (2) C_3H_4
 (3) C_6H_5 (4) C_7H_8
34. CaCO_3 is 90% pure, volume of CO_2 collected at STP when 10 gm of CaCO_3 is decomposed is:
 (1) 2.016 liter (2) 1.008 liter
 (3) 10.08 liter (4) 20.16 liter
35. 10 mL of gaseous hydrocarbon on combustion give 40 mL of $\text{CO}_2(\text{g})$ and 50 mL of $\text{H}_2\text{O}(\text{vap.})$. The hydrocarbon is –
 (1) C_4H_5 (2) C_8H_{10}
 (3) C_4H_8 (4) C_4H_{10}
36. Assuming that petrol is iso-octane (C_8H_{18}) and has density 0.8 g/mL, 1.425 litre of petrol on complete combustion will consume oxygen –
 (1) 50 L (2) 125 L
 (3) 125 mol (4) 50 mol
37. 50 gm CaCO_3 will react with gm of 20% HCl by weight –
 (1) 36.5 gm (2) 73 gm
 (3) 109.5 gm (4) 182.5 gm
38. One litre of CO_2 is passed over red-hot coke. The volume becomes 1.4 litre at same temperature and pressure. The composition at end of reaction is–
 (1) 0.6 litre CO
 (2) 0.8 litre CO_2
 (3) 0.6 litre CO_2 and 0.8 litre CO
 (4) None
39. 1 mol of KClO_3 is thermally decomposed and excess of aluminium is burnt in the gaseous product. How many moles of Al_2O_3 are formed?
 (1) 1 (2) 2 (3) 1.5 (4) 3
40. 40 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 4g magnesium oxide what will be the percentage purity of magnesium carbonate in the sample
 (1) 21% (2) 60%
 (3) 84% (4) 75%
41. The oxide of an element possesses the molecular formula M_2O_3 . If the equivalent mass of the metal is 9, the molecular mass of the oxide will be –
 (1) 27 (2) 75
 (3) 102 (4) 18
42. Mixture of two metals having mass 2 gm (molar mass: A = 15, B = 30) are bivalent and dissolve in HCl and evolve 2.24 L H_2 at STP. What is mass of A present in mixture?
 (1) 1 gm (2) 1.5 gm
 (3) 0.5 gm (4) 0.75 gm
43. The density of a solution prepared by dissolving 120 g of urea (mol. mass = 60 u) in 1000 g of water is 1.15 g/mL. The molarity of this solution is:
 (1) 2.05 M (2) 0.50 M
 (3) 1.78 M (4) 1.02 M

Exercise 3

1. Match the no. of moles listed in Column-I with their relevant amounts listed in Column-II.

	Column-I (No. of moles)		Column-II (Amount)
(A)	0.1 mole	(p)	4480 mL of CO ₂ at STP
(B)	0.2 mole	(q)	0.1 g atom of iron
(C)	0.25 mole	(r)	1.5×10^{23} molecules of oxygen gas
(D)	0.5 mole	(s)	9 mL of water

- (1) A-(q), B-(p), C-(r), D-(s)
 (2) A-(p), B-(r), C-(q), D-(s)
 (3) A-(r), B-(q), C-(s), D-(p)
 (4) A-(s), B-(p), C-(r), D-(q)

2. Match % of carbon in column I with the compound in column II.

	Column-I		Column-II
(A)	52.2%	(1)	CH ₄
(B)	75%	(2)	CO ₂
(C)	27.3%	(3)	C ₂ H ₆ O
(D)	12%	(4)	CaCO ₃

- (1) A-(3), B-(1), C-(2), D-(4)
 (2) A-(1), B-(4), C-(2), D-(3)
 (3) A-(3), B-(2), C-(4), D-(1)
 (4) A-(4), B-(3), C-(1), D-(2)

3. Match the column I to Column II

Column I	Column II
(A) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ 1.0g 1.0g ?	(p) 0.56 g
(B) $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ 1.0g 1.0g ?	(q) 1.333 g
(C) $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$ 1.0g ?	(r) 1.125 g
(D) $\text{C} + 2\text{H}_2 \rightarrow \text{CH}_4$ 1.0g 1.0g ?	(s) 1.214 g

- (1) A → r, B → p, C → s, D → q
 (2) A → r, B → s, C → p, D → q
 (3) A → p, B → r, C → q, D → s
 (4) A → s, B → p, C → r, D → q

4. Match column-I with column-II.

	Column-I (physical quantity)		Column-II (Unit)
(A)	Molarity	(p)	Mol
(B)	Mole fraction	(q)	Unitless
(C)	Mole	(r)	Mol L ⁻¹
(D)	Molality	(s)	mol kg ⁻¹

- (1) A-(r), B-(q), C-(s), D-(p)
 (2) A-(r), B-(p), C-(q), D-(s)
 (3) A-(r), B-(q), C-(p), D-(s)
 (4) A-(q), B-(r), C-(p), D-(s)

5. Match column-I with column-II.

	Column-I		Column-II
(A)	Molarity	(p)	Dependent on temperature
(B)	Molality	(q)	$\frac{w \times 1000}{M_w \times W}$
(C)	Mole fraction	(r)	$\frac{n_A}{n_A + n_B}$
(D)	Mass%	(s)	$\frac{W_{\text{solute}}}{W_{\text{solution}}} \times 100$

- (1) A-(p), B-(r), C-(q), D-(s)
 (2) A-(p), B-(q), C-(r), D-(s)
 (3) A-(r), B-(s), C-(p), D-(q)
 (4) A-(p), B-(s), C-(r), D-(q)

6. Assertion: 1 amu equals to 1.66×10^{-24} g.

Reason: 1.66×10^{-24} g equals to $\frac{1}{12}$ th of

mass of one atom of C¹² atom.

- (1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
 (2) Both Assertion and Assertion are true and Reason is not the correct explanation of Assertion.
 (3) Assertion is true and Reason is false
 (4) Assertion is false and Reason is true

7. **Assertion:** 1 g of O_2 and 1 g molecule of O_2 have equal number of molecules.
Reason: Mass of 1 mole atom is equal to its gram atomic mass.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Assertion are true and Reason is not the correct explanation of Assertion.
(3) Assertion is true and Reason is false
(4) Assertion is false and Reason is true
8. **Assertion:** 22.4 L of N_2 at NTP and 5.6 L O_2 at NTP contain equal number of molecules.
Reason: Under similar conditions of temperature and pressure, equal volume of all gases contain equal number of molecules.
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Assertion are true and Reason is not the correct explanation of Assertion.
(3) Assertion is true and Reason is false
(4) Assertion is false and Reason is true
9. **Assertion:** Volume occupied by 1 mol $H_2O_{(l)}$ is equals to 22400 cc at NTP.
Reason: 1 mol of any substance occupies 22.4 L volume at N.T.P.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false
10. **Statement I:** The terms empirical formula bear the simplest ratio of no. of atoms
Statement II: Molecular formula of a compound defines the relative number of constituent atom in the simplest ratio.
(1) Both Statement I and Statement II are false
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are true
11. **Assertion:** The mathematical approach to obtain the amount of the products or reactants is called stoichiometry.
Reason: The stoichiometry calculations are derived from chemical equations.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false.
12. **Statement I:** $2A + 3B \rightarrow C$
4/3 moles of 'C' are produced when 3 moles of 'A' & 4 moles of 'B' are added.
Statement II: 'B' is the limiting reactant for the given data.
(1) Statement-I is true, statement-II is true and statement-II is correct explanation for statement-I.
(2) Statement-I is true, statement-II is true and statement-II is NOT the correct explanation for statement-I.
(3) Statement-I is false, statement-II is true.
(4) Statement-I is true, statement-II is false.
13. **Assertion:** 44 g of CO_2 , 28 g of CO have same volume at STP.
Reason: Both CO_2 and CO are formed by C and oxygen.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false

14. **Assertion:** The mass of the products formed in a reaction depends upon the limiting reactant.
Reason: Limiting reactant reacts completely in the reaction.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false
15. **Statement I:** Equivalent wt. of Cu in both CuO and Cu₂O is 31.75.
Statement II: Equivalent wt. of an element is constant.
(1) Both Statement I and Statement II are false
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are true
16. **Assertion:** In the reaction $A + 2B \rightarrow C + 2D$ 1 gram equivalent of A reacts with 2-gram equivalent of B.
Reason: Reactants react according to different equivalent.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false
17. **Assertion:** The balancing of chemical equations is based on law of conservation of mass.
Reason: total mass of reactants is equal to total mass of products in a chemical reaction.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false
18. **Assertion:** Law of conservation of mass hold good for nuclear reaction.
Reason: Law states that mass can be neither created nor destroyed in a chemical reaction.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false
19. **Assertion:** Pure water obtained from different sources such as, river, well, spring, sea etc. always contains hydrogen and oxygen combined in the ratio 1 : 8 by mass.
Reason: A chemical compound always contains elements combined together in same proportion by mass.
(1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) If Assertion is True but the Reason is False.
(4) If both Assertion & Reason are false

Exercise 4 (Previous Year's Questions)

- The number of atoms in 0.1 mol of a triatomic gas is: ($N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$). [AIPMT-2010]
 (1) 1.800×10^{22} (2) 6.026×10^{22}
 (3) 1.806×10^{23} (4) 3.600×10^{23}
- Which has the maximum number of molecules among the following? [AIPMT MAINS-2011]
 (1) 64 g SO_2 (2) 44 g CO_2
 (3) 48 g O_3 (4) 8 g H_2
- An excess of AgNO_3 is added to 100 mL of a 0.01 M solution of dichlorotetraaquachromium (III) chloride. The number of moles of AgCl precipitated would be: [NEET UG-2013]
 (1) 0.01 (2) 0.001
 (3) 0.002 (4) 0.003
- 1.0 g of magnesium is burnt with 0.56 g O_2 in a closed vessel. Which reactant is left in excess and how much?
 (At. wt. Mg = 24; O = 16) [AIPMT-2014]
 (1) Mg, 0.16 g (2) O_2 , 0.16 g
 (3) Mg, 0.44 g (4) O_2 , 0.28 g
- When 22.4 litres of $\text{H}_2(\text{g})$ is mixed with 11.2 litres of $\text{Cl}_2(\text{g})$, each at S.T.P., the moles of $\text{HCl}(\text{g})$ formed is equal to:- [AIPMT-2014]
 (1) 1 mol of $\text{HCl}(\text{g})$ (2) 2 mol of $\text{HCl}(\text{g})$
 (3) 0.5 mol of $\text{HCl}(\text{g})$ (4) 1.5 mol of $\text{HCl}(\text{g})$
- Equal masses of H_2 , O_2 and methane have been taken in a container of volume V at temperature 27°C in identical conditions. The ratio of the volumes of gases $\text{H}_2 : \text{O}_2 : \text{methane}$ would be : [AIPMT-2014]
 (1) 8 : 16 : 1 (2) 16 : 8 : 1
 (3) 16 : 1 : 2 (4) 8 : 1 : 2
- The number of water molecules are maximum in- [AIPMT RE-2015]
 (1) 18 grams of water
 (2) 18 moles of water
 (3) 18 molecules of water
 (4) 1.8 gram of water
- A mixture of gases contain H_2 and O_2 gases in the ratio of 1 : 4 (w/w). What is the molar ratio of the two gases in the mixture? [AIPMT-2015]
 (1) 4 : 1 (2) 16 : 1
 (3) 2 : 1 (4) 1 : 4
- What is the mass of the precipitate formed when 50 mL of 16.9% solution of AgNO_3 is mixed with 50 mL of 5.8% NaCl solution? [AIPMT RE-2015]
 (Ag = 107.8, N = 14, O = 16, Na = 23, Cl = 35.5)
 (1) 7 g (2) 14g
 (3) 28 g (4) 3.5 g
- 20.0 g of a magnesium carbonate sample decomposes on heating to give carbon dioxide and 8.0 g magnesium oxide. What will be the percentage purity of magnesium carbonate in the sample?
 (At. Wt.: Mg = 24) [AIPMT RE-2015]
 (1) 60 (2) 84
 (3) 75 (4) 96
- Suppose the elements X and Y combine to form two compounds XY_2 and X_3Y_2 . When 0.1 mole of XY_2 weighs 10 g and 0.05 mole of X_3Y_2 weighs 9 g, the atomic weights of X and Y are [NEET-2016]
 (1) 30,20 (2) 40,30
 (3) 60,40 (4) 20,30
- Which of the following is dependent on temperature? [NEET-2017]
 (1) Molality (2) Molarity
 (3) Mole fraction (4) Weight percentage
- In which case is the number of molecules of water are maximum? [NEET-2018]
 (1) 0.00224 L of water vapours at 1 atm & 273 K
 (2) 0.18 g of water
 (3) 18 mL of water
 (4) 10^{-3} mol of water

14. A mixture of 2.3 g formic acid and 4.5 g oxalic acid is treated with conc. H_2SO_4 . The evolved gaseous mixture is passed through KOH pellets. Weight (in g) of the remaining product at STP will be: **[NEET-2018]**
 $\text{HCOOH} \xrightarrow{\text{con. H}_2\text{SO}_4} \text{CO}$
 $\text{COOH} \xrightarrow{\text{ConC. H}_2\text{SO}_4} \text{CO} + \text{CO}_2$
 (1) 2.8 (2) 3.0 (3) 1.4 (4) 4.4
15. The number of moles of hydrogen molecules required to produce 20 moles of ammonia through Haber's process is – **[NEET-2019]**
 (1) 30 (2) 40 (3) 10 (4) 20
16. Which one of the following has maximum number of atoms? **[NEET-2020]**
 (1) 1 g of Mg(s) [Atomic mass of Mg = 24]
 (2) 1 g of O_2 (g) [Atomic mass of O = 16]
 (3) 1 g of Li(s) [Atomic mass of Li = 7]
 (4) 1 g of Ag(s) [Atomic mass of Ag = 108]
17. A compound possesses 8% sulphur by mass. The least molecular mass is – **[NEET-2020]**
 (1) 200 (2) 400 (3) 155 (4) 355
18. One mole of carbon atom weighs 12 g, the number of atoms in it is equal to, (Mass of carbon – 12 is 1.9926×10^{-23} g): **[NEET 2020 Exam (Covid Re-exam)]**
 (1) 1.2×10^{23} (2) 6.022×10^{22}
 (3) 12×10^{22} (4) 6.022×10^{23}
19. An organic compound contains 80% (by wt.) carbon and remaining percentage of hydrogen. The right option for the empirical formula of this compound is: [Atomic wt. of C is 12, H is 1] **[NEET-2021]**
 (1) CH (2) CH_2 (3) CH_3 (4) CH_4
20. What mass of 95% pure CaCO_3 will be required to neutralize 50 mL of 0.5 M HCl solution according to the following reaction? **[NEET-2022]**
 $\text{CaCO}_3(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{CaCl}_2(\text{aq}) + \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}$
 [Calculate upto second place of decimal point]
 (1) 1.32 g (2) 3.65 g
 (3) 9.50 g (4) 1.25 g
21. The right option for the mass of CO_2 produced by heating 20 g of 20% pure limestone is (Atomic mass of Ca = 40) **[NEET-2023]**
 (1) 1.76 g (2) 2.64 g
 (3) 1.32 g (4) 1.12 g
22. Weight (g) of two moles of the organic compound, which is obtained by heating sodium ethanoate with sodium hydroxide in presence of calcium oxide is: **[NEET-2023]**
 (1) 32 (2) 30 (3) 18 (4) 16
23. 1 gram of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to: **[NEET-2024]**
 (1) Zero mg (2) 200 mg
 (3) 750 mg (4) 250 mg
24. The highest number of helium atoms is in **[NEET-2024]**
 (1) 4g of helium
 (2) 2.271098 L of helium at STP
 (3) 4 mol of helium
 (4) 4 u of helium
25. A compound X contains 32% of A, 20% of B and remaining percentage of C. Then, the empirical formula of X is: **[NEET-2024]**
 (Given atomic masses of A = 64; B = 40; C = 32u)
 (1) AB_2C_2 (2) ABC_4 (3) A_2BC_2 (4) ABC_3
26. The amount of glucose required to prepare 250 mL of $\frac{M}{20}$ aqueous solution is: **[RE-NEET-2024]**
 (Molar mass of glucose: 180 g mol^{-1})
 (1) 2.25g (2) 4.5g
 (3) 0.44g (4) 1.125g
27. 1.0 g of H_2 has same number of molecules as in: **[RE-NEET-2024]**
 (1) 14 g of N_2 (2) 18 g of H_2O
 (3) 16 g of CO (4) 28 g of N_2

ANSWER KEY

Exercise 1.1

Que.	1	2	3	4	5	
Ans.	4	3	1	2	4	

Exercise 1.2

Que.	1	2	3	4	
Ans.	3	1	3	3	

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	1	4	2	2	4	2	3	4	1	4	4	2	1	3	4	2	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38		
Ans.	3	2	2	4	3	4	2	3	1	2	2	4	1	1	2	4	4	4		

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
Ans.	2	3	3	2	2	3	2	3	3	4	4	1	1	3	2	2	1	4	1	

Exercise 1.5

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	1	2	3	2	3	4	4	1	4	3	3	1	1	3	4	4	2	4
Que.	21	22	23	24	25	26	27	28												
Ans.	1	4	1	3	1	2	1	3												

Exercise 1.6

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	1	3	2	2	3	2	2	2	3	3	3	2	2	2	2	1	4	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	4	3	1	3	3	3	2	2	2	4	2	1	2	4	4	2	3	2	3	3

Exercise 1.7

Que.	1	2	3	4	5	6	7	8	9	10	11	12	
Ans.	1	1	3	3	1	3	4	1	4	4	2	4	

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	1	2	3	3	3	2	3	4	1	2	2	3	1	4	3	3	2	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	2	3	1	2	1	3	3	2	4	2	1	1	4	1	4	3	4	3	1	1
Que.	41	42	43																	
Ans.	3	1	1																	

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	
Ans.	1	1	2	3	2	1	4	4	4	2	1	1	2	1	1	4	1	4	1	

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	2	1	1	3	2	1	1	2	2	2	3	1	1	3	2	4	4	1
Que.	21	22	23	24	25	26	27													
Ans.	1	1	4	3	4	1	1													

Chapter 2

STRUCTURE OF ATOM

Chapter Summary

- Introduction
- Dalton's Atomic Theory
- Fundamental Particles (Electron, Proton, Neutron)
- Thomson's Atomic Model
- Rutherford's Atomic Model
- Some Important Definitions
- Electromagnetic Waves (EM Waves) or Radiant Energy.
- Electromagnetic Spectrum
- Planck's Quantum Theory
- Bohr's Atomic Model
- Application of Bohr's Atomic Model
- Energy Definitions
- Spectrum
- Hydrogen Spectrum
- Wave Mechanical Model of an Atom
 - Dual nature of Matter (de-Broglie Concept)
 - Heisenberg Uncertainty Principle
- Quantum Mechanical Model of Atom (Schrodinger Equation)
- Quantum Number
- Electronic Configuration
- Extra Stability of Half Filled and Completely Filled Orbitals

Introduction

- (a) The word atom was first introduced by Ostwald (1803 - 1807) in scientific world.
- (b) According to him matter is ultimately made up of extremely small indivisible particles called atoms.
- (c) It takes part in chemical reactions.
- (d) Atom is neither created nor destroyed

Dalton's Atomic Theory

Dalton proposed the atomic theory on the basis of the law of conservation of mass and law of definite proportions. He also proposed the law of multiple proportion as a logical consequence of this theory. The salient features of this theory is:

- (a) Each element is composed by extremely small particles called atoms.
- (b) Atoms of a particular element are all alike but differ with the atoms of other elements.
- (c) Atom of each element is an ultimate particle, and has a characteristic mass but is structureless.
- (d) Atom is indestructible i.e. it can neither be destroyed nor created by simple chemical reactions.
- (e) Atom of an element takes part in chemical reaction to form molecule.
- (f) Atoms of different elements combine in fixed ratio of small whole numbers to form compound (now called molecules).

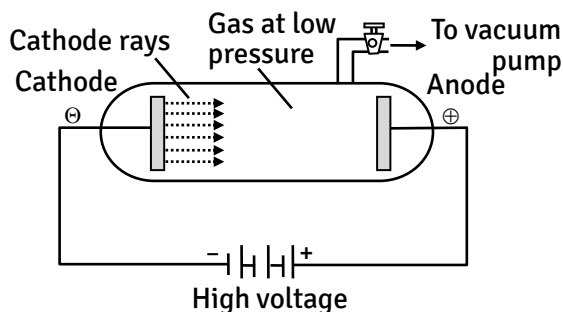
Fundamental Particles

1. Electron:

- (a) Electron was discovered by Sir J.J. Thomson
- (b) The charge on the electron is -1.6×10^{-19} coulomb (Millikan)
- (c) The molar mass of electron is 5.48×10^{-4} gm/mole

- (d) The mass of electron in motion is expressed as $m' = \frac{m}{\left(1 - \frac{v^2}{c^2}\right)^{3/2}}$ where m' = mass of the electron in motion, m = rest mass, v = velocity of the electron, c = velocity of light
- (e) In 1897, J.J. Thomson determined the e/m value (charge/mass) of the electron by studying the deflections of cathode rays in electric and magnetic fields. The value of e/m has been found to be -1.7588×10^8 coulomb
- (f) The first precise measurement of the charge on the electron was made by Robert A. Millikan. in 1909 by oil drop experiment. Its value was found to be -1.6022×10^{-19} coulomb.
- (g) The mass of electron can be calculated from the value of e/m and the value of e which is 9.1096×10^{-31} Kg.

Cathode rays:



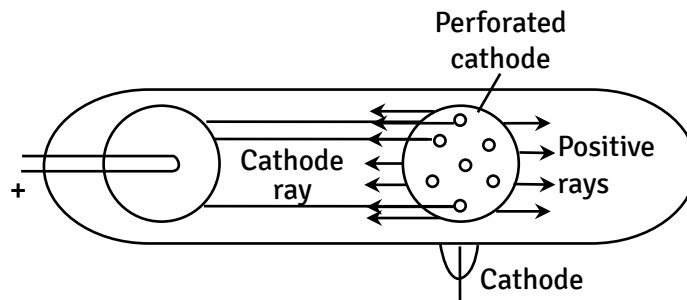
- (a) The electron was discovered as a result of the studies of the passage of electricity through gases at extremely low pressures known as discharge tube experiments.
- (b) When a high voltage of the order of 10,000 volts or more was impressed across the electrodes, some sort of invisible rays moved from the negative electrode to the positive electrodes these rays are called as cathode rays.

Properties of Cathode rays:

- (i) Path of travelling is straight from the cathode with a very high velocity. As it produces shadow of an object placed in its path
- (ii) Cathode rays produce mechanical effects. If a small pedal wheel is placed between the electrodes, it rotates. This indicates that the cathode rays consist of material part
- (iii) When electric and magnetic fields are applied to the cathode rays in the discharge tube, the rays are deflected thus establishing that they consist of charged particles.
- (iv) Cathode rays produce X-rays when they strike against hard metals like tungsten, copper etc.
- (v) When the cathode rays are allowed to strike a thin metal foil, it gets heated up. Thus the cathode rays possess heating effect.
- (vi) They produce a green glow when strike the glass wall beyond the anode. Light is emitted when they strike the zinc sulphide screen.
- (vii) Cathode rays penetrate through thin sheets of aluminium and other metals.
- (viii) They affect the photographic plates
- (ix) The ratio of charge to mass i.e. charge/mass is same for all the cathode rays irrespective of the gas used in the tube.

2. Proton:

- (a) Proton was discovered by Goldstein
- (b) Proton carries a charge of $+1.602 \times 10^{-19}$ coulomb, i.e., one-unit positive charge.
- (c) Mass of proton is 1.672×10^{-27} kg or 1.0072 amu
- (d) A proton is defined as a sub-atomic particle which has a mass nearly 1 amu and a charge of + 1 unit

Positive Rays-Discovery of Proton:

- (a) The existence of positively charged particles in an atom was shown by E. Goldstein in 1886
- (b) He repeated the same discharge tube experiments by using a perforated cathode.
- (c) It was observed that when a high potential difference was applied between the electrodes, not only cathode rays were produced but also a new type of rays was produced simultaneously from anode moving towards cathode and passed through the holes or canal of the cathode. These termed as canal ray or anode ray.

Properties of Anode Rays:

- (i) These rays travel in straight lines and cast shadow of the object placed in their path.
- (ii) The anode rays are deflected by the magnetic and electric fields like cathode rays but direction is different that mean these rays are positively charged.
- (iii) These rays have kinetic energy and produces heating effect also.
- (iv) The e/m ratio of for these rays is smaller than that of electrons
- (v) Unlike cathode rays, their e/m value is dependent upon the nature of the gas taken in the tube.
- (vi) These rays produce flashes of light on ZnS screen
- (vii) These rays can pass through thin metal foils
- (viii) They are capable to produce ionisation in gases
- (ix) They can produce physical and chemical changes.

3. Neutron:

- (a) It has been found that for all atoms except hydrogen atomic mass is more than the atomic number. Thus Rutherford (1920) suggested that in an atom, there must be present at least a third type of fundamental particle.
- (b) It should be electrically neutral and possess mass nearly equal to that of proton. He proposed its name as neutron.
- (c) Chadwick (1932), bombarded beryllium with a stream of α -particles and observed electrically and magnetically neutral radiations.
- (d) There were neutral particles which was called neutron. Nuclear reaction is as follows

$${}_4\text{Be}^9 + {}_2\text{He}^4 \longrightarrow {}_6\text{C}^{12} + {}_0\text{n}^1$$
- (e) A neutron is a subatomic particle which has a mass 1.675×10^{-24} g, approximately 1 amu, or nearly equal to the mass of proton on hydrogen atom and carrying no electrical charge.

Fundamental Particles Table:

Atoms are made up of three fundamental particles. The charge and mass of these fundamental particles are as follows:

		Electron	Proton	Neutron
	Symbol	e or ${}_{-1}\text{e}^0$ or e^-	P or ${}_{1}\text{p}^1$	n or ${}_{0}\text{n}^1$
Mass	Kg	9.109534×10^{-31}	$1.6726485 \times 10^{-27}$	$1.6749543 \times 10^{-27}$
	amu	5.4858026×10^{-4}	1.007276471	1.008665012
	Relative	$\frac{1}{1837}$	1	1
Charge	Actual (in C)	$1.6021892 \times 10^{-19}$	$1.6021892 \times 10^{-19}$	0
	Relative	-1	1	0

One unit charge = 4.80298×10^{-10} esu

= 1.60210×10^{-19} coulombs one amu = $\frac{1}{12}$ th $\times$ mass of one atom of ${}_{6}\text{C}^{12}$

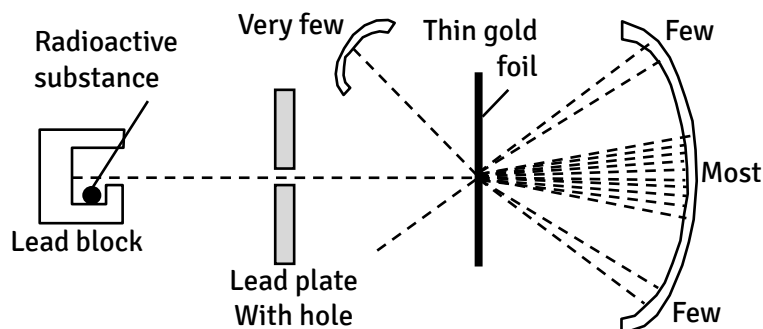
Thomson's Atomic Model

It states the arrangement of electrons and protons in an atom. The main principles are

- After discovery of electron and proton attempts were made to find out their arrangement in an atom. The first simple model was proposed by J.J. Thomson known as Thomson's atomic model.
- He proposed that the positive charge is spread over a sphere of the size of the atom (i.e. 10^{-8} cm radius) in which electrons are embedded to make the atom as whole neutral.
- This model could not explain the experimental results of Rutherford's α -particle scattering; therefore, it was rejected.

Rutherford's Atomic Model**Rutherford gold foil experiment:**

Rutherford carried out experiment on the bombardment of atoms by high speed positively



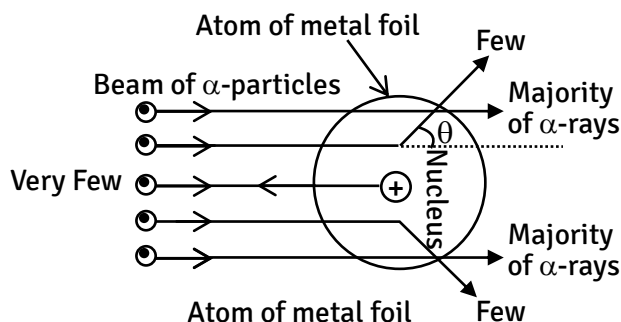
charged α -particles emitted from radium and gave the following observations, which was based on his experiment.

Observation:

- Most of the α - particles (nearly 99%) continued with their straight path.
- Some of the α - particles passed very close to the centre of the atom and deflected by small angles.
- Very few particles thrown back (180°).

Conclusions:

- Most of the α -particles were continued their straight path that means most of the space of the atom is empty.
- The centre of an atom has a positively charged body called nucleus which repel positively charged α particles and thus explained the scattering phenomenon.
- Whole mass of an atom is concentrated in its nucleus and very few throwbacks mean the size of the nucleus and very few throwbacks mean the size of the nucleus is very small 10^{-13} cm. It showed that the nucleus is 10^{-5} times small in size as compared to the total size of atom.

**Applications of Rutherford Model:**

On the basis of scattering experiments, Rutherford proposed model of the atom, which is known as nuclear atomic model. According to this model –

- An atom consists of a heavy positively charged nucleus where all the protons and neutrons are present. Protons & neutrons are collectively referred to as nucleons. Almost whole of the mass of the atom is contributed by these nucleons. The magnitude of the +ve charge on the nucleus is different for different atoms.
- The volume of the nucleus is very small and is only a minute fraction of the total volume of the atom. Nucleus has a diameter of the order of 10^{-12} to 10^{-13} cm and the atom has a diameter of the order of 10^{-8} cm

$$\frac{D_A}{D_N} = \frac{\text{Diameter of the atom}}{\text{Diameter of the nucleus}} = \frac{10^{-8}}{10^{-13}} = 10^5, D_A = 10^5 D_N$$

Thus diameter (size) of the atom is 10^5 times the diameter of the nucleus.

- The radius of a nucleus is proportional to the cube root of the number of nucleons within it.

$$R \propto A^{1/3} \Rightarrow R = R_0 A^{1/3} \text{ cm}$$

Where $R_0 = 1.33 \times 10^{-13}$ (a constant) and $A = \text{mass number (p + n)}$ and $R = \text{radius of the nucleus}$.

$$R = 1.33 \times 10^{-13} A^{1/3} \text{ cm}$$

- There is an empty space around the nucleus called extra nuclear part. In this part electrons are present. The number of electrons in an atom is always equal to number of protons present in the nucleus. As the nuclear part of atom is responsible for the mass of the atom, the extra nuclear part is responsible for its volume.

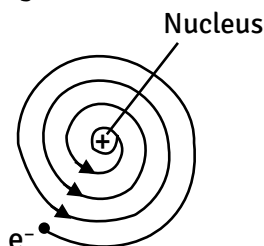
The volume of the atom is about 10^{15} times the volume of the nucleus.

$$\frac{\text{volume of the atom}}{\text{volume of the nucleus}} = \frac{(10^{-8})^3}{(10^{-13})^3} = 10^{15}$$

- Electrons revolve around the nucleus in closed orbits with high speeds. The centrifugal force acting on the revolving e^- is being counter balanced by the force of attraction between the electrons and the nucleus.
- This model was similar to the solar system, the nucleus representing the sun and revolving electrons as planets.

Drawbacks of Rutherford model:

- (1) This theory could not explain stability of atom. According to Maxwell electron loses its energy continuously in the form of electromagnetic radiations. As a result of this, the e^-



should lose energy at every turn and move closer and closer to the nucleus following a spiral path. The ultimate result will be that it will fall into the nucleus, thereby making the atom unstable.

- (2) If the electrons lose energy continuously, the observed spectrum should be continuous but the actual observed spectrum consists of well-defined lines of definite frequencies. Hence, the loss of energy by electron is not continuous in an atom.

Some Important Definitions

Mass Number: It is represented by capital A. The sum of number of Neutrons and protons is called the mass Number of the element. i.e. $A = \text{number of protons} + \text{number of Neutrons}$

Atomic Number: It is represented by Z. The number of protons present in the Nucleus is called atomic number of an element. It is also known as nuclear charge.

For neutral atom: Number of proton = Number of electron

For +ve charged atom: Number of $e^- = Z - (\text{charge on atom})$, $Z = \text{number of protons only}$.

For -ve charged atom: Number of $e^- = Z + (\text{charge on atom})$, $Z = \text{number of protons only}$.

Ex. ${}_{17}\text{Cl}^{35} \rightarrow n = 18, p = 17, e^- = 17$

Two different elements cannot have the same Atomic Number

Number of Neutrons = Mass number – Atomic number

$= A - Z = (p + n) - p = n$

Representation of element $\rightarrow {}_Z\text{X}^A$ (where X $\rightarrow$ symbol of element)

Isotopes: Given by Soddy, are the atoms of a given element which have the same atomic number but different mass number i.e. They have same nuclear charge but different number of Neutrons.

Ex. $\rightarrow$	${}_6\text{C}^{12}$	${}_6\text{C}^{13}$	${}_6\text{C}^{14}$
	$e^- = 6$	$e^- = 6$	$e^- = 6$
	$p = 6$	$p = 6$	$p = 6$
	$n = 6$	$n = 7$	$n = 8$

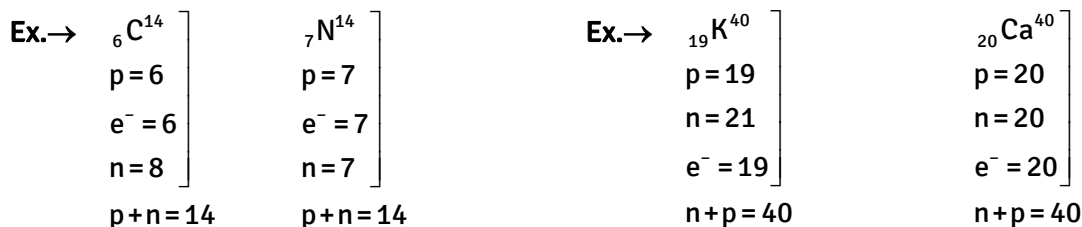
Ex. $\rightarrow$	(Protium)	Deuterium	Tritium)
	${}_1\text{H}^1$	${}_1\text{H}^2$	${}_1\text{H}^3$
	$e^- = 1$	$e^- = 1$	$e^- = 1$
	$p = 1$	$p = 1$	$p = 1$
	$n = 0$	$n = 1$	$n = 2$

${}_1\text{H}^1$ is the only normal hydrogen which have $n = 0$ i.e. no neutrons

Deuterium is also called as heavy hydrogen. It represents by D

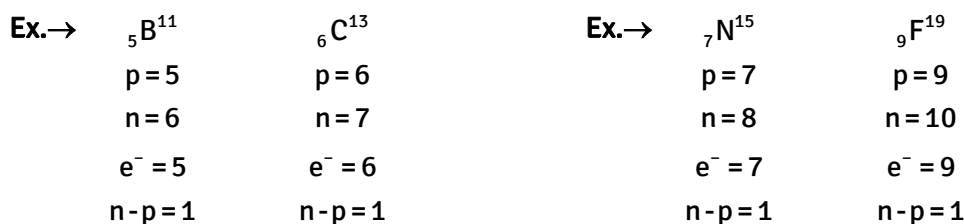
- Isotopes have same chemical property but different physical property.
- Isotopes do not have the same value of e/m .

Isobars: Given by Aston, isobars are the atoms of different element which have the same mass number but different atomic number i.e. They have different number of Electron, Protons & Neutrons But sum of number of neutrons & Protons remains same.



- Isobars do not have the same chemical & physical property.

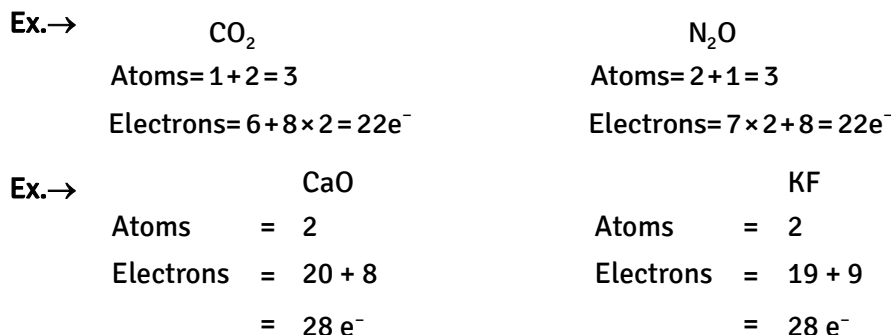
Isodiaphers: They are the atoms of different element which have the same difference of the number of Neutrons & Protons.



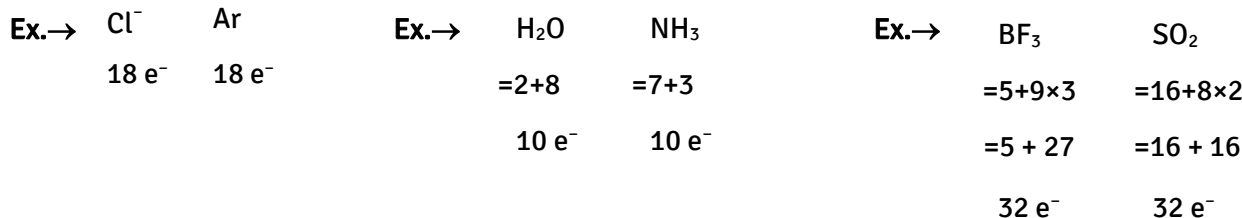
Isotones/ Isoneutronic species / Isotonic: They are the atoms of different element which have the same number of neutrons.



Isosters: They are the molecules which have the same number of atoms & electrons.



Isoelectronic Species: They are the atoms, molecules or ions which have the same number of electrons.



- Ex.1:** The specific charge of a proton is $9.6 \times 10^7 \text{ C kg}^{-1}$, then for an α -particles it will be-
 (A) $2.4 \times 10^7 \text{ C kg}^{-1}$ (B) $4.8 \times 10^7 \text{ C kg}^{-1}$ (C) $19.2 \times 10^7 \text{ C kg}^{-1}$ (D) $38.4 \times 10^7 \text{ C kg}^{-1}$

Sol: (B) $\frac{e}{m}$ for P = 9.6×10^7 ; $\frac{e}{m}$ for $\alpha = \frac{2e}{4m}$

$$= \frac{e}{2m} = \frac{9.6 \times 10^7}{2} = 4.8 \times 10^7$$

Ex.2: Element with mass no. 81 contains 31.7% more neutrons as compared to protons. Assign atomic symbol.

Sol: Mass number = 81, i.e, $p + n = 81$

If protons = x , then neutrons = $x + \frac{31.7}{100} \times x = 1.317x$

$$\therefore x + 1.317x = 81 \text{ or } 2.317x = 81 \text{ or } x = \frac{81}{2.317} = 35$$

Hence, the symbol is ($z = 35$) Br

Ex.3: Ion with mass no. 37 possesses 1 unit of (-)ve charge. If ion contains 11.1% more 'n' than e^- , find symbol?

Sol: Suppose number of electrons in the ion = x

Then number of neutrons = $x + \frac{11.1}{100}x = 1.111x$

No. of electrons in the neutral atom = $x - 1$

$$\therefore \text{No. Protons} = x - 1$$

Mass number = No. neutrons + No. of protons

$$\therefore 37 = 1.111x + x - 1$$

$$2.111x = 38 \text{ or } x = 18$$

$$\therefore \text{No. of protons} = \text{Atomic no.} = x - 1 = 18 - 1 = 17$$

Hence, the symbols of the ion will be ${}_{17}^{35}\text{Cl}^{-1}$.

Ex.4: If no. of protons in X^{-2} is 16. then no. of e^- in X^{+2} will be-

(A) 14 (B) 16 (C) 18 (D) None

Sol. No. of proton in X^{-2} is = 6

No. of electron in X^{+2} is = 14

Ex.5: In C^{12} atom if mass of e^- is doubled and mass of proton is halved, then calculate the percentage change in mass no. of C^{12} .

Sol.

e^-	P^+	n^0	
6	6	6	$A \rightarrow 12$
12	3	6	$A \rightarrow 9$
$\% \text{ change} = \frac{3}{12} \times 100 = 25\%$			

Ex.6: An element have three isotopes and their isotopic weight are 11, 12, 13 unit and their percentage of occurrence in nature is 85, 10, 5 respectively then calculate the ave. atomic weight of element.

Sol. Average Atomic weight

$$= \frac{11 \times 85 + 12 \times 10 + 13 \times 5}{85 + 10 + 5} = \frac{935 + 120 + 65}{100}$$

$$\text{Avg. wt.} = \frac{1120}{100} = 11.2$$

Exercise 1.1

1. An increasing order (lowest first) for the values of e/m for electron (e), proton (p), neutron (n) and alpha (α) particle is:

- (1) e, p, n, α (2) n, α , p, e
(3) n, p, e, α (4) n, p, α , e

2. Select iso electronic set:

- (a) Na^+ , H_3O^+ , NH_4^+
(b) O_3^{2-} , NO_3^- , HCO_3^-
(c) P^{3-} , HCl , C_2H_6 , PH_3
(d) N^{3-} , O^{2-} , F

- (1) a, b, d (2) b, c, d
(3) a, b, c, d (4) a, b, c

3. The ratio of specific charge of a proton and an α -particle is:

- (1) 2: 1 (2) 1: 2
(3) 1: 4 (4) 1: 1

4. $^{13}_6\text{C}$ and $^{12}_6\text{C}$ differ from each in respect of number of:

- (1) electrons
(2) protons
(3) neutrons
(4) none of these

5. Which one of the following constitutes a group of the isoelectronic species?

- (1) C_2^{2-} , O_2^- , CO , NO
(2) NO^+ , C_2^{2-} , CN^- , N_2
(3) CN^- , N_2 , O_2^{2-} , C_2^{2-}
(4) N_2 , O_2^- , NO^+ , CO

6. Match the following correct terms:

	List I		List II
(i)	$^{54}_{26}\text{Fe}$, $^{56}_{26}\text{Fe}$, $^{57}_{26}\text{Fe}$, $^{28}_{26}\text{Fe}$	(a)	Isotopes
(ii)	^3_1H , ^3_2He	(b)	Isotones
(iii)	$^{76}_{32}\text{Ge}$, $^{77}_{33}\text{As}$	(c)	Isodiaphers
(iv)	$^{235}_{92}\text{U}$, $^{231}_{90}\text{Th}$	(d)	Isobars
(v)	^1_1H , ^2_1D , ^3_1T		

Match the above correct terms:

- (1) (i) $\rightarrow$ a, (ii) $\rightarrow$ d, (iii) $\rightarrow$ b, (iv) $\rightarrow$ c, (v) $\rightarrow$ a
(2) (i) $\rightarrow$ a, (ii) $\rightarrow$ d, (iii) $\rightarrow$ d, (iv) $\rightarrow$ c, v $\rightarrow$ a
(3) (v) $\rightarrow$ a, (iv) $\rightarrow$ c, (iii) $\rightarrow$ d, (ii) $\rightarrow$ b, (i) $\rightarrow$ a
(4) None of them

7. Isotopes of an element have

- (1) similar chemical properties but different physical properties
(2) similar chemical and physical properties
(3) similar physical properties but different chemical properties.
(4) different chemical and physical properties.

8. In an atom $^{27}_{13}\text{Al}$ number of protons is (A), electron (B) and neutron (C) hence ratio will be [in order C : B : A]

- (1) 13 : 14 : 13 (2) 13 : 13 : 14
(3) 14 : 13 : 13 (4) 14 : 13 : 14

9. Let mass of electron is half, mass of proton is two times and mass of neutron is three fourth of original. The find out new atomic wt. of O^{16} atom:

- (1) increases by 37.5% (2) Remain constant
(3) increases by 12.5% (4) decreases by 25%

10. The relative abundance of two rubidium isotopes of atomic weights 85 and 87 are 75% and 25% respectively. The average atomic wt. of rubidium is:

- (1) 75.5 (2) 85.5
(3) 86.5 (4) 87.5

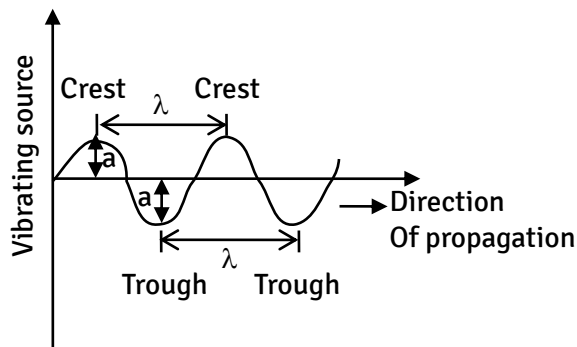
11. Naturally occurring boron has two isotopes whose atomic weights are 10.00(I) and 11.00 (II). Atomic weight of natural boron is 10.80. The percentage of isotopes (I) and (II) respectively is:

- (1) 20 and 80 (2) 10 and 20
(3) 15 and 75 (4) 30 and 70

Electromagnetic Waves (EM Waves) or Radiant Energy

According to this theory the energy transmitted from one body to another in the form of waves and these waves travel in the space with the same speed as light (3×10^8 m/s) and these waves are known as Electromagnetic waves or Radiant energy. The radiant energy does not need any medium for propagation.

Ex: Radio waves, micro waves, Infra-red rays, visible rays, ultraviolet rays, x-rays, gamma rays. The radiant Energy have electric and magnetic fields and travel at right angle to these fields. The upper most point of the wave is called crest and the lower most portion is called trough. Some of the terms employed in dealing with the waves are described below.



Wavelength (λ) (Lambda): It is defined as the distance between two nearest crest or trough. It is measured in terms of a $\text{\AA}$ (Angstrom), pm (Picometre), nm (nanometer), cm (centimeter), m (metre)

$$1\text{\AA} = 10^{-10}\text{m}, \quad 1\text{ pm} = 10^{-12}\text{m}, \\ 1\text{ nm} = 10^{-9}\text{m}, \quad 1\text{ cm} = 10^{-2}\text{m}$$

Frequency (ν) (nu): Frequency of a wave is defined as the number of waves which pass through a point in 1 second.

It is measured in terms of Hertz (Hz), second^{-1} or cycle per second (cps) ($1\text{Hertz} = 1\text{second}^{-1}$)

Time period (T): Time taken by a wave to pass through one point. $T = \frac{1}{\nu}$ second

Velocity (c): Velocity of a wave is defined as distance covered by a wave in 1 second

$$C = \lambda/T = \lambda\nu \quad \text{or} \quad \nu = c/\lambda \quad \text{or} \\ C = \nu (S^{-1}) \times \lambda(m) \quad \text{or} \quad c = \nu \lambda (ms^{-1})$$

Since c is constants i.e. frequency is inversely proportional to λ

Wave number ($\bar{\nu}$) (nu bar): It is the reciprocal of the wave length that is number of waves present in Unit length.

$$\bar{\nu} = \frac{1}{\lambda}$$

It is measured in terms of cm^{-1} , m^{-1} etc.,

Amplitude (a): The amplitude of a wave is defined as the height of crest or trough of through.

Important note: $\boxed{\nu = \frac{c}{\lambda} = c\bar{\nu}} \quad \left(\bar{\nu} = \frac{1}{\lambda} \right)$

Electromagnetic Spectrum

Electromagnetic wave or radiation is not a single wavelength radiation, but a mixture of various wavelength or frequencies. All the frequencies have same speed.

If all the components of Electromagnetic Radiation (EMR) are arranged in order of decreasing or increasing wavelengths or frequencies, the pattern obtained is known as Electromagnetic Spectrum. The following table shows all the components of light.

S. No.	Name	Wave length	Frequency (Hz)	Source
(1)	Radio wave	$3 \times 10^7 - 3 \times 10^{14}$	$1 \times 10^5 - 1 \times 10^9$	Alternating current of high frequency
(2)	Microwave	$6 \times 10^6 - 3 \times 10^7$	$1 \times 10^9 - 5 \times 10^{11}$	Klystron tube
(3)	Infrared (IR)	$7600 - 6 \times 10^6$	$5 \times 10^{11} - 3.95 \times 10^{16}$	Incandescent objects
(4)	Visible	$3800 - 7600$	$3.95 \times 10^{16} - 7.9 \times 10^{14}$	Electric bulbs, sun rays
(5)	Ultraviolet (UV)	$150 - 3800$	$2 \times 10^{16} - 3 \times 10^{19}$	Sun rays, arc lamps with mercury vapours
(6)	X-Rays	$0.1 - 150$	$2 \times 10^{16} - 3 \times 10^{19}$	Cathode rays striking metal plate
(7)	γ -Rays	$0.01 - 0.1$	$3 \times 10^{19} - 3 \times 10^{20}$	Secondary effect of radioactive decay

Planck's Quantum Theory

According to Planck's quantum theory:

1. The radiant energy emitted or absorbed by a body not continuously but discontinuously in the form of small discrete packets of energy and these packets are called quantum.
2. In case of light, the smallest packet of energy is called as 'photon' but in general case the smallest packet of energy called as quantum.
3. The energy of each quantum is directly proportional to frequency of the radiation i.e.

$$E \propto \nu \Rightarrow E = h\nu \text{ or } E = \frac{hc}{\lambda} \left\{ \because \nu = \frac{c}{\lambda} \right\}$$

h is proportionality constant or Planck's constant

$$h = 6.626 \times 10^{-37} \text{ kJ s or } 6.626 \times 10^{-34} \text{ J s}$$

or $6.626 \times 10^{-27} \text{ erg s}$

4. Total amount of energy transmitted from one body to another will be some integral multiple of energy of a quantum. $E = nh\nu$ where n is an integer and n = number of quantum

$$E = nh\nu = \frac{nhc}{\lambda} = nhc\bar{\nu}$$

Bohr's Atomic Model

Bohr developed a model for hydrogen and hydrogen like atoms one-electron species (hydrogenic species). He applied quantum theory in considering the energy of an electron bound to the nucleus.

Important postulates of Bohr's Model:

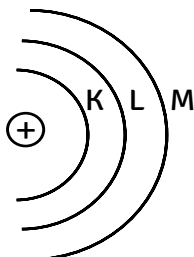
- (1) Electrons revolve around the nucleus along certain circular paths known as "orbit" or "shells"

K, L, M, are shells

K → 1st orbit

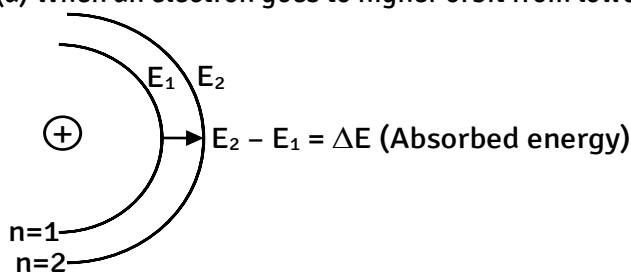
L → 2nd orbit

M → 3rd orbit

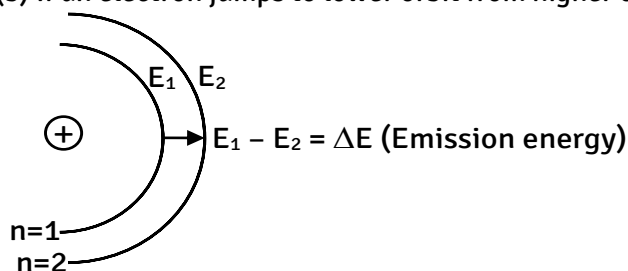


- (2) Electron is associated with a fixed energy in a particular orbit. The change in electronic energy is possible only when; electron changes its orbit number.

(a) When an electron goes to higher orbit from lower orbit, then energy is absorbed.

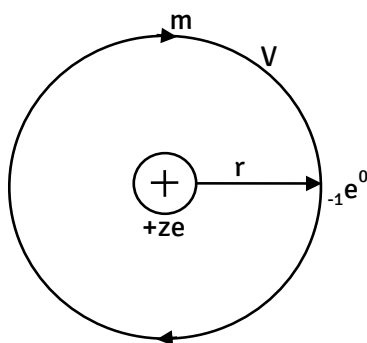


(b) If an electron jumps to lower orbit from higher orbit, then energy is radiated.



- (3) The electrostatic force of attraction acting between the electron and the nucleus is counter balance by

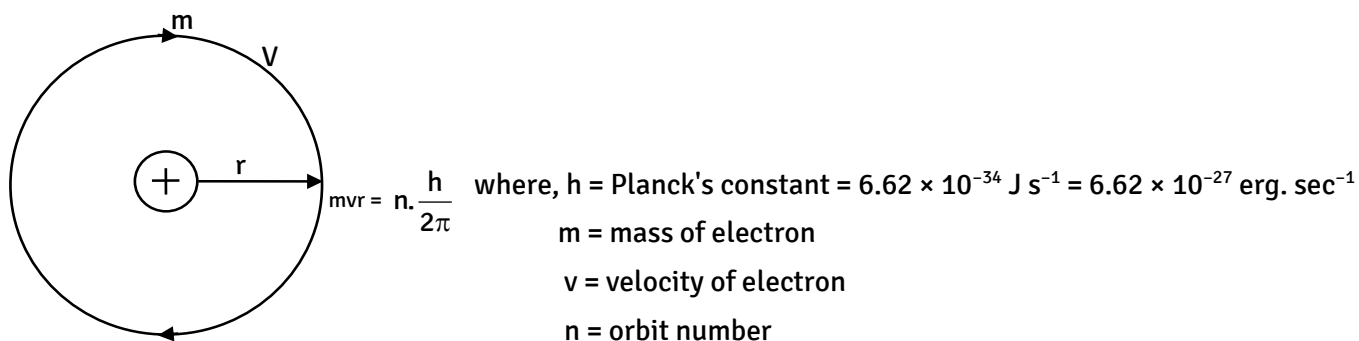
the centrifugal force $\frac{mV^2}{r} = K \frac{Ze^2}{r^2}$



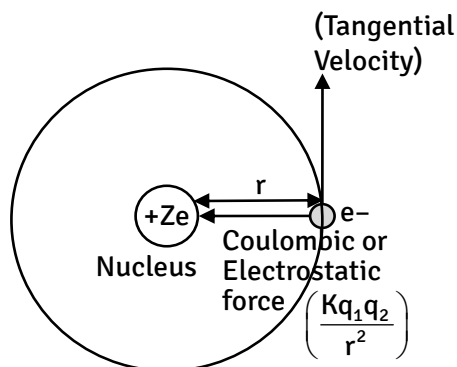
$$\text{or } mv^2 = K \frac{Ze^2}{r^2} \dots\dots\dots(i)$$

$$\frac{1}{4\pi\epsilon_0} \cdot \frac{Ze^2}{r} \dots\dots\dots(ii)$$

- (4) Electrons can revolve only in those orbits in which angular momentum is integral multiple of $\frac{h}{2\pi}$

**Note:**

If the energy supplied to hydrogen atom is less than 13.6 eV, it will accept or absorb only those quanta which can take it to a certain higher energy level i.e., all those photons having energy less than or more than a particular energy level will not be absorbed by hydrogen atom. But if energy supplied to hydrogen atom is more than 13.6 eV then all photons are absorbed and excess energy appear as kinetic energy of emitted photo electron.

Application of Bohr's Model**1. Radius of various orbits (shell):****Derivation:**

$$r = \frac{n^2 h^2}{4\pi^2 m K Z e^2}$$

$$\text{Coulombic force} = \frac{K q_1 q_2}{r^2} = \frac{K Z e \cdot e}{r^2} = \frac{K Z e^2}{r^2}$$

(Where K is Constant) $K = 9 \times 10^9 \text{ Nm}^2/\text{coulomb}^2$

As we know that Coulombic force = Centrifugal force

$$\frac{K Z e^2}{r^2} = \frac{m v^2}{r} \text{ or } v^2 = \frac{K Z e^2}{m r} \dots\dots\dots(1); \quad \text{As we know that } \frac{n h}{2\pi} \text{ or } v = \frac{n h}{2\pi m r} \dots\dots\dots(2)$$

Now putting the value of v from eqⁿ.(2) to eqⁿ.(1)

$$\left(\frac{n h}{2\pi m r} \right)^2 = \frac{K Z e^2}{m r} \text{ or } \frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{K Z e^2}{m r}; \quad \boxed{r = \frac{n^2 h^2}{4\pi^2 m K Z e^2}} \dots\dots\dots(3)$$

Putting the value of π , h , m , k & e (constants) in the above eqⁿ. (3)

$$r = 0.529 \times 10^{-8} \times \frac{n^2}{Z} \text{ cm} \{ 1 \text{ \AA} = 10^{-10} \text{ m} = 10^{-8} \text{ cm} \}$$

$$r_n = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

This formula is only applicable for hydrogen and hydrogen like species i.e. species contains single electron.

2. Velocity of an electron:

now putting the value of Angular momentum

$$m.v.r. = \frac{nh}{2\pi};$$

$$v = \frac{nh}{2\pi mr}$$

$$v = \frac{nh4\pi^2 m K Z e^2}{2\pi m n^2 h^2};$$

$$v = \frac{2\pi K Z e^2}{nh}$$

now putting the value of π , K , e & h

$$v = 2.188 \times 10^8 \times \frac{Z}{n} \text{ cms}^{-1}$$

3. Energy of an electron:

Let the total energy of an electron be E . It is the sum of kinetic and potential Energy.

i.e. $E = \text{K.E.} + \text{P.E.}$

$$E = \left(\frac{1}{2} m v^2 \right) + \left(-\frac{K q_1 q_2}{r} \right) \left[\text{P.E.} = -\frac{K z e^2}{r} \right]$$

$$E = \frac{1}{2} m v^2 + \frac{K \cdot Z e \cdot (-e)}{r} = \frac{1}{2} m v^2 - \frac{K Z e^2}{r}$$

now putting the value of $m v^2$ from eq. (1)

$$E = \frac{K Z e^2}{2r} - \frac{K Z e^2}{r} = -\frac{K Z e^2}{2r}$$

now putting the value of r from eq. (3)

$$E_n = -\frac{K Z e^2 \times 4\pi^2 m K Z e^2}{2n^2 h^2}$$

or

$$E_n = -\frac{2\pi^2 m \times K^2 Z^2 e^4}{n^2 h^2}$$

now putting the value of π , K , e , m , h we get:

$$E_n = -\frac{21.69 \times 10^{-19} \times Z^2}{n^2} \text{ J/atom}$$

or

$$E_n = -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}$$

$$E_n = -1312 \times \frac{Z^2}{n^2} \text{ kJmol}^{-1}$$

or

$$E_n = -313.6 \times \frac{Z^2}{n^2} \text{ kcalmol}^{-1}$$

This formula is applicable for hydrogen atom & hydrogen like species i.e. single electron species. Since n can have only integral values, it follows that total energy of the e^- is quantised.

The -ve sign indicates that the e^- is under attraction towards nucleus.

Some extra points:

$$(i) K.E. = \frac{KZe^2}{2r} \quad \text{i.e.} \quad K.E. \propto \frac{1}{r}$$

On increasing radius, K.E. decreases.

$$(ii) P.E. = -\frac{KZe^2}{r} \quad \text{i.e.} \quad P.E. \propto -\frac{1}{r}$$

On increasing radius, P.E. increases.

$$(iii) E = -\frac{KZe^2}{2r} \quad \text{i.e.} \quad E \propto -\frac{1}{r}$$

Results: On increasing radius, total energy increases.

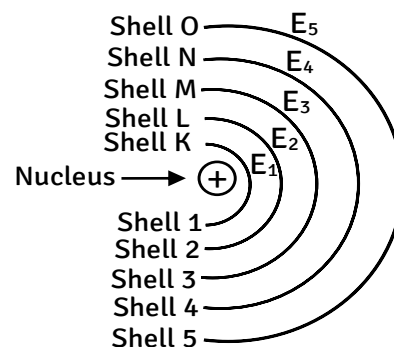
$$P.E. = 2KE \quad KE = -E \quad P.E. = 2E$$

Energy difference between two energy levels

$$E_{n_2} - E_{n_1} = -13.6 \times Z^2 \left[\frac{1}{n_2^2} - \frac{1}{n_1^2} \right]$$

Energy level for H atom can be represented as follows:

$n = 6$ or P	$E_6 = -0.38 \text{ eV}$	
$n = 5$ or O	$E_5 = -0.54 \text{ eV}$	
$n = 4$ or N	$E_4 = -0.85 \text{ eV}$	$E_5 - E_4 = 0.36 \text{ eV}$
$n = 3$ or M	$E_3 = -1.51 \text{ eV}$	$E_4 - E_3 = 0.66 \text{ eV}$
$n = 2$ or L	$E_2 = -3.4 \text{ eV}$	$E_3 - E_2 = 1.89 \text{ eV}$
$n = 1$ or K	$E_1 = -13.6 \text{ eV}$	$E_2 - E_1 = 10.2 \text{ eV}$
i.e. $(E_2 - E_1) > (E_3 - E_2) > (E_4 - E_3) > (E_5 - E_4) \dots\dots$		



Energy Definitions

(a) Ground state (G. S.):

The lowest energy state of an atom or ion or molecule

G. S. for H-atom $n = 1$

He⁺ ion $n = 1$

(b) Excited state (E. S.):

The energy states above the ground state are referred to as an excited

$n = 2$ 1^{st} E.S.

$n = 3$ 2^{nd} E.S.

$n = 4$ 3^{rd} E.S.

$$\text{Total E.S.} = (n - 1)$$

(c) Excitation Potential (E. P.):

The amount of energy needed to promote an electron from ground state to 'n'

level 1st E. P. $E_2 - E_1 = -3.4 - (-13.6) = 10.2 \text{ eV}$

2nd E. P. $E_3 - E_1 = -1.51 - (-13.6) = 12.09$

(d) Ionisation potential (I. P.) or Ionisation Energy (I.E.):

Corresponding energy required to remove the electron from G.S. to the infinite.

$$I.P. = E_{\infty} - E_1 = 0 - (-13.6 \times Z^2) = 13.6 \times Z^2 \text{ eV}$$

(e) Separation Energy (S. E.):

The amount of energy needed to remove electron from the excited state to the infinity.

$$\text{S.E.} = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - 0 \right]$$

$$\text{S.E.} = \frac{13.6 \times Z^2}{n_1^2}$$

Ex.7: Calculate the frequency, wave number of the microwaves with wavelength $4 \times 10^7 \text{ nm}$.

Sol. Wavelength (λ) = $4 \times 10^7 \text{ nm} = 4 \times 10^{-2} \text{ m}$ ($1 \text{ nm} = 10^{-9} \text{ m}$)

$$\text{Wave number } (\bar{\nu}) = \frac{1}{\lambda} = \frac{1}{4 \times 10^{-2} \text{ m}}$$

$$\bar{\nu} = 25 \text{ m}^{-1}$$

$$\text{Frequency } (\nu) = \frac{c}{\lambda}; \quad \nu = \frac{3 \times 10^8 \text{ ms}^{-1}}{4 \times 10^{-2} \text{ m}} = 7.5 \times 10^9 \text{ s}^{-1}$$

$$\nu = 7.5 \times 10^9 \text{ s}^{-1}$$

Ex.8: If the radius of 2nd Bohr orbit of hydrogen atom is r_2 . The radius of third Bohr orbit will be:

- (A) $\frac{4}{9} r_2$ (B) $4r_2$ (C) $\frac{9}{4} r_2$ (D) $9r_2$

Sol: (C)

$$r \propto \frac{n^2}{Z}$$

$$\therefore \frac{r_2}{r_3} = \frac{2^2}{3^2} \Rightarrow r_3 = \frac{9}{4} r_2$$

Ex.9: Calculate the radii of 2nd Bohr orbit of Li^{2+} .

Sol. Radii

$$(r_n) = \frac{52.9(n^2)}{Z} \text{ pm}; \quad Z \text{ of } \text{Li}^{2+} = 3$$

$$r_2 = \frac{52.9 \times (2^2)}{3} \text{ pm}; \quad r_2 = 70.53 \text{ pm}$$

Ex.10: Calculate the energy of an electron in the first Bohr orbit of He^+ .

$$\text{Sol.} \quad E_n = -2.18 \times 10^{-18} \left(\frac{Z^2}{n^2} \right) \text{ J} = -13.6 \frac{Z^2}{n^2} \text{ eV} = 54.4 \text{ eV}$$

$$Z = 2 \quad n = 1$$

$$E_1 = \frac{-2.18 \times 10^{-18} \times (2)^2}{(1)^2}; \quad E_1 = -8.72 \times 10^{-18} \text{ J}$$

Ex.11: (i) Energy associated with 1st orbit in H atom is $-2.18 \times 10^{-18} \text{ J atom}^{-1}$. What is the energy associated with fifth orbit?

(ii) Calculate the radius of Bohr's fifth orbit for hydrogen atom.

$$\text{Sol:} \quad (i) E_n = -\frac{2.18 \times 10^{-18}}{n^2} \text{ J} \therefore E_5 = -\frac{2.18 \times 10^{-18}}{5^2} = -8.72 \times 10^{-20} \text{ J}$$

$$(ii) \text{ For H - atom, } r_n = 0.529 \times n^2 \text{ \AA} \therefore r_5 = 0.529 \times 5^2 = 13.225 \text{ \AA}$$

Exercise 1.2

- The frequency of line spectrum of sodium is $5.09 \times 10^{14} \text{ sec}^{-1}$. Its wavelength (in nm) will be [$c = 3 \times 10^8 \text{ m/sec}$]
 (1) 510 nm (2) 420 nm
 (3) 589 nm (4) 622 nm
- The ratio of the energy of a photon of 2000 Å wavelength radiation to that of 4000 Å radiation is
 (1) 1 / 4 (2) 4 (3) 1 / 2 (4) 2
- The number of photons of light having wavelength 100 nm which can provide 1.00 J energy is nearly
 (1) 10^7 photons (2) 5×10^{18} photons
 (3) 5×10^{17} photons (4) 5×10^7 photons
- A 1 kW radio transmitter operates at a frequency of 880 Hz. How many photons per second does it emit?
 (1) 1.71×10^{21} (2) 1.71×10^{33}
 (3) 6.02×10^{23} (4) 2.85×10^{26}
- How many times does light travel faster in vacuum than an electron in Bohr first orbit of hydrogen atom?
 (1) 13.7 times (2) 67 times
 (3) 137 times (4) 97 times
- A gas absorbs a photon of 300 nm and then re-emits two photons. One photon has a wavelength 600 nm. The wavelength of second photon is –
 (1) 300 nm (2) 400 nm
 (3) 500 nm (4) 600 nm
- Bohr's model can explain –
 (1) The spectrum of only hydrogen atom
 (2) The spectrum of the atoms of all the elements
 (3) The spectrum of only sodium atom
 (4) The spectrum of atomic or ionic species having one electron
- On Bohr's stationary orbits
 (1) Electrons do not move
 (2) Electrons move emitting radiations
 (3) Energy of the electron remains constant
 (4) Angular momentum of the electron is $h/2\pi$
- Angular momentum for P-shell electron is
 (1) $\frac{3h}{\pi}$ (2) Zero
 (3) $\frac{\sqrt{2}h}{2\pi}$ (4) None
- The ratio of radii of the third and fifth orbits of Li^{+2} will be
 (1) 9 : 25 (2) 25 : 9
 (3) 2 : 3 (4) 4 : 5
- The diameter of the second orbit of hydrogen atom should be:
 (1) 2.12 Å (2) 4.23 Å
 (3) 2.10 Å (4) 4.01 Å
- The radius of a shell for H -atom is 4.761 Å. The value of n is :
 (1) 3 (2) 9
 (3) 5 (4) 4
- On the basis of Bohr's model, the radius of the 3rd orbit is –
 (1) Equal to the radius of first orbit
 (2) Three-times the radius of the first orbit
 (3) Five-times the radius of the first orbit
 (4) Nine times the radius of first orbit
- What should be the circumference of the second orbit of hydrogen atom?
 (1) 13.3 Å (2) 3.7 Å
 (3) 26.6 Å (4) 3.4 Å
- The ratio of the radii of first orbits of H, He^+ and Li^{+2} is :
 (1) 1 : 2 : 3 (2) 6 : 3 : 1
 (3) 9 : 4 : 1 (4) 6 : 3 : 2

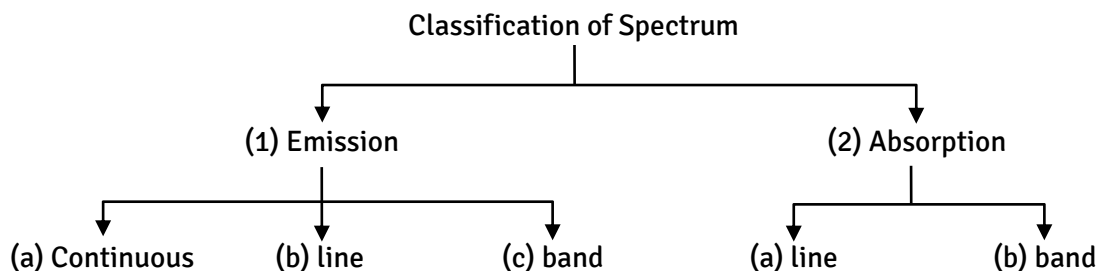
16. The approximate quantum number of a circular orbit of diameter 20.6 nm of the hydrogen atom according to Bohr's theory is:
(1) 10 (2) 14 (3) 12 (4) 16
17. The ratio of the radii of two Bohr orbit's of H - atom is 4 : 1 what would be their nomenclature:
(1) K & L (2) L & K
(3) N & L (4) 2 & 3 both
18. The radius of which of the following orbit is same as that of the first Bohr's orbit of hydrogen atom:
(1) $\text{He}^+(n = 2)$ (2) $\text{Li}^{2+}(n = 2)$
(3) $\text{Li}^{2+}(n = 3)$ (4) $\text{Be}^{3+}(n = 2)$
19. The expression for calculation of velocity is
(1) $v = \left(\frac{KZe^2}{mr} \right)$ (2) $v = \frac{2\pi KZe^2}{nh}$
(3) $v = \frac{nh}{2\pi KZe^2}$ (4) Both (1) & (2)
20. The velocity of electron in third excited state of Be^{3+} will be
(1) $\frac{3}{4}(2.188 \times 10^8) \text{ ms}^{-1}$
(2) $\frac{3}{4}(2.188 \times 10^6) \text{ ms}^{-1}$
(3) $(2.188 \times 10^6) \text{ kms}^{-1}$
(4) $(2.188 \times 10^3) \text{ kms}^{-1}$
21. What should be the velocity of the electrons present in the first, second and third orbits of H, He^+ and Li^{+2} , respectively?
(1) $2.188 \times 10^8 \text{ cm per second}$
(2) $5.47 \times 10^7 \text{ cm per second}$
(3) $4.376 \times 10^8 \text{ cm per second}$
(4) $2.188 \times 10^6 \text{ cm per second}$
22. If v_1 , v_2 , v_3 and v_4 are velocities of the electron present in the first orbit of H, He^+ , Li^{+2} and Be^{+3} , then which of the following should be their increasing order?
(1) $v_1 < v_2 < v_3 < v_4$ (2) $v_4 = v_3 = v_2 = v_1$
(3) $v_1 < v_2 < v_3 > v_4$ (4) $v_1 > v_2 < v_3 > v_4$
23. The correct statement(s) about Bohr's orbits of hydrogen atom is/are
(1) $r = \left[\frac{n^2 h^2}{4^2 m k z e^2} \right]$
(2) K.E. of the electron = $-1/2$ (P. E. of the electron)
(3) Angular momentum (L) = $n \left(\frac{h}{2\pi} \right)$
(4) All the above
24. Which orbits of H, He^+ and Li^{+2} have identical energies?
(1) Second orbits of all the three
(2) First orbit of H, second orbit of He^+ and third orbit of Li^{+2}
(3) Third orbit of all the three
(4) Fourth orbit of H, First orbit of He^+ and Fifth orbit of Li^{+2}
25. What should be the ratio of energies of the fifth orbits of Li^{+2} and He^+ ?
(1) 4 : 9 (2) 9 : 4
(3) 12 : 16 (4) 7 : 2
26. The energy of H -atom in n^{th} orbit is E_n then energy in n^{th} orbit of singly ionised helium atom will be:
(1) $4E_n$ (2) $E_n/4$
(3) $2E_n$ (4) $E_n/2$
27. What should be the correct order of energies of the first orbits of H, He^+ and Li^{+2} , if these are represented as E_1 , E_2 and E_3 respectively?
(1) $E_1 < E_2 < E_3$ (2) $E_3 < E_2 < E_1$
(3) $E_1 < E_2 > E_3$ (4) $E_1 = E_2 = E_3$
28. In hydrogen atom energy of first excited state is: -3.4 eV. Then find out the K.E. of the electron in the same orbit of hydrogen atom:
(1) 3.4 eV (2) +6.8 eV
(3) -13.6 eV (4) +13.6 eV

29. If potential energy of an electron in hydrogen atom is $-x$ eV, then its kinetic energy will be –
 (1) x eV (2) $-xeV$
 (3) $2x$ eV (4) $x/2$ eV
30. The energy of electron in first Bohr's orbit of H -atom is -13.6 eV. What will be its potential energy in $n = 4^{\text{th}}$ orbit
 (1) -13.6 eV (2) -3.4 eV
 (3) -0.85 eV (4) -1.70 eV
31. Potential energy is -27.2 eV in second orbit of He^+ then calculate, double of total energy in first excited state of Hydrogen atom:
 (1) -13.6 eV (2) -54.4 eV
 (3) -6.8 eV (4) -27.2 eV
32. The potential energies of first, second and third Bohr's orbits of He^+ cation are E_1 , E_2 and E_3 . The correct sequence of these energies is –
 (1) $E_1 > E_2 > E_3$ (2) $E_1 = E_2 > E_3$
 (3) $E_1 = E_2 = E_3$ (4) $E_3 > E_2 > E_3$
33. In a hydrogen atom, if the energy of electron in the ground state is -13.6 eV, then that in the 2^{nd} excited state is:
 (1) -1.51 eV (2) -3.4 eV
 (3) -6.04 eV (4) -13.6 eV
34. Energy required to remove an e^- from M shell of H -atom is 1.51 eV, then energy of first excited state will be:
 (1) -1.51 eV (2) $+1.51$ eV
 (3) -3.4 eV (4) -13.6 eV
35. The energy of second Bohr orbit of the hydrogen atom is -328 kJ/mol. Hence the energy of fourth Bohr orbit should be-
 (1) -41 kJ/mol. (2) -1312 kJ/mol.
 (3) -164 kJ/mol. (4) -82 kJ/mol.
36. The ratio of $E_2 - E_1$ to $E_4 - E_3$ for the hydrogen atom is approximately equal to –
 (1) 10 (2) 15 (3) 17 (4) 12
37. A single electron orbits a stationary nucleus ($z = 5$). The energy required to excite the electron from the third to the fourth Bohr orbit will be:
 (1) 4.5 eV (2) 8.53 eV
 (3) 25 eV (4) 16.53 eV
38. Ionization energy for hydrogen atom in ergs, Joules and eV respectively is
 (1) 21.8×10^{-12} , 218×10^{-20} , 13.6
 (2) 13.6 , 218×10^{-20} , 21.8×10^{-13}
 (3) 21.8×10^{-20} , 13.6 , 218×10^{-13}
 (4) 21.8×10^{-13} , 13.6 , 21.8×10^{-20}
39. If the I.P. of Li^{+2} is 122.4 eV. Find out I.P. of C^{+5} –
 (1) 122.4×4 eV (2) 122.4×2 eV
 (3) 122.4×3 eV (4) 122.4×5 eV
40. The ionization energy of a hydrogen atom is 13.6 eV. The energy of the third-lowest electronic level in doubly ionized lithium ($Z = 3$) is –
 (1) -28.7 eV (2) -54.4 eV
 (3) -122.4 eV (4) -13.6 eV
41. If the ionization energy of hydrogen is 313.8 K cal per mole, then the energy of electron in 2^{nd} excited state will be
 (1) -113.2 K cal./mole
 (2) -78.45 K cal./mole
 (3) -313.8 cal./mole
 (4) -35 K cal./mole
42. Supposing the ionization energy of hydrogen atom is 640 eV. Point out the main shell having energy equal to -40 eV
 (1) $n = 2$ (2) $n = 3$
 (3) $n = 4$ (4) $n = 5$
43. The energy of an electron in the first Bohr orbit of H atom is -13.6 eV. The possible energy value(s) of the excited state for electrons in Bohr orbits of Li^{2+} is:
 (1) -61.2 eV (2) -13.6 eV
 (3) -122.4 eV (4) $+6.8$ eV

44. Maximum frequency of emission is obtained for the transition:
 (1) $n = 2$ to $n = 1$ (2) $n = 6$ to $n = 2$
 (3) $n = 1$ to $n = 2$ (4) $n = 2$ to $n = 6$
45. If the radii of first orbits of H, He^+ , Li^{+2} and Be^{+3} are r_1 , r_2 , r_3 and r_4 respectively, then their correct decreasing order will be:
 (1) $r_1 > r_2 > r_3 > r_4$ (2) $r_3 < r_2 > r_4 < r_1$
 (3) $r_1 < r_2 < r_3 > r_4$ (4) Radius of all are equal
46. An electron in an atom jumps in such a way that its kinetic energy changes from x to $\frac{x}{4}$. The change in potential energy will be:
 (1) $+\frac{3}{2}x$ (2) $-\frac{3}{8}x$
 (3) $+\frac{3}{4}x$ (4) $-\frac{3}{4}x$
47. The distance between 4th and 3rd Bohr orbits of He^+ is:
 (1) $2.645 \times 10^{-10}\text{m}$ (2) $1.322 \times 10^{-10}\text{m}$
 (3) $1.851 \times 10^{-10}\text{m}$ (4) None
48. The energy levels for ${}_ZA^{(z-1)}$ can be given by:
 (1) E_n for $A^{(z-1)} = z^2 \times E_n$ for H
 (2) E_n for $A^{(z-1)} = z \times E_n$ for H
 (3) E_n for $A^{(z-1)} = \frac{1}{z^2} \times E_n$ for H
 (4) E_n for $A^{(z-1)} = \frac{1}{z} \times E_n$ for H
49. The angular momentum of electron of H-atom is proportional to:
 (1) r^2 (2) $\frac{1}{r}$
 (3) $\sqrt{r}$ (4) $\frac{1}{\sqrt{r}}$

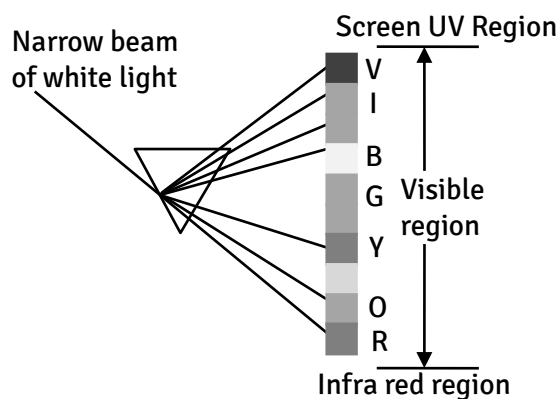
Spectrum

When a radiation is passed through a spectroscope (prism) for the dispersion of the radiation, the pattern (photograph) obtained on the screen (photographic plate) is called as spectrum of the given radiation

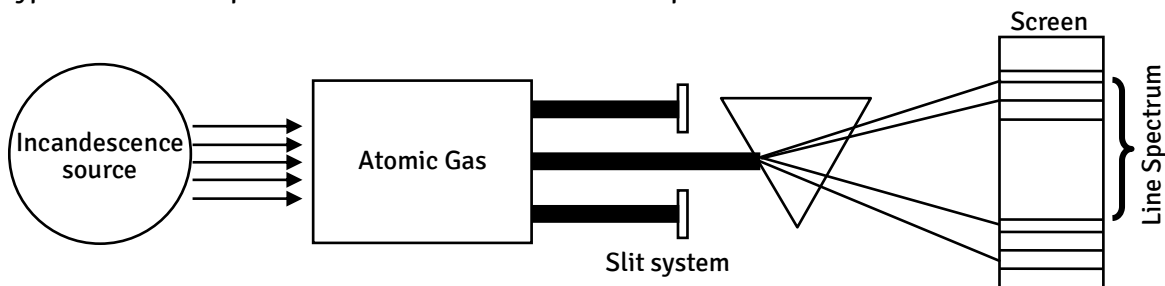


- (1) **Emissions spectrum:** When the radiation emitted from Incandescence source (e.g. from the candle, sun, tube light, burner, bulb, or by-passing electric discharge through a gas at low pressure, by heating some substance at high temperature) is passed directly through the prism and the pattern obtained on the screen is known as emission spectrum.

- (a) **Emission continuous spectrum or continuous spectrum:**
 When a narrow beam of white light is passed through a prism, it is dispersed into 7 colours from violet to Red.



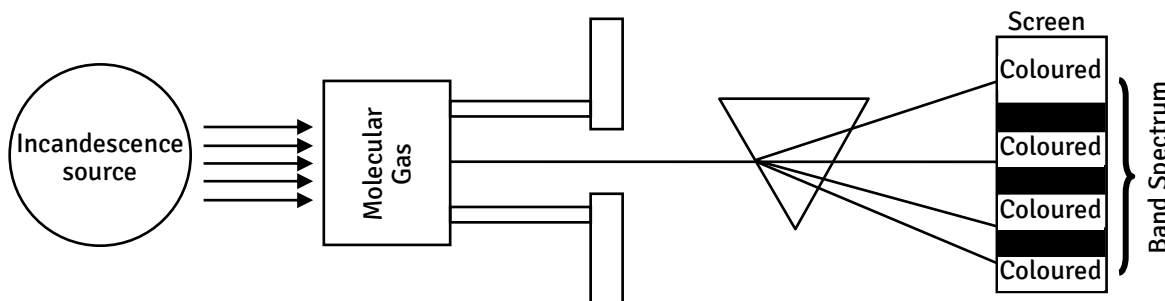
- (b) **Emission line spectrum:** When an atomic gas is raised to incandescence source or subjected to electrical excitation, it first absorbs energy & then gives it out as radiation. On examining these radiation through a spectro scope a spectrum is obtained which have well defined lines, each corresponding to a definite wave length & these lines are separated from each other by dark space. This type of Emission spectrum is called as Emission line spectrum.



Special Note:

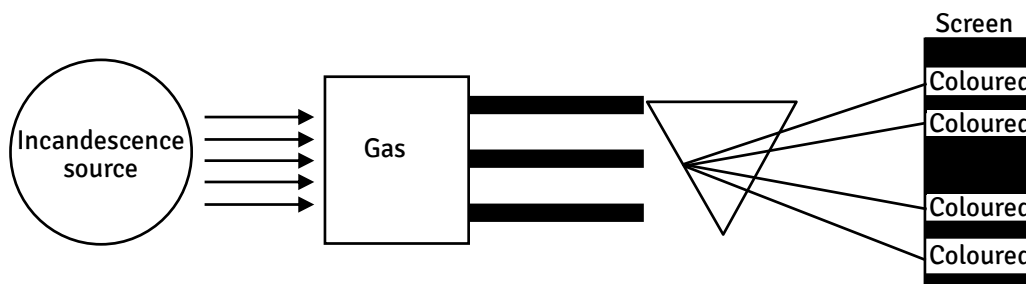
1. No two Elements will have identical line spectrum since no two elements have identical energy level therefore the line spectrum of the elements are described as finger prints differing from each other like the finger prints of the human beings.
2. Since line spectrum is obtained by the Emission of Energy through the atoms of the element therefore line spectrum is also called as atomic spectrum.

- (c) **Emission band spectrum:** If molecular form of the gas is used, it first absorbs energy for not only electron transition but for rotational, vibrational and electron translation then emits radiations. On examining these radiations through a spectroscope a spectrum is obtained on the screen, which consists group of closely packed lines called Bands, therefore this type of Emission spectrum is called as emission band spectrum. Bands are separated from each other by dark space.



Note: Since band spectrum are caused by molecules therefore band spectrum are also called as molecular spectrum.

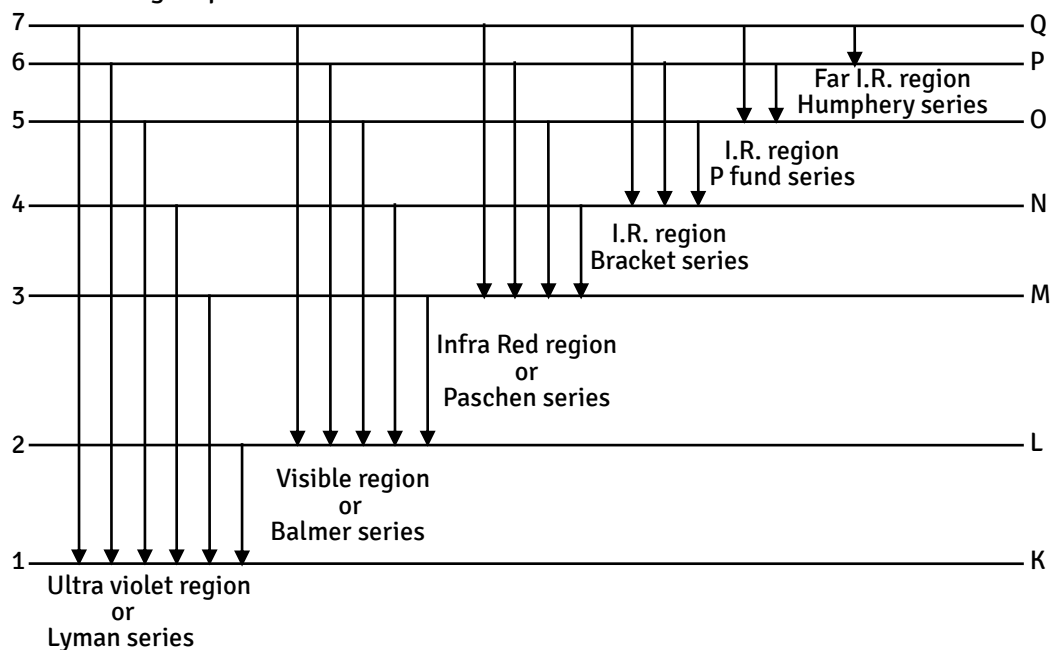
- (2) **Absorption spectrum:** When white light is first passed through a solution or vapours of a chemical substance or gas and then analyzed by spectroscope, it is observed that some dark lines are obtained in otherwise continuous spectrum. This type of spectrum is called as Absorption spectrum.



- If white light is passed through atomic gas, then the obtained spectrum is called as Absorption line spectrum.
- If white light is passed through molecular gas, then the obtained spectrum is called as Absorption band spectrum.

Hydrogen Line Spectrum or Hydrogen Spectrum

When an electric excitation is applied on hydrogen atomic gas at Low pressure, a bluish light is emitted. when a ray of this light is passed through a prism, a spectrum of several isolated sharp line is obtained. The wavelength of various lines show that spectrum lines lie in visible, Ultraviolet and Infra red region. These lines are grouped into different series.



Series	Discovered by	regions	$n_2 \rightarrow n_1$
lyman	lyman	U.V. region	$n_2 = 2, 3, 4, \dots / n_1 = 1$
Balmer	Balmer	Visible region	$n_2 = 3, 4, 5, \dots / n_1 = 2$
Paschen	Paschen	Infra-red (I. R.)	$n_2 = 4, 5, 6, \dots / n_1 = 3$
Bracket	Bracket	I.R. region	$n_2 = 5, 6, 7, \dots / n_1 = 4$
Pfund	Pfund	I.R. region	$n_2 = 6, 7, \dots / n_1 = 5$
Humphery	Humphery	far I.R. region	$n_2 = 7, \dots / n_1 = 6$

Rydberg formula: In 1890, Rydberg gave a very simplest theoretical Equation for the calculation of the wavelength of various lines of hydrogen like spectrum and the equation is

$$\bar{\nu} = \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where R = Rydberg constant = $109678 \text{ cm}^{-1} \simeq 109700 \text{ cm}^{-1}$

$$\frac{1}{R} = 9.12 \times 10^{-6} \text{ cm} = 912 \text{ \AA}$$

n_1 and n_2 are orbits and for a particular series n_1 is constant and n_2 varies.

for Lyman

for Lyman $n_1 = 1, n_2 = 2, 3, 4, \dots$

for Balmer $n_1 = 2, n_2 = 3, 4, 5, \dots$

for Paschen $n_1 = 3, n_2 = 4, 5, 6, \dots$

for Brackett $n_1 = 4, n_2 = 5, 6, 7, \dots$

for Pfund $n_1 = 5, n_2 = 6, 7, 8, \dots$

for Humphery $n_1 = 6, n_2 = 7, 8, 9, \dots$

Derivations of Rydberg formula:

$$E = E_{n_2} - E_{n_1}$$

$$E = \frac{-2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2} - \left[\frac{-2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} \right] = \frac{2\pi^2 m K^2 Z^2 e^4}{n_1^2 h^2} - \frac{2\pi^2 m K^2 Z^2 e^4}{n_2^2 h^2} \quad \left(E = h\nu = \frac{hc}{\lambda} \right)$$

$$\frac{hc}{\lambda} = \frac{2\pi^2 m K^2 Z^2 e^4}{h^2} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \quad \text{or} \quad \frac{1}{\lambda} = \frac{2\pi^2 m K^2 e^4 Z^2}{ch^3} \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

where $\frac{2\pi^2 m K^2 e^4}{ch^3}$ is a constant which is equal to rydberg constant (R).

$$\frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

Calculation of number of spectral lines:

(a) Total number of spectral lines

$$= 1 + 2 + \dots (n_2 - n_1) = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$$

if $n_1 = 1$ (ground state)

Total number of spectral lines

$$= \frac{(n_2 - 1)n_2}{2} = \frac{n(n-1)}{2}$$

(b) Number of spectral lines which falls in a particular series ($n_2 - n_1$)

where n_2 = higher energy state, n_1 = lower energy state

Limitation of the Bohr's model:

1. Bohr's theory does not explain the spectrum of multi electron atom.
2. Why the Angular momentum of the revolving electron is equal to $\frac{nh}{2\pi}$, has not been explained by Bohr's theory.
3. Bohr interrelate quantum theory of radiation and classical law of physics without any theoretical explanation.
4. Bohr's theory does not explain the fine structure of the spectral lines. Fine structure of the spectral line is obtained when spectrum is viewed by spectroscope of more resolution power.
5. Bohr theory does not explain the splitting of spectral lines in the presence of magnetic field (Zeeman's effect) or electric field (Stark's effect)

Wave Mechanical Model of an Atom

This model consists three model.

1. Dual nature of Matter (de-Broglie concept)
2. Heisenberg Uncertainty principle.

1. The Dual nature of matter (The wave nature of electron):

- (i) In 1924, a French physicist, Louis de-Broglie suggested that if the nature of light is both that of a particle and of a wave, then this dual behaviour should be true for the matter also.
- (ii) The wave nature of light rays and X-rays is proved on the basis of their interference and diffraction and many facts related to radiations can only be explained when the beam of light rays is regarded as composed of energy corpuscles or photons whose velocity is 3×10^{10} cm/s.
- (iii) According to de-Broglie, the wavelength λ of an electron is inversely proportional to its momentum p .

$$\lambda \propto \frac{1}{p} \text{ or } \lambda = \frac{h}{p}$$

(Here h = Planck's constant, p = momentum of electron)

Momentum (p) = Mass (m) $\times$ Velocity (v) $\lambda = \frac{h}{mv} \therefore$

- (iv) The above relation can be confined as follows by using Einstein's equation, Planck's quantum theory and wave theory of light.

Einstein's equation, $E = mc^2$ where E is energy, m is mass of a body and c is its velocity.

$\therefore E = hv$ (According to Planck's quantum theory) and $c = v\lambda$ (According to wave theory of light)

$$\therefore v = \frac{c}{\lambda} \Rightarrow E = h \times \frac{c}{\lambda}$$

But according to Einstein's equation $E = mc^2$

$$E = mc^2 = h \times \frac{c}{\lambda} \quad \text{or} \quad mc = \frac{h}{\lambda}$$

$$\text{or} \quad p = \frac{h}{\lambda} \quad \text{or} \quad \lambda = \frac{h}{p}$$

- (v) It is clear from the above equation that the value of λ decreases on increasing either m or v or both. The wavelength of many fast-moving objects like an aeroplane or a cricket ball, is very low because of their high mass.

Bohr's Theory and de-Broglie concept:

- (i) According to de-Broglie, the nature of an electron moving around the nucleus is like a wave that flows in circular orbits around the nucleus.
- (ii) If an electron is regarded as a wave, the quantum condition as given by Bohr in his theory is readily fulfilled.
- (iii) If the radius of a circular orbit is r , then its circumference will be $2\pi r$.

- (iv) We know that according to Bohr theory, $mvr = \frac{nh}{2\pi}$

$$\text{or} \quad 2\pi r = \frac{nh}{mv} (\because mv = p \text{ momentum}) \quad \text{or} \quad 2\pi r = \frac{nh}{p} \left(\frac{h}{p} = \lambda \text{ de-Broglie equation} \right)$$

$\therefore 2\pi r = n\lambda$ (where n = total number of waves 1, 2, 3, 4, 5, ∞ and λ = Wavelength)

$$(v) \therefore 2\pi r = \frac{nh}{mv} \text{ or } mvr = \frac{nh}{2\pi}$$

$\therefore mvr$ = Angular momentum

Thus mvr = Angular momentum, which is a integral multiple of $\frac{h}{2\pi}$.

- (vi) It is clear from the above description that according to de-Broglie there is similarity between wave theory and Bohr theory.

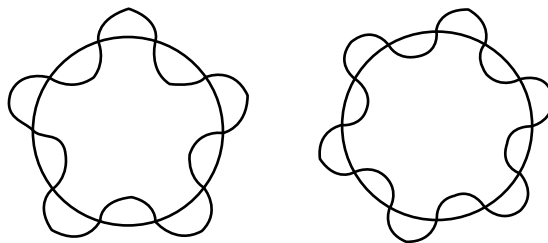


Figure: similarity between de-Broglie waves and Bohr's orbit number of waves made by electron in one complete revolution = n (orbit number)

2. Heisenberg uncertainty principle:

In 1927, Werner Heisenberg presented a principle known as Heisenberg uncertainty principle which states as: "It is impossible to measure simultaneously the exact position and exact momentum of a body as small as an electron."

The uncertainty of measurement of position (Δx) and the uncertainty of momentum (Δp) or $\Delta(mv)$, are related by Heisenberg's relationship as:

$$x.p \geq \frac{h}{4\pi}$$

or $x.(mv) \geq \frac{h}{4\pi}$

(Where h is Planck's constant.)

For an electron of mass $m(9.10 \times 10^{-28}g)$, the product of uncertainty is quite large.

$$x.v \geq \frac{h}{4\pi m}$$

$$x.v \geq \frac{6.624 \times 10^{-27}}{4 \times 3.14 \times 9.10 \times 10^{-28}} = 0.57 \text{ erg second per gram approximately}$$

$\Delta x \Delta v$ is called uncertainty product

When $\Delta x = 0$, $\Delta v = \infty$ and vice-versa.

In the case of bigger particles (having considerable mass), the value of uncertainty product is negligible. If the position is known quite accurately, i.e., Δx is very small, Δv becomes large and vice-versa.

$$Et \geq \frac{h}{4\pi} \text{ (for energy and time)}$$

On the basis of this principle, therefore, Bohr's picture of an electron in an atom, which gives a fixed position in a fixed orbit best we can think of in terms of probability of locating an electron with a probable velocity in a given region of space at a given time. The space or a three-dimensional region around the nucleus where there is maximum probability of finding an electron of a specific energy is called an atomic orbital.

Points to Ponder

- de-Broglie wavelength in terms of kinetic energy.

$$\text{Kinetic Energy (K.E.)} = \frac{1}{2}mv^2 \quad \text{or} \quad m \times \text{K.E.} = \frac{1}{2}m^2v^2 \quad \text{or} \quad m^2v^2 = 2m \text{ K.E.} \quad \text{or} \quad mv = \sqrt{2m \text{ K.E.}}$$

$$\text{But } \lambda = \frac{h}{mv} \quad \therefore \quad \lambda = \frac{h}{\sqrt{2m \text{ K.E.}}} \quad (\because mv = \sqrt{2m \text{ K.E.}})$$

- When a charged particle carrying Q coulomb is accelerated by applying potential difference V then $\text{K.E.} = Q \times V$ Joule

$$\text{But } \lambda = \frac{h}{\sqrt{2m \text{ K.E.}}} \quad \therefore \quad \lambda = \frac{h}{\sqrt{2m QV}}$$

$$\text{For electron } \left(\lambda = \sqrt{\frac{150}{V}} \text{ \AA} \right) = \frac{12.25}{\sqrt{V}} \text{ \AA}$$

- The wave nature of electron was verified experimentally by Davisson and Germer.
- de-Broglie hypothesis is applicable to macroscopic as well as microscopic objects but it has no physical significance for macroscopic objects.

- Remember $\frac{h}{4\pi} = 0.527 \times 10^{-34} \text{ J sec}$

Ex.12: If electron make transition from 7th excited state to 2nd orbit in H atom sample find the max. number of spectral lines observed.

$$\text{Sol: } \Delta n = 8 - 2 = 6 \text{ spectral lines} = 6 \left(\frac{6+1}{2} \right) = 6 \times \frac{7}{2} = 21$$

Ex.13: An electron in isolated hydrogen atom is in 4th excited state, then, upon de excitation:

- (A) the maximum number of possible photons will be 10
 (B) the maximum number of possible photons will be 6
 (C) it can emit two photons in ultraviolet region
 (D) It can emit one photon in visible region

Sol: (A)

Ex.14: In a hydrogen like sample, ions are in a particular excited state, if electrons make transition upto 1st excited state, then it produces maximum 15 different types of spectral lines then electrons were in:

- (A) 5th state (B) 6th state (C) 7th state (D) 8th state

Sol: (C)

$$\frac{\Delta n(\Delta n + 1)}{2} = 15 \Rightarrow \Delta n = 5 \Rightarrow n - 2 = 5 \Rightarrow n = 7$$

Ex.15: What are the frequency and wavelength of a photon emitted during a transition from $n = 5$ to $n = 2$ in H atom?

Sol: According to Rydberg formula, $\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$$\text{Here, } R = 109,677 \text{ cm}^{-1}, n_2 = 5, n_1 = 2$$

$$\therefore \bar{\nu} = 109,677 \left(\frac{1}{2^2} - \frac{1}{5^2} \right) \text{ cm}^{-1} = 109,677 \times \frac{21}{100} \text{ cm}^{-1} = 23032.2 \text{ cm}^{-1}$$

$$\lambda = \frac{1}{\bar{\nu}} = \frac{1}{23032.2 \text{ cm}^{-1}} = 434 \times 10^{-7} \text{ cm} = 434 \times 10^{-9} \text{ m} = 434 \text{ nm}$$

$$\nu = \frac{c}{\lambda} = \frac{3 \times 10^8 \text{ ms}^{-1}}{434 \times 10^{-9} \text{ m}} = 6.91 \times 10^{14} \text{ s}^{-1}$$

Ex.16: Wavelength of 1st line in Balmer series is 656 nm for H-atom. Cal. wavelength of 2nd line and limiting line in Balmer series is H-atom.

Sol: According to Rydberg formula, $\bar{\nu} = \frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

For the Balmer series, $n_1 = 2$ and for the 1st line, $n_2 = 3$

$$\therefore \frac{1}{656} = R \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R \times \left(\frac{1}{4} - \frac{1}{9} \right) = R \times \frac{5}{36} = \frac{5R}{36} \quad \dots(i)$$

For the second line, $n_1 = 2$, $n_2 = 4$

$$\therefore \frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R \left(\frac{1}{4} - \frac{1}{16} \right) = R \times \frac{3}{16} = \frac{3R}{16} \quad \dots(ii)$$

Dividing eqn. (i) by eqn. (ii), we get $\frac{\lambda}{656} = \frac{5}{36} \times \frac{16}{3}$ or $\lambda = 485.9 \text{ nm}$

$$\text{For the limiting line, } n_1 = 2, n_2 = \infty \therefore \frac{1}{\lambda} = R \left(\frac{1}{2^2} - \frac{1}{\infty^2} \right) = \frac{R}{4} \quad \dots(iii)$$

Dividing eqn. (i) by eqn.(ii), we get $\frac{\lambda}{656} = \frac{5}{36} \times 4$ or $\lambda = 364.4 \text{ nm}$

Ex.17: Calculate the frequency of the spectral line corresponding to $n_1 = 2$ and $n_2 = 4$. To which spectral series does this line belong?

Sol. $\bar{\nu} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$

$$n_1 = 2 \quad n_2 = 4$$

$$\bar{\nu} = \frac{1}{\lambda}$$

$R = \text{Rydberg constant} = 109677 \text{ cm}^{-1}$

Substituting the values in the equation

$$\frac{1}{\lambda} = 109677 \left(\frac{1}{(2)^2} - \frac{1}{(4)^2} \right) \text{ cm}^{-1}$$

$$\frac{1}{\lambda} = 20,564.43 \text{ cm}^{-1}; \quad \lambda = 4.86 \times 10^{-5} \text{ cm}$$

$$\lambda = 4.86 \times 10^{-7} \text{ m}; \quad \text{Frequency } (\nu) = \frac{c}{\lambda}$$

$$\nu = \frac{3 \times 10^8 \text{ ms}^{-1}}{4.86 \times 10^{-7} \text{ m}} = 6.172 \times 10^{14} \text{ s}^{-1}$$

So, the frequency of spectral line is $6.172 \times 10^{14} \text{ s}^{-1}$.

Since, $n_1 = 2$, so it belongs to Balmer series.

Ex.18: Calculate wavelength of 150 gm rubber ball with 50 m/sec velocity

Sol. $\lambda = \frac{h}{mv}$

Given $v = 50 \text{ m sec}^{-1}$

$m = 150 \text{ g}$

$$\therefore \lambda = \frac{6.625 \times 10^{-34}}{150 \times 50} = 0.088 \times 10^{-33} \text{ m}$$

Ex.19: Velocity of helium atom at 300K is 2.40×10^2 meter per sec. What is its wave length? (mass number of helium is 4):

- (A) 0.416 nm (B) 0.83 nm (C) 803 Å (D) 8000 Å

Sol: (A)

$$\lambda = \frac{h}{mv}$$

$$\text{Mass of helium} = \frac{4.0 \times 10^{-3}}{6.023 \times 10^{23}} \text{ kg. and } h = 6.62 \times 10^{-34}$$

$$\lambda = 6.62 \times 10^{-34} \times \frac{6.023 \times 10^{23}}{4.0 \times 10^{-3}} \times \frac{1}{2.4 \times 10^2} = 0.416 \times 10^{-9} \text{ meter}$$

$$\lambda = 0.416 \text{ nm}$$

Ex.20: K.E. of the electron is 4.55×10^{-25} J. Its de Broglie wave length is -

- (A) 4700 Å (B) 8300 Å (C) 7200 Å (D) 7400 Å

Sol: (C)

$$\lambda = \frac{h}{\sqrt{2mKE}}$$

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 4.55 \times 10^{-25}}} \Rightarrow \lambda = \frac{6.626 \times 10^{-34}}{9.1 \times 10^{-28}}$$

$$\lambda = 0.72 \times 10^{-6} \times 10^{10} \text{ Å}$$

$$\lambda = 7200 \text{ Å}$$

Ex.21: For particles having same kinetic energy, the de Broglie wavelength is -

- (A) Directly proportional to its velocity (B) Inversely proportional to its velocity
(C) Independent of velocity and mass (D) Unpredictable

Sol: (A)

$$\lambda = \frac{h}{mv}; \quad KE = \frac{1}{2}mv^2$$

$$mv = \frac{2KE}{v}$$

$$\therefore \lambda = \frac{h}{\frac{2KE}{v}} \Rightarrow \lambda = h \left(\frac{v}{2KE} \right)$$

Ex.22: Number of waves made by a Bohr electron in one complete revolution in 3rd orbit is:

- (A) 2 (B) 3 (C) 4 (D) 1

Sol: (B)

$$\text{Circumference of 3rd orbit} = 2\pi r_3$$

According to Bohr, angular momentum of electron in 3rd orbit is

$$mvr_3 = 3 \frac{h}{2\pi} \quad \text{or} \quad \frac{h}{mv} = \frac{2\pi r_3}{3} \quad \text{by De-Broglie equation}$$

$$\lambda = \frac{h}{mv}$$

$$\therefore \lambda = \frac{2\pi r_3}{3} \Rightarrow 2\pi r_3 = 3\lambda$$

i.e. circumference of 3rd orbit is three times the wavelength of electron or number of waves made by Bohr electron in one complete revolution in 3rd orbit is three.

Ex.23: Calculate the uncertainty in the velocity of an electron when the uncertainty in its position is 1.012×10^{-12} m.

Sol. According to Heisenberg's uncertainty principle,

$$\Delta x \times \Delta p \geq \frac{h}{4\pi}$$

$$p = mv$$

$$\therefore mv \geq \frac{h}{4\pi\Delta x}; \quad v \geq \frac{h}{4\pi m \Delta x}$$

$$\text{Mass of an electron} = 9.1 \times 10^{-31} \text{ kg}$$

$$\pi = 3.142 \text{ or } \frac{22}{7}$$

$$\Delta x = 1.012 \times 10^{-12} \text{ m}$$

$$\therefore \Delta v = \frac{6.626 \times 10^{-34} \text{ Js}}{4 \times 3.142 \times (9.1 \times 10^{-31} \text{ kg}) \times (1.012 \times 10^{-12} \text{ m})}$$

$$(J = \text{kg m}^2 \text{ s}^{-1})$$

$$\Delta v = 5.72 \times 10^7 \text{ ms}^{-1}$$

Ex.24: If uncertainty in position and momentum are equal, the uncertainty in velocity is -

- (A) $\sqrt{\frac{h}{2\pi}}$ (B) $\frac{1}{2m} \sqrt{\frac{h}{\pi}}$ (C) $\sqrt{\frac{h}{\pi}}$ (D) None of these

Sol: (B)

$$\Delta x \cdot m \Delta v = \frac{h}{4\pi} \quad \text{or} \quad (m \Delta v)^2 = \frac{h}{4\pi}$$

$$m \Delta v = \sqrt{\frac{h}{4\pi}} \quad \text{or} \quad \Delta v \geq \sqrt{\frac{h}{4\pi m^2}} = \frac{1}{2m} \sqrt{\frac{h}{\pi}}$$

Ex.25: The uncertainty in position and velocity of a particle are 10^{-10} m and $5.27 \times 10^{-24} \text{ ms}^{-1}$ respectively. Calculate the mass of the particle is ($h = 6.625 \times 10^{-34} \text{ J-s}$)

- (A) 0.099 (B) 0.99 g (C) 0.92 kg (D) None of these

Sol: (A)

According to Heisenberg's uncertainty principle,

$$m = \frac{h}{4\pi \Delta x \cdot \Delta v}$$

$$\text{or } = \frac{6.625 \times 10^{-34}}{4 \times 3.143 \times 10^{-10} \times 5.27 \times 10^{-24}} = 0.099 \text{ kg.}$$

Ex.26: On the basis of Heisenberg's uncertainty principle, show that the electron cannot exist within the nucleus.

Sol: Diameter of the nucleus is of the order of 10^{-15} m. The maximum uncertainty in the position of electron within the nucleus will be 10^{-15} m. Now, from Heisenberg's uncertainty principle.

$$\Delta x \cdot m \Delta v \geq \frac{h}{4\pi} \quad \text{or} \quad \Delta v_{\min} = \frac{h}{4\pi m \Delta x}$$

Exercise 1.3

- Splitting of spectral lines under the influence of magnetic field is called:
 - Zeeman effect
 - Stark effect
 - Photoelectric effect
 - None of these
- Which of the following series of lines in the atomic emission spectrum of hydrogen is in the visible region?
 - Lyman
 - Paschen
 - Balmer
 - Brackett
- The spectrum of He^+ is expected to be similar to that of:
 - Li^+
 - H
 - Na
 - He
- The wavelength of the third line of the Balmer series for a hydrogen atom is –
 - $\frac{21}{100R_H}$
 - $\frac{100}{21R_H}$
 - $\frac{21R_H}{100}$
 - $\frac{100R_H}{21}$
- Wave-length of the first line of Paschen Series for hydrogen atom is – ($R = 109700 \text{ cm}^{-1}$)
 - 18780 Å
 - 2854 Å
 - 3452 Å
 - 6243 Å
- Which of the following should be the expression for the last line of Paschen series?
 - $\frac{1}{\lambda} = R \left(\frac{1}{9} - \frac{1}{\infty^2} \right)$
 - $\frac{1}{\lambda} = R \left(\frac{1}{4} - \frac{1}{9} \right)$
 - $\frac{1}{\lambda} = R \left(\frac{1}{9} - \frac{1}{16} \right)$
 - $\frac{1}{\lambda} = R \left(\frac{1}{16} - \frac{1}{\infty} \right)$
- The wave number of the first line of Balmer series of hydrogen is 15200 cm^{-1} . The wave number of the first Balmer line of Li^{2+} ion is-
 - 15200 cm^{-1}
 - 60800 cm^{-1}
 - 76000 cm^{-1}
 - 136800 cm^{-1}
- The wavelength of photon obtained by electron transition between two levels in H atom and singly ionised He are λ_1 and λ_2 respectively, then:
 - $\lambda_2 = \lambda_1$
 - $\lambda_2 = 2\lambda_1$
 - $\lambda_2 = \lambda_1/2$
 - $\lambda_2 = \lambda_1/4$
- Wave number of a spectral line for a given transition is $x \text{ cm}^{-1}$ for He^+ , then its value for Be^{3+} for the same transition is
 - $4x \text{ cm}^{-1}$
 - $x \text{ cm}^{-1}$
 - $x/4 \text{ cm}^{-1}$
 - $2x \text{ cm}^{-1}$
- What should be the frequency of radiation of the emission spectrum when the electron present in hydrogen atom undergoes transition from $n = 3$ energy level to the ground state?
 - $3 \times 10^{15} \text{ second}^{-1}$
 - $3 \times 10^5 \text{ second}^{-1}$
 - $3 \times 10^{10} \text{ second}^{-1}$
 - $3 \times 10^8 \text{ second}^{-1}$
- The wavelength of the radiation emitted, when in a hydrogen atom electron falls from infinity to stationary state 1, would be (Rydberg constant = $1.097 \times 10^7 \text{ m}^{-1}$)
 - 91 nm
 - 192 nm
 - 406 nm
 - $9.1 \times 10^{-8} \text{ nm}$
- The ratio of minimum wavelengths of Lyman & Balmer series will be
 - 1.25
 - 0.25
 - 5
 - 10
- If the shortest wavelength of H atom in Lyman series is x , then longest wavelength in Balmer series of He^+ is
 - $\frac{9x}{5}$
 - $\frac{36x}{5}$
 - $\frac{x}{4}$
 - $\frac{5x}{9}$

14. Find out ratio of following for photon
 $(v_{\max})_{\text{Lyman}} : (v_{\max})_{\text{Brakett}}$
 (1) 1 : 16 (2) 16 : 1
 (3) 4 : 1 (4) 1 : 4
15. What transition in He^+ will have the same λ as the 1st line in Lyman series of H-atom :
 (1) $5 \rightarrow 3$ (2) $3 \rightarrow 2$
 (3) $6 \rightarrow 4$ (4) $4 \rightarrow 2$
16. The difference in the wavelength of the 1st line of Lyman series and 2nd line of Balmer series in a hydrogen atom is
 (1) $\frac{9}{2R}$ (2) $\frac{4}{R}$
 (3) $\frac{88}{15R}$ (4) None
17. If H-atom is supplied with 12.1 eV energy and electron returns to the ground state after excitation the number of spectral lines in Balmer series would be: (use energy of ground state of H-atom = -13.6 eV)
 (1) 1 (2) 2 (3) 3 (4) 4
18. Which transition emits photon of maximum frequency:
 (1) second spectral line of Balmer series
 (2) second spectral line of Paschen series
 (3) fifth spectral line of Humphery series
 (4) first spectral line of Lyman series
19. Which electronic level would allow the hydrogen atom to absorb a photon but not to emit a photon?
 (1) 3s (2) 3p
 (3) 2s (4) 1s
20. Which is the de-Broglie equation?
 (1) $h = p\lambda$ (2) $h = p\lambda^{-1}$
 (3) $h = \lambda p^{-1}$ (4) $h = p + \lambda$
21. Which of the following has the largest de Broglie wavelength given that all have equal velocity?
 (1) CO_2 molecule (2) NH_3 molecule
 (3) Electron (4) Proton
22. The correct order of wavelength of Hydrogen (${}_1\text{H}^1$), Deuterium (${}_1\text{H}^2$) and Tritium (${}_1\text{H}^3$) moving with same kinetic energy is
 (1) $\lambda_{\text{H}} > \lambda_{\text{D}} > \lambda_{\text{T}}$ (2) $\lambda_{\text{H}} = \lambda_{\text{D}} = \lambda_{\text{T}}$
 (3) $\lambda_{\text{H}} < \lambda_{\text{D}} < \lambda_{\text{T}}$ (4) $\lambda_{\text{H}} < \lambda_{\text{D}} > \lambda_{\text{T}}$
23. A 200 g cricket ball is thrown with a speed of $3.0 \times 10^3 \text{ cm s}^{-1}$. What will be its de Broglie's wavelength - [$h = 6.6 \times 10^{-27} \text{ g cm}^2 \text{ s}^{-1}$]
 (1) $1.1 \times 10^{-32} \text{ cm}$ (2) $2.2 \times 10^{-32} \text{ cm}$
 (3) $0.55 \times 10^{-32} \text{ cm}$ (4) $11.0 \times 10^{-32} \text{ cm}$
24. The wavelength associated with a golf ball weighing 200 g and moving at a speed of 5m/h is of the order:
 (1) 10^{-10} m (2) 10^{-20} m
 (3) 10^{-30} m (4) 10^{-40} m
25. An electron has kinetic energy $2.8 \times 10^{-23} \text{ J}$. De-Broglie wavelength will be nearly ($m_e = 9.1 \times 10^{-31} \text{ kg}$)
 (1) $9.28 \times 10^{-24} \text{ m}$ (2) $9.28 \times 10^{-7} \text{ m}$
 (3) $9.28 \times 10^{-8} \text{ m}$ (4) $9.28 \times 10^{-10} \text{ m}$
26. No. of wave in fourth orbit :-
 (1) 4 (2) 5 (3) 0 (4) 1
27. What is the de-Broglie wavelength associated with the hydrogen electron in its third orbit?
 (1) $9.96 \times 10^{-10} \text{ cm}$ (2) $9.96 \times 10^{-8} \text{ cm}$
 (3) $9.96 \times 10^4 \text{ cm}$ (4) $9.96 \times 10^8 \text{ cm}$

28. If the de-Broglie wavelength of the fourth Bohr orbit of hydrogen atom is $4\text{\AA}$, the circumference of the orbit will be:
 (1) $4\text{\AA}$ (2) 4 nm
 (3) $16\text{\AA}$ (4) 16 nm
29. If the radius of first Bohr orbit of hydrogen atom is 'x' then de Broglie wavelength of electron in 3rd orbit is nearly –
 (1) $2\pi x$ (2) $6\pi x$
 (3) $9x$ (4) $\frac{x}{3}$
30. A particle X moving with a certain velocity has a de-Broglie wavelength of $1\text{\AA}$. If particle Y has a mass of 25% that of X and velocity 75% that of X, de-Broglie wavelength of Y will be –
 (1) $3\text{\AA}$ (2) $5.33\text{\AA}$
 (3) $6.88\text{\AA}$ (4) $48\text{\AA}$
31. Which of the following is the most correct expression for Heisenberg's uncertainty principle
 (1) $x.p = \frac{h}{4\pi}$ (2) $x.p \geq \frac{h}{4\pi}$
 (3) $x.p \leq \frac{h}{4\pi}$ (4) $x.v = \frac{h}{4\pi}$
32. For the electron if the uncertainty in velocity is Δv , the uncertainty in its position (Δx) is given below:
 (1) $\frac{h}{2\pi m v}$ (2) $\frac{2\pi}{h m v}$
 (3) $\frac{h}{4\pi m v}$ (4) $\frac{2\pi m}{h v}$
33. Heisenberg Uncertainty principle is not valid for
 (1) Moving electron
 (2) Motor car
 (3) Stationary particles
 (4) 2 & 3 both
34. The uncertainty in momentum of moving particle is $1.0 \times 10^{-15} \text{ kg m s}^{-1}$ the minimum uncertainty in its position would be
 (1) $5.28 \times 10^{-20} \text{ m}$ (2) $5.28 \times 10^{-49} \text{ m}$
 (3) $6.63 \times 10^{-49} \text{ m}$ (4) $6.63 \times 10^{-22} \text{ m}$
35. What electronic transition in Li^{2+} produces the radiation of the same wavelength as the first line in the Lyman series of hydrogen?
 (1) $n = 4$ to $n = 2$ (2) $n = 9$ to $n = 6$
 (3) $n = 9$ to $n = 3$ (4) $n = 6$ to $n = 3$
36. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2 \text{ eV}$. The same energy change is observed in the second Balmer transition of:
 (1) Li^{2+} (2) Li^+ (3) He^+ (4) Be^{3+}

Quantum Mechanical Model of Atom

- In 1926 Erwin Schrodinger was developed this model.
- This atomic model is based on wave and particle nature of electron.
- This model describes that in electronic field of (+ve)ly charged nucleus the electron has 3D wave.
- Wave motion of an electron was explained by the equation given by Schrodinger. The differential equation is:

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \psi = 0$$

where x, y and z are cartesian co-ordinates of the electron

m = mass of the electron ; E = total energy of electron

V = potential energy of electron ; h = Planck's constant

Ψ = wave function of the electron

Significance of Ψ :

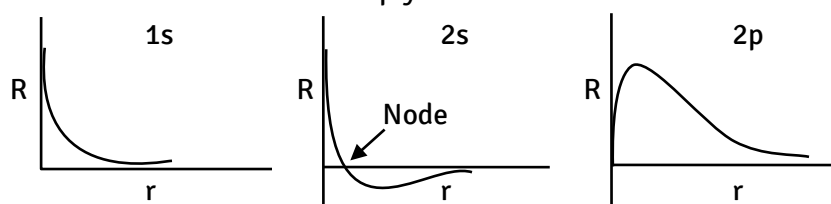
Coordination's x, y and z expresses amplitude function which is also termed as wave function. Depending upon the coordinate's values, wave function can have positive or negative values.

Significance of Ψ^2 :

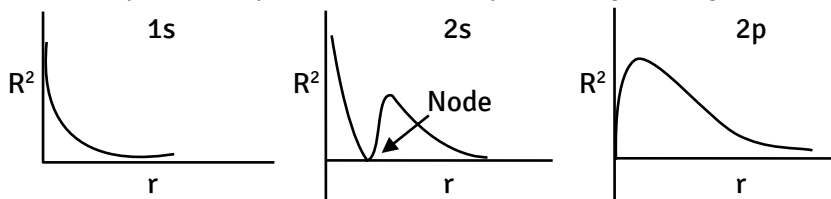
Ψ^2 is a probability factor. It describes the probability for finding an electron within a small space. The space where there is a maximum probability for finding an electron is termed as orbital.

Variations of Radial Wave Function (R)**(i) Plots of radial wave function R against the distance r:**

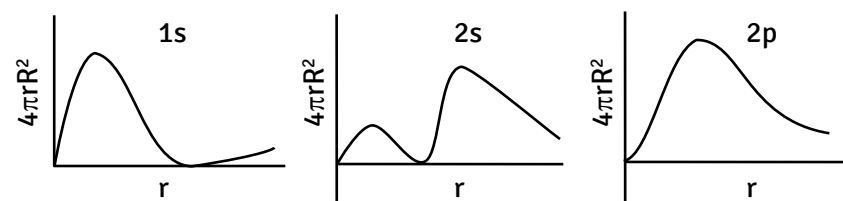
The variation of the radial part of the orbital wave functions for 1s, 2s and 2p orbitals. The radial function value changes from positive to zero then to negative. The region where this function reduces to zero is called nodal surfaces or simply nodes.

**(ii) Radial probability density (R^2):**

Square of radial wave function, R^2 , is the measure of the probability of finding the electron in a unit volume around a particular point and is called probability density.

**(iii) Radial distribution function, ($4\pi r^2 R^2$):**

Probability density of finding electron at a point at a distance r from the nucleus. It is more useful for discussing the probability for finding the electron in a spherical shell between the spheres of radii $r + dr$ and r as the atoms have spherical symmetry.

**Quantum Number**

As we know to search a particular person in this world 4 things are needed: -

- Country to which the person belongs.
- The city in that country where the person is residing
- The area in that city
- House number

Similarly, to locate the position of an electron in the atom 4 identification number are required and these identification number are called as quantum number.

1. Principal quantum number (n) → Shell (Orbit)
2. Azimuthal quantum number (l) → Sub shell
3. Magnetic quantum number (m) → Orbital
4. Spin quantum number (s) → Spin of e^-

1. Principal quantum number: Given By → Bohr's

It represents the name, size and energy of the shell to which e^- belongs

- The value of n lies between 1 to ∞
- i.e $n = 1, 2, 3, 4, \dots, \infty$ corresponding name of shells are K, L, M, N, O, -----
- Greater the value of n , greater is the distance from the nucleus.

$$r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$$

$$r_1 < r_2 < r_3 < r_4 < r_5 \text{ -----}$$

- Greater the value of n , greater is the energy of shell

$$E = -13.6 \times \frac{Z^2}{n^2} \text{ eV / atom}$$

$$E_1 < E_2 < E_3 < E_4 \text{ -----}$$

- The angular momentum of a revolving electron is $mvr = \frac{nh}{2\pi}$

Where n = Principal quantum number.

- The number of electrons in a particular shell is equal to $2n^2$

2. Azimuthal quantum number / Angular quantum number / Secondary quantum number / Subsidiary quantum number:

Represented by ' ℓ ' (Given by - Sommerfeld)

It represents the shape of the subshell and orbital and orbital angular momentum Value of ℓ between 0 to $(n - 1)$

i.e. $\ell = 0, 1, 2, \dots, (n-1)$

$\ell = 0$ (s Subshell)

$\ell = 1$ (p Subshell)

$\ell = 2$ (d Subshell)

$\ell = 3$ (f Subshell)

Ex. If $n = 1$ then $\ell = 0 \Rightarrow 1$ s i.e. in $n = 1$ shell, only one subshell 's' is present.

If $n = 2$ then $\ell = 0, 1 \Rightarrow 2s, 2p$ i.e. in $n = 2$ shell, two subshell 's' & 'p' are present.

If $n = 3$ then $\ell = 0, 1, 2 \Rightarrow 3s, 3p, 3d$ i.e. in $n = 3$ shell, three subshell 's', 'p' & 'd' are present.

If $n = 4$ then $\ell = 0, 1, 2, 3 \Rightarrow 4s, 4p, 4d, 4f$ i.e. in $n = 4$ shell, four subshell 's', 'p', 'd' & 'f' are present.

- If the value of n is same then the order of energy of the various subshell will be $s < p < d < f$

Ex. $4s < 4p < 4d < 4f, 3s < 3p < 3d, 2s < 2p$

- If Value of ℓ is same but value of n is different then the order of energy will be.

Ex. $1s < 2s < 3s < 4s < 5s < 6s$

$3d < 4d < 5d < 6d$

$4p < 5p < 6p$

- The orbital angular momentum = $\sqrt{l(l+1)} \cdot \frac{h}{2\pi}$ or $\sqrt{l(l+1)} \cdot \hbar$ $\left\{ \hbar = \frac{h}{2\pi} \right\}$ {h is called as 'hash'}

Orbital angular momentum: For s subshell = 0

- For p Subshell = $\sqrt{2} \frac{h}{2\pi}$ or $\sqrt{2}\hbar$
- The number of electrons in a particular subshell is equal to $2(2\ell+1)$
- for s subshell number of electrons = $2e^-$
- for p subshell number of electrons = $6e^-$
- for d subshell number of electrons = $10e^-$
- for f subshell number of electrons = $14e^-$
- Shape of the subshell:

s $\rightarrow$ spherical

p $\rightarrow$ dumb bell shape

d $\rightarrow$ double dumb bell shape

f $\rightarrow$ complex shape

3. Magnetic quantum number / Orientation quantum number (m): given by $\rightarrow$ lind
It represents the shape of different orbitals and the orientation of electron cloud (orbital)

- Under the influence of magnetic field each subshell is further subdivided into orbitals (The electron cloud is known as orbital)

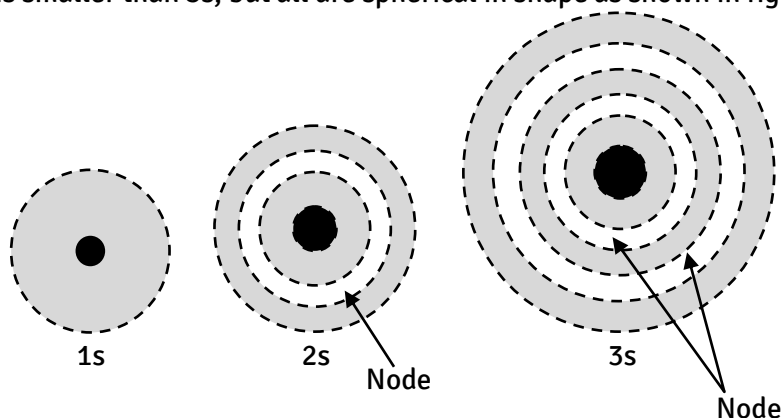
Magnetic quantum number describe this different distribution of electron cloud.

- Value of m = all integral value from $-\ell$ to $+\ell$ including zero.
- i.e. Value of $m = -\ell \rightarrow 0 \rightarrow +\ell$

Case-I: If $\ell = 0$ then $m = 0$ it implies that s subshell has only one orbital called as s orbital.

Shapes of s-orbitals:

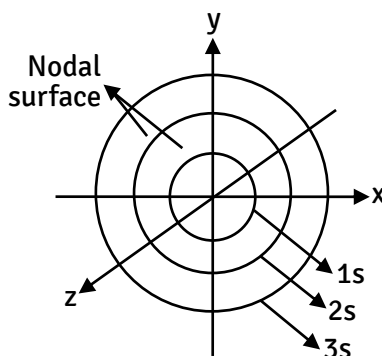
The s-orbitals are spherically symmetrical about the nucleus, i.e., the probability of finding s electron is same in all directions from the nucleus. The size of the orbital depends on the value of principal quantum number, there is one spherically symmetrical orbital. The 1s orbital is smaller than 2s-orbital and 2s-orbital is smaller than 3s, but all are spherical in shape as shown in figure.



Although the s-orbitals belonging to different shells are spherically symmetrical, yet they differ in certain respects as explained below:

- The probability of 1s electron is found to be maximum near the nucleus and decreases as the distance from the nucleus increases. In case of 2s electrons, the probability is again maximum near the nucleus and then decreases to zero and increases again and then decreases as the distance from the nucleus increases. The intermediate region (a spherical shell) where the probability is zero is called a nodal surface of simply node. Thus, 2s-orbital differs from 1s-orbital in having one node within it. Similarly, 3s has two nodes. In general, any ns orbital has (n-1) nodes.
- The size and energy of the s-orbital increases as the principal quantum number increases, i.e., the size and energy of s-orbital increases in the order $1s < 2s < 3s$

The s orbital of higher energy levels are also symmetrically spherical and can be represented as follows:



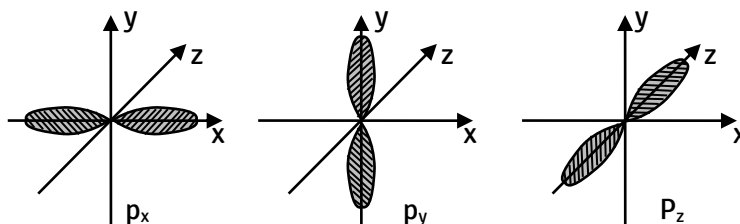
Case-II: If $\ell = 1$ (p - subshell)

$$\text{then } m = \begin{array}{c|c|c} -1 & 0 & +1 \\ \hline p_x & p_z & p_y \end{array}$$

It implies that, p subshell have three orbitals called as p_x , p_y and p_z .

Shape of p-orbitals:

There are three p-orbitals, commonly referred to as p_x , p_y and p_z . These three p-orbitals, possess equivalent energy and therefore, have same relation with the nucleus. They, however, differ in their direction & distribution of the charge.

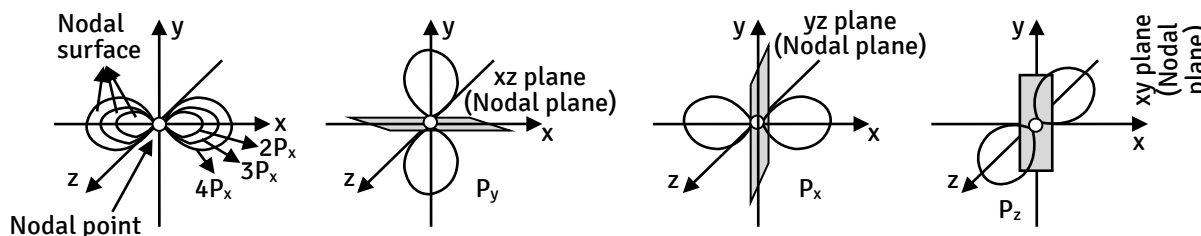


These three p-orbitals are situated at right angle to another and are directed along x, y and z axes (figure)

- Each p orbital has dumb bell shape (2 lobes which are separated from each other by a point of zero probability called nodal point or node or nucleus).
- The two lobes of each orbital are separated by a plane of zero electron density called nodal plane.
- Each p orbital of higher energy level are also dumb bell shape but they have nodal surface.

Nodal surface:	Orbital	Nodal surface
	2 p_x	0
	3 p_x	1
	4 p_x	2
	Np_x	(n-2)

Nodal Plane:	Orbital	Nodal plane	
	p_x	yz	plane
	p_y	xz	plane
	p_z	xy	plane

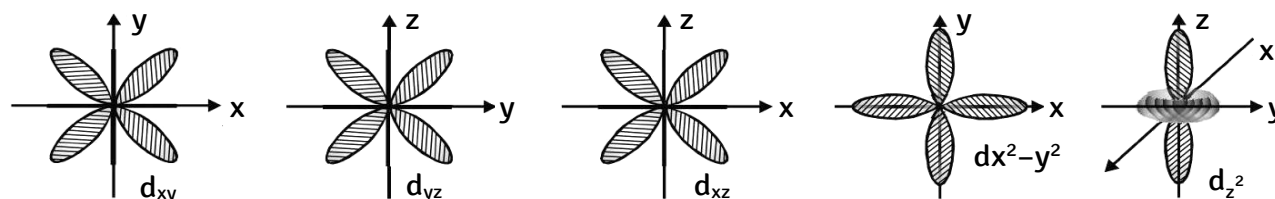
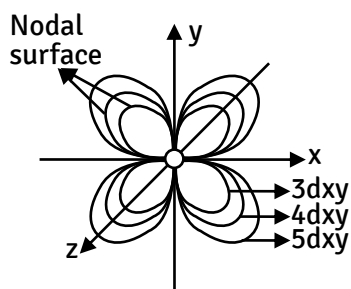
**Case III:**

When $\ell = 2$, 'm' has five values $-2, -1, 0, +1, +2$. It implies that d subshell of any energy shell has five orbitals. All the five orbitals are not identical in shape. Four of the d orbitals $d_{xy}, d_{yz}, d_{xz}, d_{x^2-y^2}$ contain four lobes while fifth orbital d_{z^2} consists of only two lobes. The lobes d_{xy} orbital lie between x and y axes. Similar is the case for d_{yz} and d_{xz} . Four lobes of $d_{x^2-y^2}$ orbital are lying along x and y axes while the two lobes of d_{z^2} orbital are lying along z axes and contain a ring of negative charge surrounding the nucleus in xy plane. Geometry of d orbital is Double Dumb bell

$$m = \begin{vmatrix} -2 & -1 & 0 & +1 & +2 \\ d_{xy} & d_{yz} & d_{z^2} & d_{xz} & d_{x^2-y^2} \end{vmatrix}$$

Shape of d-orbitals:

- It implies that d subshell has 5 orbitals i.e. five electron cloud and can be represented as follows.

**In d orbital:**

$$\text{Number of nodal surface} = n - \ell - 1$$

$$\text{number of nodal plane} = \ell$$

$$\text{total node} = n - \ell - 1 + \ell = (n - 1)$$

- Nodal plane:

$d_{xy} \rightarrow xz$ & yz nodal plane ; $d_{xz} \rightarrow xy$ & yz nodal plane

$d_{yz} \rightarrow dzx$ & yx nodal plane ; $d_{x^2-y^2} \rightarrow 2$, nodal plane ; $d_{z^2} \rightarrow 0$, nodal plane

Note: Orbitals of d subshell are Equivalent in energy.

f subshell: When $\ell = 3$ (f subshell)

Then

$m = -3$	-2	-1	0	$+1$	$+2$	$+3$
f_{x^3}	f_{y^3}	f_{xyz}	f_{z^3}	$f_{x(y^2-z^2)}$	$f_{y(z^2-x^2)}$	$f_{z(x^2-y^2)}$

The structure of f-orbital is very complex

- The number of values of f-orbitals in f subshell show that f subshell has 7 orbitals which are equivalent in Energy.

s subshell $\rightarrow$ $\boxed{s}$

p subshell $\rightarrow$ $\boxed{p_x} \boxed{p_y} \boxed{p_z}$

d subshell $\rightarrow$ $\boxed{d_{xy}} \boxed{d_{yz}} \boxed{d_{z^2}} \boxed{d_{xz}} \boxed{d_{x^2-y^2}}$

f subshell $\rightarrow$ $\boxed{f_{x^3}} \boxed{f_{y^3}} \boxed{f_{z^3}} \boxed{f_{xyz}} \boxed{f_{y(z^2-x^2)}} \boxed{f_{z(x^2-y^2)}}$

4. Spin Quantum number (s): (Given by $\rightarrow$ Gold schmidt)

It represents the direction of electron Spin around its own axis

- for clock wise spin/spin up ($\uparrow$) electron $\rightarrow \pm 1/2$
- for anticlock wise spin/spin down ($\downarrow$) electron $\rightarrow \mp 1/2$

Spin angular momentum of an $e^- = \sqrt{s(s+1)} \cdot \frac{h}{2\pi}$ or $\sqrt{s(s+1)} \cdot \hbar$

- Each orbital can accommodate 2 electrons with opposite spin or spin paired.

Correct $\boxed{\uparrow\downarrow}$ Spin paired e^- Wrong $\boxed{\uparrow\uparrow}$ Spin parallel e^-

Electronic Configuration

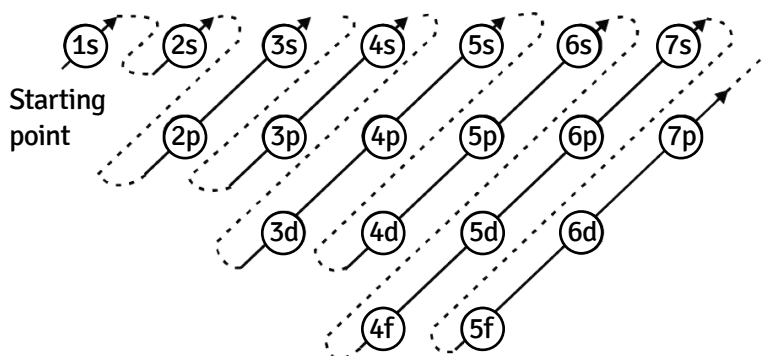
Filling of electron in different energy subshell is electronic configuration.

Rules for filling Subshell:

1. Aufbau Principle
2. $(n + \ell)$ rule
3. Hund's maximum multiplicity principle
4. Pauli's exclusion principle

1. Aufbau Principle: Aufbau is a German word and its meaning 'Building up'

- Aufbau principle gives a sequence in which various subshell are filled up depending on the relative order of the Energies of various subshell.
- Principle:** The subshell with minimum energy is filled up first when this subshell obtained maximum quota of electrons then the next subshell of higher energy starts filling.
- The sequence in which various subshell are filled are as follows.



$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^6, 6s^2, 4f^{14}, 5d^{10}, 6p^6, 7s^2, 5f^{14}, 6d^{10}, \dots$

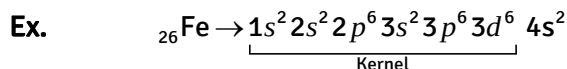
For Example:

${}_1\text{H}$	→	$1s^1$
${}_2\text{He}$	→	$1s^2$
${}_3\text{Li}$	→	$1s^2, 2s^1$
${}_4\text{Be}$	→	$1s^2, 2s^2$
${}_5\text{B}$	→	$1s^2, 2s^2, 2p^1$
${}_6\text{C}$	→	$1s^2, 2s^2, 2p^2$
${}_7\text{N}$	→	$1s^2, 2s^2, 2p^3$
${}_8\text{O}$	→	$1s^2, 2s^2, 2p^4$
${}_9\text{F}$	→	$1s^2, 2s^2, 2p^5$
${}_{10}\text{Ne}$	→	$1s^2, 2s^2, 2p^6$
${}_{11}\text{Na}$	→	$1s^2, 2s^2, 2p^6, 3s^1$
${}_{12}\text{Mg}$	→	$1s^2, 2s^2, 2p^6, 3s^2$
${}_{13}\text{Al}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^1$
${}_{14}\text{Si}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^2$
${}_{15}\text{P}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^3$
${}_{16}\text{S}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^4$
${}_{17}\text{Cl}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^5$
${}_{18}\text{Ar}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6$
${}_{19}\text{K}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$
${}_{20}\text{Ca}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2$
${}_{21}\text{Sc}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^1$
${}_{22}\text{Ti}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^2$
${}_{23}\text{V}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^3$
${}_{24}\text{Cr}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^5$ [Exception]
${}_{25}\text{Mn}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^5$
${}_{26}\text{Fe}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$
${}_{27}\text{Co}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^7$
${}_{28}\text{Ni}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^8$
${}_{29}\text{Cu}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$ [Exception]
${}_{30}\text{Zn}$	→	$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}$

Electronic configuration can be written by following different methods:

- ${}_{26}\text{Fe} \rightarrow$
 - (1) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^6$
 - (2) $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^6, 4s^2$
 - (3) $1s^2, 2s^2p^6, 3s^2p^6d^6, 4s^2$
- ${}_{26}\text{Fe} \rightarrow$

$$1s^2 \underbrace{2s^2 2p^6}_{(n-2)} \underbrace{3s^2 3p^6 3d^6}_{(n-1)} \underbrace{4s^2}_n$$
- $n \rightarrow$ Outer most Shell or Ultimate Shell or Valence Shell
In this Shell e^- are Called as Valance electron or this is called core charge
- $(n-1) \rightarrow$ Penultimate Shell or core or pre valence Shell
- $(n-2) \rightarrow$ Pre Penultimate Shell
- If we remove the last n Shell (ultimate Shell) then the remaining shell collectively be called as Kernal.



2. **$(n+l)$ rule:** According to it the sequence in which various subshell are filled up can also be determined with the help of $(n+l)$ value for a given subshell.

Principle of $(n+l)$ rule: The subshell with lowest $(n+l)$ value is filled up first, when two or more subshell have same $(n+l)$ value then the subshell with lowest value of n is filled up first.

In case of H-atom: Energy only depends on principle quantum number

$$1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f < \dots\dots\dots$$

Sub Shell	n	l	n+l	
1s	1	0	1	
2s	2	0	2	
2p	2	1	3	(1)
3s	3	0	3	(2)
3p	3	1	4	(1)
4s	4	0	4	(2)
3d	3	2	5	(1)
4p	4	1	5	(2)
5s	5	0	5	(3)
4d	4	2	6	(1)
5p	5	1	6	(2)
6s	6	0	6	(3)

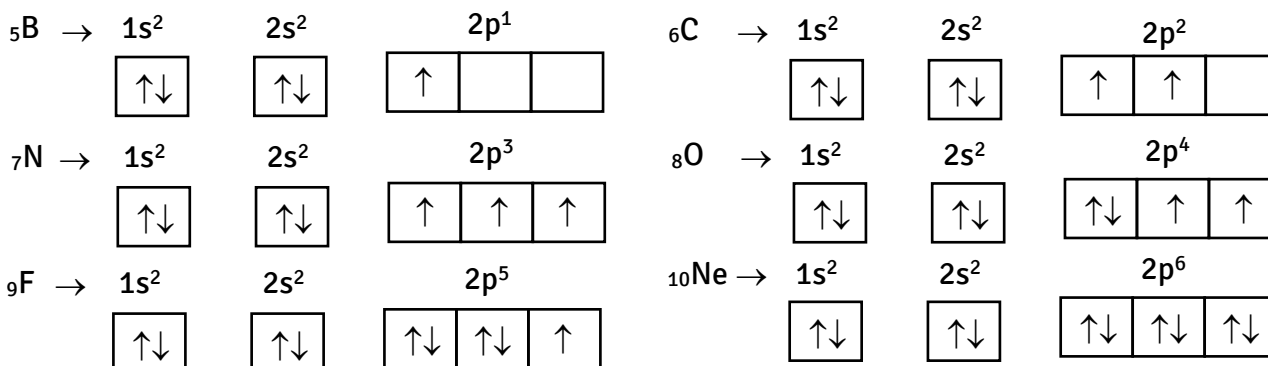
Order: $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6, 5s^2, 4d^{10}, 5p^6, 6s^2, 4f^{14}, 5d^{10}, 6p^6, 7s^2, 5f^{14}, 6d^{10}, \dots$

3. **Hund's Maximum Multiplicity Rule:** (Multiplicity: Many of the same kind)

- According to Hund's rule electrons are distributed among the Orbitals of subshell in such a way as to give maximum number of unpaired electrons with parallel spin.

- Thus, the Orbital available in the subshell are first filled singly with parallel spin electron before they begin to pair this means that pairing of electrons occurs with the introduction of second electron in 's' subshell, fourth electron in 'p' subshell, 6th electron in 'd' Subshell & 8th e⁻ in 'f' subshell.

Ex.:

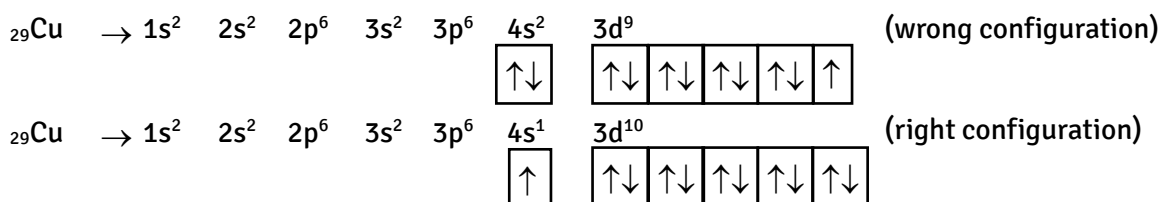


4. **Pauli's Exclusion principle:** In 1925 Pauli stated that no two electrons in an atom can have same values of all four quantum numbers. i.e. An orbital can accommodate maximum 2 electrons with opposite spin.

Ex.:

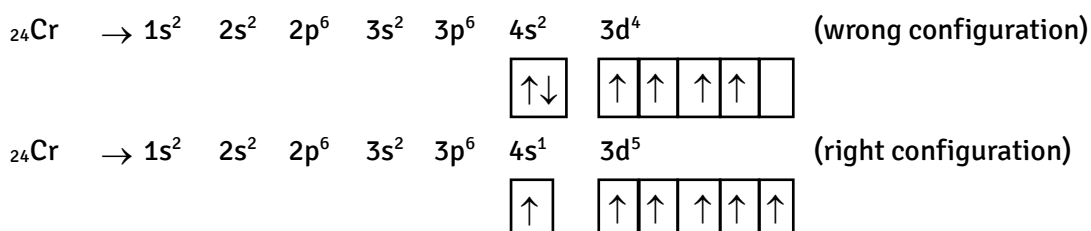
${}_6\text{C}^{12}$	$\rightarrow$	$1s^2$	$2s^2$	$2p^2$
		$\begin{array}{ c } \hline \uparrow\downarrow \\ \hline \end{array}$	$\begin{array}{ c } \hline \uparrow\downarrow \\ \hline \end{array}$	$\begin{array}{ c c c } \hline \uparrow & \uparrow & \\ \hline \end{array}$
				$p_x \ p_z \ p_y$
n		1	2	2
ℓ		0	0	1
m		0	0	+1, -1, 0
s		$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, -\frac{1}{2}$	$+\frac{1}{2}, +\frac{1}{2}$

Ex.:

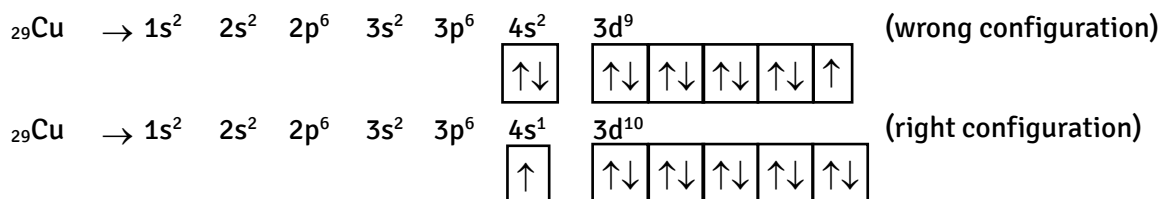


Exception of Aufbau principle: In some cases, it is seen that the electronic configuration is slightly different from the arrangement given by Aufbau principle. A simple reason behind this is that half-filled & full filled subshell have got extra stability.

Ex.:



Ex.:



Ex.27: Write down the values of azimuthal quantum number possible for electron present in 3rd shell.

Sol: 3rd shell means $n = 3$

Azimuthal quantum number (l) has values 0 to $(n - 1)$. So, the possible values of azimuthal quantum number are 0, 1, 2.

Ex.28: Sub-shell that comes after f is called g. No. of orbitals in g-sub-shell.

Sol: $l = 4$ No. of Orbitals = 9

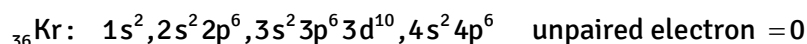
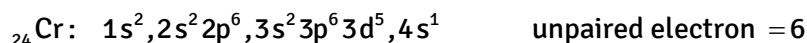
Ex.29: Which of the following sets of quantum number are incorrect?

- (A) $n = 3, l = 2, m = +1, s = +1/2$ (B) $n = 3, l = 3, m = +3, s = +1/2$
 (C) $n = 4, l = 0, m = 0, s = -1/2$ (D) $n = 5, l = 4, m = +4, s = -1/2$

Sol: (B)

Ex.30: Calculate total spin, magnetic moment for the atoms having at. no. 7, 24 and 36.

Sol: The electronic configuration is:



Total spin for an atom = $\pm 1/2 \times \text{no. of unpaired electron}$

For ${}_7\text{N}$, it is $= \pm 3/2$; For ${}_{24}\text{Cr}$, it is $= \pm 3$; For ${}_{36}\text{Kr}$, it is $= 0$

Also magnetic moment = $\sqrt{n(n+2)}$

For ${}_7\text{N}$, it is $= \sqrt{15}$; For ${}_{24}\text{Cr}$, it is $= \sqrt{48}$; For ${}_{36}\text{Kr}$, it is $= \sqrt{0}$

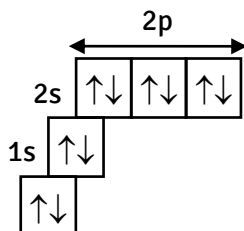
Exercise 1.4

- The azimuthal quantum number describes what?
 (1) orbital shape (2) electron spin
 (3) energy level (4) electron velocity
- Which quantum number is sufficient to determine the energy of the electron in hydrogen atom?
 (1) l (2) n (3) m_s (4) m_l
- Which of the following statements concerning the four quantum numbers is false:
 (1) n gives idea of the size of an orbital
 (2) l gives the shape of an orbital
 (3) m gives the energy of the electron in the orbital
 (4) s gives the direction of spin of the electron in an orbital
- The maximum number of electrons in a principal shell is –
 (1) $2n$ (2) $2n^2$
 (3) 2 (4) $\sqrt{2}$
- How many electrons can fit into the orbitals that comprise the 3rd shell ($n=3$)?
 (1) 2 (2) 8 (3) 18 (4) 32
- The number of orbitals in $n = 3$ are –
 (1) 1 (2) 3
 (3) 5 (4) 9
- For azimuthal quantum number $l = 3$, the maximum number of electrons will be :
 (1) 2 (2) 6 (3) zero (4) 14

8. For principal quantum number $n = 4$, the total number of orbitals having $\ell = 3$ is
 (1) 3 (2) 7 (3) 5 (4) 9
9. The total number of electrons that can be accommodated in all the orbitals having principal quantum number 2 and azimuthal quantum number 1 is
 (1) 2 (2) 4 (3) 6 (4) 8
10. An electron has a spin quantum number $+1/2$ and a magnetic quantum number -1 . It cannot be present in –
 (1) d – Orbital (2) f – Orbital
 (3) s – Orbital (4) p – Orbital
11. A sub-shell with $n = 6$, $\ell = 2$ can accommodate a maximum of:
 (1) 12 electrons (2) 36 electrons
 (3) 10 electrons (4) 72 electrons
12. Which of the following sets of quantum numbers is correct for an electron in 4f-orbital?
 (1) $n = 4, \ell = 3, m = 4, s = +\frac{1}{2}$
 (2) $n = 4, \ell = 4, m = -4, s = -\frac{1}{2}$
 (3) $n = 4, \ell = 3, m = +1, s = +\frac{1}{2}$
 (4) $n = 3, \ell = 2, m = -2, s = +\frac{1}{2}$
13. Which orbital represents the following set of quantum numbers $n = 3, \ell = 0, m = 0, s = +1/2$
 (1) 3p (2) 2s (3) 3s (4) 2p
14. The orbital angular momentum of an electron in 2s orbital is –
 (1) $+\frac{1}{2} \cdot \frac{h}{2\pi}$ (2) Zero
 (3) $\frac{h}{2\pi}$ (4) $\sqrt{2} \cdot \frac{h}{2\pi}$
15. The maximum probability of finding electron in the d_{xy} orbital is:
 (1) along the x-axis
 (2) along the y-axis
 (3) at an angle of 45° from the x & y axis
 (4) at an angle of 90° from the x & y axis
16. Which d - orbital has different shape from rest of all d- orbitals?
 (1) $d_{x^2-y^2}$ (2) d_{z^2}
 (3) d_{xy} (4) d_{xz}
17. The quantum numbers $+\frac{1}{2}$ and $-\frac{1}{2}$ for the electron spin represent
 (1) rotation of the electron in clockwise and anticlockwise direction respectively
 (2) rotation of the electron in anticlockwise and clockwise direction respectively
 (3) magnetic moment of the electron pointing up and down respectively
 (4) two quantum mechanical spin states which have no classical analogue.
18. Which orbital has 1 nodal plane?
 (1) s (2) p (3) d (4) f
19. Angular and Spherical nodes in 3s are-
 (1) 1, 1 (2) 1, 0
 (3) 2, 0 (4) 0, 2
20. The number of nodal planes in a p_x orbital is:
 (1) 1 (2) 2 (3) 3 (4) 0
21. For $4p_y$ orbital: There are nodal plane = and azimuthal quantum number ℓ
 (1) 1, 0 (2) 0, 1
 (3) 1, 1 (4) 2, 1
22. Four sets of values of quantum numbers (n, ℓ, m and s) are given below. Which set does not provide a permissible solution of the wave equation?
 (1) 3, 2, -2, $\frac{1}{2}$ (2) 3, 3, 1, $-\frac{1}{2}$
 (3) 3, 2, 1, $\frac{1}{2}$ (4) 3, 1, 1, $\frac{1}{2}$
23. The atomic orbitals are progressively filled in order of increasing energy. This principle is called.
 (1) Hund's rule (2) Aufbau principle
 (3) Exclusion principle (4) de- Broglie rule.

24. When 4d orbital is complete, the newly entering electrons goes into
 (1) 5f (2) 5s
 (3) 5p (4) 6d

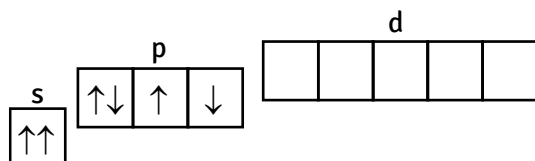
25. Which of the following elements is represented by the electronic configuration:



- (1) Nitrogen (2) Fluorine
 (3) Oxygen (4) Neon

26. The number of unpaired electrons in carbon atom is –
 (1) 2 (2) 4
 (3) 1 (4) 3

27.



The above configuration is not correct as it violates

- (1) Only Hund's rule
 (2) Only Pauli's exclusion principle
 (3) (n + 1) rule
 (4) (Hund + Pauli) rule

28. For the energy levels in an atom, which one of the following statements is correct?
 (1) There are seven principal electron energy levels
 (2) The second principal energy level has four orbitals and contains a maximum of eight electrons
 (3) The M energy level can have a maximum of 32 electrons
 (4) The 4s sub-energy level has higher energy than 3d sub-energy level

29. Which of the following sets of quantum numbers represents the highest energy of an atom?

- (1) $n = 3, \ell = 1, m = 1, s = +\frac{1}{2}$
 (2) $n = 3, \ell = 2, m = 1, s = +\frac{1}{2}$
 (3) $n = 4, \ell = 0, m = 0, s = +\frac{1}{2}$
 (4) $n = 3, \ell = 0, m = 0, s = +\frac{1}{2}$

30. The energy of an electron of $2p_y$ orbital is –
 (1) Greater than $2p_x$ orbital
 (2) Less than $2p_z$ orbital
 (3) Equal to 2s orbital
 (4) Same as that of $2p_x$ and $2p_z$ orbitals

31. The electronic configuration together with the quantum number of last electron for lithium is –

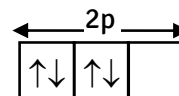
- (1) $1s^2 2s^1$ 2,0,0,+ $\frac{1}{2}$
 (2) $1s^2 2s^1$ 2,0,0,+ $\frac{1}{2}$ or $-\frac{1}{2}$
 (3) $1s^2 2s^0 2p^1$ 2,1,0, $\pm\frac{1}{2}$
 (4) $2s^2 2s^1$ 2,1,0, $\pm\frac{1}{2}$

32. The maximum number of unpaired electrons are in
 (1) Fe^{2+} (2) Fe^{3+} (3) Fe^{4+} (4) Fe

33. The number of unpaired electrons in Zn^{+2} :
 (1) 0 (2) 1 (3) 2 (4) 3

34. The number of d-electrons retained in Fe^{2+} (At. no. of Fe = 26) ion is:
 (1) 6 (2) 3 (3) 4 (4) 5

35. If the electronic structure of oxygen atom is written as $1s^2, 2s^2$



it would violate

- (1) Hund's rule
 (2) Pauli's exclusion principle
 (3) Both Hund's and Pauli's principles
 (4) None of these

36. The 19th electron of chromium has which of the following sets of quantum numbers?
- (1) $n \rightarrow 3, \ell \rightarrow 0, m \rightarrow 0, s \rightarrow \frac{1}{2}$
 (2) $n \rightarrow 3, \ell \rightarrow 2, m \rightarrow -2, s \rightarrow \frac{1}{2}$
 (3) $n \rightarrow 4, \ell \rightarrow 0, m \rightarrow 0, s \rightarrow \frac{1}{2}$
 (4) $n \rightarrow 4, \ell \rightarrow 1, m \rightarrow -1, s \rightarrow \frac{1}{2}$
37. The magnetic quantum number for valence electron of sodium atom is –
 (1) 3 (2) 2 (3) 1 (4) 0
38. The correct set of quantum numbers to the unpaired electron of fluorine atom –
 (1) $n = 3, \ell = 0, m = 0$ (2) $n = 3, \ell = 1, m = 1$
 (3) $n = 2, \ell = 0, m = 0$ (4) $n = 2, \ell = 1, m = 1$
39. Consider the ground state of Cr atom ($Z = 24$). The numbers of electrons with the azimuthal quantum numbers, $\ell = 1$ and 2 are, respectively:
 (1) 12 and 4 (2) 12 and 5
 (3) 16 and 4 (4) 16 and 5
40. $N = 1s^7$ is wrong according to
 (1) Pauli's exclusion principle.
 (2) Hund's rule of maximum multiplicity.
 (3) Heisenberg's uncertainty principle.
 (4) Aufbau principle.
41. If the number of electrons in p-orbitals are two, then which one of the following is in accordance with Hund's rule –
 (1) $p_x^2 p_y^0 p_z^0$ (2) $p_x^0 p_y^2 p_z^0$
 (3) $p_x^0 p_y^0 p_z^2$ (4) $p_x^1 p_y^1 p_z^0$
42. Which of the following ions has the maximum value of magnetic moment?
 (1) Cu^+ (2) Cu^{2+} (3) Fe^{2+} (4) Fe^{3+}
43. An improbable configuration is:
 (1) $[\text{Ar}]3d^4, 4s^2$ (2) $[\text{Ar}]3d^5, 4s^1$
 (3) $[\text{Ar}]3d^6, 4s^2$ (4) $[\text{Ar}]3d^{10}, 4s^1$
44. The electronic configurations of Cr^{2+} and Cu^{2+} are abnormal –
 (1) Due to extra stability of exactly half filled and exactly fully filled sub shells
 (2) Because they belong to d-block
 (3) Both the above
 (4) None of the above
45. In chromium atom, in ground state, the number of occupied orbitals is
 (1) 14 (2) 15 (3) 7 (4) 12
46. Which of the following statements in relation to the hydrogen atom is correct?
 (1) 3s, 3p and 3d orbitals all have the same energy
 (2) 3s and 3p orbitals are of lower energy than 3d orbital
 (3) 3p orbital is lower in energy than 3d orbital
 (4) 3s orbital is lower in energy than 3p orbital
47. Which of the following transition neither shows absorption nor emission of energy in case of Hydrogen atom:
 (1) $3p_x \rightarrow 3s$ (2) $3d_{xy} \rightarrow 3d_{yz}$
 (3) $3s \rightarrow 3d_{xy}$ (4) All the above
48. Arrange the orbitals of H-atom in the increasing order of their energy -
 $3p_x, 2s, 4d_{xy}, 3s, 4p_z, 3p_y, 4s$
 (1) $2s < 3s = 3p_x = 3p_y < 4s = 4p_z = 4d_{xy}$
 (2) $2s < 3s < 3p_x = 3p_y < 4s = 4p_z = 4d_{xy}$
 (3) $2s < 3s < 3p_x = 3p_y < 4s = 4p_z = 4d_{xy}$
 (4) $2s < 3s < 3p_x = 3p_y < 4s < 4p_z < 4d_{xy}$
49. The maximum number of electrons with clockwise spin that can be accommodated in a f-sub-shell is
 (1) 14 (2) 7 (3) 5 (4) 10

50. A compound of vanadium has a magnetic moment of 1.73 BM. The electronic configuration of vanadium ion in the compound is:
 (1) [Ar] 3d² (2) [Ar] 3d¹
 (3) [Ar] 3d³ (4) [Ar] 3d⁰4s¹
51. If Hund's rule is followed, magnetic moment of Fe²⁺, Mn⁺ and Cr all having 24 electrons will be in order –
 (1) Fe²⁺ < Mn⁺ < Cr (2) Fe²⁺ < Cr = Mn⁺
 (3) Fe²⁺ = Mn⁺ < Cr (4) Mn²⁺ = Cr < Fe²⁺
52. Which of the following set of quantum number is impossible for an electron?
 (1) n = 1, l = 0, m_l = 0, m_s = + $\frac{1}{2}$
 (2) n = 9, l = 7, m_l = -6, m_s = - $\frac{1}{2}$
 (3) n = 2, l = 1, m_l = 0, m_s = + $\frac{1}{2}$
 (4) n = 3, l = 2, m_l = -3, m_s = + $\frac{1}{2}$
53. Which of the following statements is correct for an electron having azimuthal quantum number l = 2?
 (1) The electron may be in the lowest energy shell.
 (2) The electron is in a spherical orbital.
 (3) The electron must have spin m_s = + $\frac{1}{2}$
 (4) The electron may have a magnetic quantum number = - 1
54. Ψ^2 (psi) the wave function represents the probability of finding electron, its values depend:
 (1) inside the nucleus
 (2) far from the nucleus
 (3) near the nucleus
 (4) upon the type of orbital
55. Which set of quantum numbers is possible for the last electron of Mg⁺ ion?
 (1) n = 3, l = 2, m = 0, s = +1/2
 (2) n = 2, l = 3, m = 0, s = +1/2
 (3) n = 1, l = 0, m = 0, s = +1/2
 (4) n = 3, l = 0, m = 0, s = +1/2
56. In any subshell, the maximum number of electrons having same value of spin quantum number is:
 (1) $\sqrt{\ell(\ell+1)}$ (2) l + 2
 (3) 2l + 1 (4) 4l + 2
57. The electronic configuration of silver atom in ground state is:
 (1) [Ar]3d¹⁰, 4s¹
 (2) [Xe]4f¹⁴, 5d¹⁰, 6s¹
 (3) [Kr]4d¹⁰, 5s¹
 (4) [Kr]4d⁹, 5s²
58. Which of the following statement is (are) correct?
 (1) The electronic configuration of Cr is [Ar]3d⁵, 4s¹ (Atomic no. of Cr = 24)
 (2) The magnetic quantum number may have a negative value
 (3) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type, (Atomic no. of Ag = 47)
 (4) All of the above
59. A subshell n = 5, l = 3 can accommodate:
 (1) 10 electrons (2) 14 electrons
 (3) 18 electrons (4) None of these
60. The ratio of magnetic moments of Fe (III) and Co (II) is:
 (1) $\sqrt{5} : \sqrt{7}$ (2) $\sqrt{35} : \sqrt{15}$
 (3) 7 : 3 (4) $\sqrt{24} : \sqrt{15}$

Extra Stability of Half Filled and Completely Filled Orbitals

Half-filled and completely filled sub-shells have extra stability due to the following reasons.

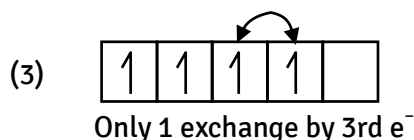
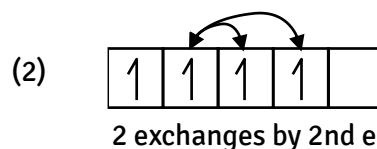
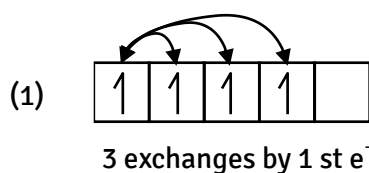
Symmetry of orbitals:

- It is a well-known fact that symmetry leads to stability.
- Thus, if the shift of an electron from one orbital to another orbital differing slightly in energy results in the symmetrical electronic configuration, it becomes more stable
- For example, p^3 , d^5 , f^7 configurations are more stable than their near ones

Exchange Energy:

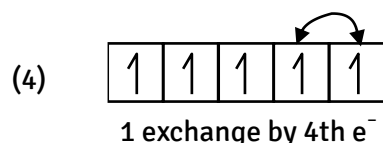
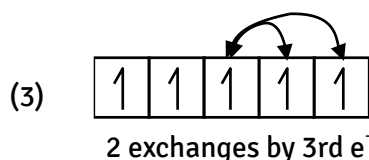
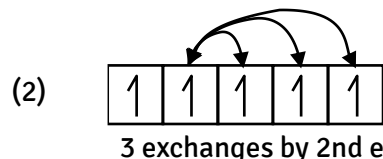
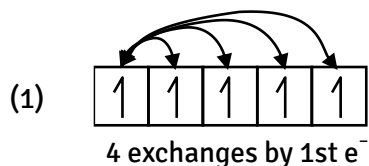
- The e^- in various subshells can exchange their positions, since e^- in the same subshell have equal energies.
- The energy is released during the exchange process within the same subshell.
- In case of half-filled and completely filled orbitals, the exchange energy is maximum and is greater than the loss of orbital energy due to the transfer of electron from a higher to a lower sublevel e.g. from 4s to 3d orbitals in case of Cu and Cr
- The greater the number of possible exchanges between the electrons of parallel spins present in the degenerate orbitals, the higher would be the amount of energy released and more will be the stability
- Let us count the number of exchange that are possible in d^4 and d^5 configuration d^4 among electrons with parallel spins:

d^4



Total number of possible exchanges = $3 + 2 + 1 = 6$

d^5



Total number of possible exchanges = $4 + 3 + 2 + 1 = 10$

Exercise 2

- The ratio of specific charge of an electron to that of a proton is:
 (1) 1 : 1 (2) 1837 : 1
 (3) 1 : 1837 (4) 2 : 1
- Mass of one mole electron is:
 (1) 9.1×10^{-31} g (2) 0.000549 amu
 (3) 9.1×10^{-28} kg. (4) 5.48×10^{-4} g
- An isotone of ${}_{32}\text{Ge}^{76}$ is:
 (i) ${}_{32}\text{Ge}^{77}$ (ii) ${}_{33}\text{As}^{77}$
 (iii) ${}_{34}\text{Se}^{77}$ (iv) ${}_{34}\text{Se}^{78}$
 (1) (ii) & (iii) (2) (i) & (ii)
 (3) (ii) & (iv) (4) (ii) & (iii) & (iv)
- A certain negative ion X^{-2} has in its nucleus 18 neutrons and 18 electrons in its extra nuclear structure. What is the mass number of the most abundant isotope of 'X'.
 (1) 35.46 (2) 32
 (3) 34 (4) 39
- Suppose 10^{-17} J of energy is needed by the interior of human eye to see an object. How many photons of green light ($\lambda = 550$ nm) are needed to generate this minimum amount of energy?
 (1) 14 (2) 28
 (3) 39 (4) 42
- The distance between 3rd and 2nd Bohr orbits of hydrogen atom is:
 (1) 0.529×10^{-8} cm (2) 2.645×10^{-8} cm
 (3) 2.116×10^{-8} cm (4) 1.058×10^{-8} cm
- In second orbit of H atom what is velocity of e^{-} ?
 (1) 2.18×10^6 m/sec
 (2) 3.27×10^6 m/sec
 (3) 10.9×10^5 m/sec
 (4) 21.8×10^6 m/sec
- The ratio of potential energy and total energy of an electron in a Bohr orbit of hydrogen-like species is:
 (1) 2 (2) -2
 (3) 1 (4) -1
- $E_n = -313.6/n^2$. If the value of $E_n = -34.84$ then to which of the following values does 'n' correspond?
 (1) 1 (2) 2
 (3) 3 (4) 4
- What should be the ratio of energies of the electrons of the first orbits of Na^{+10} and H?
 (1) 11 : 1 (2) 121 : 1
 (3) 1 : 121 (4) 1 : 11
- The binding energy of e^{-} in ground state of hydrogen atom is 13.6 eV. The energies required to eject out an electron from three lowest state of He^{+} atom will be (in eV).
 (1) 13.6, 10.2, 3.4 (2) 13.6, 3.4, 1.5
 (3) 13.6, 27.2, 40.8 (4) 54.4, 13.6, 6
- Electron energy in H atom is given by $E_n = (-2.18 \times 10^{-18})/n^2$ J. Cal. Energy required to remove an electron completely from the $n = 2$ orbit.
 (1) 3.4 eV (2) 1.51 eV
 (3) 13.6 eV (4) 10.2 eV
- The ratio of the difference in energy between the first and second Bohr orbit to that between second and third Bohr orbit in H-atom is:
 (1) 4/9 (2) 1/3
 (3) 27/5 (4) 1/2
- Given that in the H-atom the transition energy for $n = 1$ to $n = 2$ is 10.2 eV. The energy for the same transition on Be^{3+} is:
 (1) 20.4 eV (2) 163.2 eV
 (3) 30.6 eV (4) 40.8 eV
- What should be the wavelength and energy respectively of the emitted light when the electron of hydrogen atom undergoes transition from first excited state to ground state?
 (1) 1215 Å and 21.8×10^{-12} erg
 (2) 6560 Å and 16.35×10^{-12} erg
 (3) 1215 Å and 16.35×10^{-12} erg
 (4) 6560 Å and 21.8×10^{-12} erg

16. An hydrogen atom (ionisation energy 13.6 eV) jumps from third excited state to first excited state. The energy of photon emitted in this process is:
 (1) 1.89 eV (2) 2.55 eV
 (3) 12.09 eV (4) 12.75 eV
17. The wavelength of radiation emitted when an electron in a hydrogen atom makes a transition from an energy level with $n = 3$ to a level with $n = 2$ is:
 [Given that $E_n = \frac{-1312}{n^2} \text{ kJmol}^{-1}$]
 (1) $6.56 \times 10^{-7} \text{ m}$ (2) 65.6 nm
 (3) $65.6 \times 10^{-7} \text{ m}$ (4) any of the above
18. The ionization enthalpy of hydrogen atom is $1.312 \times 10^6 \text{ J mol}^{-1}$. The energy required to excite electron in an atom from $n = 1$ to $n = 2$ is -
 (1) $8.51 \times 10^5 \text{ J mol}^{-1}$ (2) $6.56 \times 10^5 \text{ J mol}^{-1}$
 (3) $7.56 \times 10^5 \text{ J mol}^{-1}$ (4) $9.84 \times 10^5 \text{ J mol}^{-1}$
19. If the ionisation potential of an atom is 20 V, its first excitation potential will be -
 (1) 5 V (2) 10 V
 (3) 15 V (4) 20 V
20. The energy required to escape the electron from ground state of H is 13.6 eV then the same for 1st excited state of H atom:
 (1) 3.4 (2) 13.6
 (3) 27.2 (4) can't say anything
21. A photon of energy 12.75 eV is completely absorbed by a hydrogen atom initially in ground state. The principle quantum number of the excited state is
 (1) 1 (2) 3
 (3) 4 (4) ∞
22. If a photon of energy 14 eV. is incident on an H -atom, what is true?
 (1) Atom will be ionized and electron will have a kinetic energy of 14 eV
 (2) Atom will be ionized and electron will have a kinetic energy of 0.4 eV
 (3) Photon passes through atom without interacting with it
 (4) More than one electron will make transitions
23. An electron of energy 10.8 eV is incident on an H -atom then:
 (1) The electron will come out with 10.8 eV energy.
 (2) The electron will be completely absorbed
 (3) 10.2 eV of the electron would be absorbed by H atom and it would come out with 0.6 eV energy.
 (4) None
24. What is maximum wavelength of line of Balmer series of Hydrogen spectrum ($R = 1.09 \times 10^7 \text{ m}^{-1}$):
 (1) 400 nm (2) 654 nm
 (3) 486 nm (4) 384 nm
25. The wave number of the limiting line in Lyman series of hydrogen is 109678 cm^{-1} . The wave number of the limiting line in Balmer series of He^+ would be:
 (1) 54839 cm^{-1} (2) 219356 cm^{-1}
 (3) 109678 cm^{-1} (4) 438712 cm^{-1}
26. Wave length of particular transition for H atom is 400 nm. What can be wavelength of He^+ for same transition:
 (1) 400 nm (2) 100 nm
 (3) 1600 nm (4) 200 nm
27. The wave number ratio for two transitions in the spectrum of atomic hydrogen is 108 : 7. The transition are :
 (1) First Lyman and first Paschen transitions
 (2) First Lyman and first Brackett transitions
 (3) First and second Balmer transitions
 (4) First Lyman and second Balmer transitions
28. The first Lyman transition in the hydrogen spectrum has $\Delta E = 10.2 \text{ eV}$. The same energy change is observed in the second Balmer transition of -
 (1) Li^{+2} (2) Li^+
 (3) He^+ (4) Be^{3+}

29. The wavelength of first line of the Lyman series for hydrogen is 1216 Å. The wavelength for the first line of this series for a 10-time ionised sodium atom ($z = 11$) will be:
 (1) 1000 Å (2) 100 Å
 (3) 10 Å (4) 1 Å
30. The limiting line in Balmer series will have a frequency of:
 (1) $3.65 \times 10^{14} \text{ sec}^{-1}$ (2) $3.29 \times 10^{15} \text{ sec}^{-1}$
 (3) $8.22 \times 10^{14} \text{ sec}^{-1}$ (4) $-8.22 \times 10^{14} \text{ sec}^{-1}$
31. What is the ratio of the de-Broglie wave lengths for electrons accelerated through 200 volts and 50 volts?
 (1) 1 : 2 (2) 2 : 1
 (3) 3 : 10 (4) 10 : 3
32. Uncertainty in the position of an electron (mass = $9.1 \times 10^{-31} \text{ kg}$) moving with a velocity $300 \text{ m} \cdot \text{sec}^{-1}$ accurate upto 0.001%, will be : ($h = 6.63 \times 10^{-34} \text{ J-s}$)
 (1) $19.2 \times 10^{-2} \text{ m}$ (2) $5.76 \times 10^{-2} \text{ m}$
 (3) $1.92 \times 10^{-2} \text{ m}$ (4) $3.84 \times 10^{-2} \text{ m}$
33. The maximum number of electrons in a p-orbital with $n = 6$ and $m = 0$ can be:
 (1) 14 (2) 6
 (3) 2 (4) 10
34. The correct Schrodinger's wave equation for an electron with E as total energy and V as potential energy is:
 (1) $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2}{mh^2} (E - V)\psi = 0$
 (2) $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi m}{h^2} (E - V)\psi = 0$
 (3) $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V)\psi = 0$
 (4) $\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{\partial^2 \psi}{\partial z^2} + \frac{8\pi m^2}{h^2} (E - V)\psi = 0$
35. Which orbital is represented by the complete wave function ψ_{420} ?
 (1) 4d (2) 3d
 (3) 4p (4) 4s
36. n , ℓ and m values of an electron in $3p_y$ orbital are:
 (1) $n = 3$; $\ell = 1$ and $m = 1$
 (2) $n = 3$; $\ell = 1$ and $m = -1$
 (3) Both 1 and 2 are correct
 (4) None of these
37. When the value of principal quantum number n is 3, the permitted value of azimuthal quantum numbers ℓ and magnetic quantum numbers 'm' are:
 (1) ℓ m
 0 0
 1 +1, 0, -1
 2 +2, +1, 0, -1, -2
 (2) ℓ m
 0 1
 2 +2, 1, -2
 3 +3, +2, +1, -2, -3
 (3) ℓ m
 0 0
 1 1, 2, 3, 0, 1, 2
 2 +3, +2, 1, -2, -3
 (4) ℓ m
 1 0, 1
 2 1, 1, 2
 3 0, 1, 2, 3
38. Which statement is not correct for $n = 5$, $m = 2$; $s = +1/2$?
 (1) $\ell = 4$ (2) $\ell = 0, 1, 2, 3$
 (3) $\ell = 3$ (4) $\ell = 2, 3, 4$
39. No. of all subshells of $n + \ell = 7$ is:
 (1) 4 (2) 5
 (3) 6 (4) 7
40. An electron has magnetic quantum number as -3, what is its principal quantum number?
 (1) 1 (2) 2
 (3) 3 (4) 4

41. An electron is in one of 4d orbital. Which of the following orbital quantum number value is not possible?
 (1) $n = 4$ (2) $\ell = 1$
 (3) $m = 1$ (4) $m = 2$
42. For the azimuthal quantum number (ℓ), the total number of magnetic quantum number is given by:
 (1) $\ell = \frac{(m+1)}{2}$ (2) $\ell = \frac{(m-1)}{2}$
 (3) $\ell = \frac{(2m+1)}{2}$ (4) $\ell = \frac{(2m-1)}{2}$
43. Spin angular momentum for electron:
 (1) $\sqrt{s(s+1)} \frac{h}{2\pi}$ (2) $\sqrt{2s(s+1)} \frac{h}{2\pi}$
 (3) $\sqrt{s(s+2)} \frac{h}{2\pi}$ (4) None
44. Maximum electrons in d - orbital are:
 (1) 2 (2) 10
 (3) 6 (4) 14
45. If $n = 3$, then which value of ' ℓ ' is correct?
 (1) 0 (2) 1
 (3) 2 (4) All of them
46. The total value of m for the electrons ($n = 4$) is:
 (1) 4 (2) 8
 (3) 16 (4) 32
47. If $\ell = 3$ then type and number of orbital is:
 (1) 3p, 3 (2) 4f, 14
 (3) 5f, 7 (4) 3d, 5
48. The number of waves made by a Bohr electron in an orbit of maximum magnetic quantum number +2:
 (1) 3 (2) 4
 (3) 2 (4) 1
49. In an atom, for how many electrons, the quantum numbers will be, $n = 3$, $\ell = 2$, $m = +2$, $s = \frac{1}{2}$?
 (1) 18 (2) 6
 (3) 24 (4) 1
50. n and ℓ values of an orbital 'A' are 3 and 2, of another orbital 'B' are 5 and 0. The energy of
 (1) B is more than A
 (2) A is more than B
 (3) A and B are of same energy
 (4) None
51. How many electrons in an atom with $Z = 104$ can have $(n + \ell) = 8$?
 (1) 18 (2) 16
 (3) 3 (4) 6
52. Sum of the paired electrons present in the orbital with $\ell = 2$ in all the species Fe^{2+} , Co^{2+} and Ni^{+2} are:
 (1) 9 (2) 12
 (3) 6 (4) 15
53. When 3d orbital is complete, the next entering electron goes into:
 (1) 4f (2) 4s
 (3) 4p (4) 4d
54. The explanation for the presence of three unpaired electrons in the nitrogen atom can be given by:
 (1) Pauli's exclusion principle
 (2) Hund's rule
 (3) Aufbau's principle
 (4) Uncertainty principle
55. Which represents the correct pattern of electron filling in Valence shell of Cr?
 (1)

$\uparrow\downarrow$	$\uparrow\downarrow$	$\uparrow$			$\uparrow$
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 (2)

$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$		$\uparrow\downarrow$
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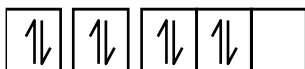
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 (4)

$\uparrow\downarrow$	$\uparrow$	$\uparrow$	$\uparrow$	$\uparrow$	
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56. Electronic configuration



has violated:

- (1) Hund's rule (2) Pauli's principle
(3) Aufbau principle (4) $(n + \ell)$ rule

57. Four quantum numbers of unpaired electron of chlorine are:

- (1) $n = 3, l = 2, m = 0, s = +\frac{1}{2}$
(2) $n = 3, l = 1, m = 0, s = +\frac{1}{2}$
(3) $n = 3, l = 1, m = +1, s = 0$
(4) $n = 3, l = 0, m = -1, s = +\frac{1}{2}$

58. The quantum number of 20th electron of Fe ($Z = 26$) ion would be :

- (1) 3, 2, -2, -1/2 (2) 3, 2, 0, 1/2
(3) 4, 0, 0, +1/2 (4) 4, 1, -1, +1/2

59. ^{36}Kr has the electronic configuration $(18 \text{ Ar}) 4s^2 3d^{10} 4p^6$. The 39th electron will go into which one of the following sub-levels?

- (1) 4 f (2) 4 d
(3) 3 p (4) 5 s

60. The total spin resulting from a d^7 configuration is:

- (1) 1/2 (2) 2 (3) 1 (4) 3/2

61. Any nf -orbital can accommodate upto :

- (1) 14 electrons
(2) Six electrons
(3) Two electrons with parallel spin
(4) Two electrons with opposite spin

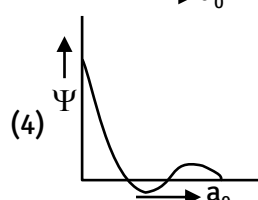
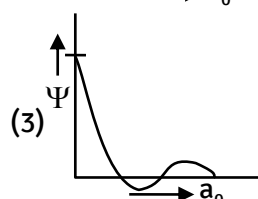
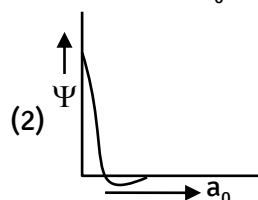
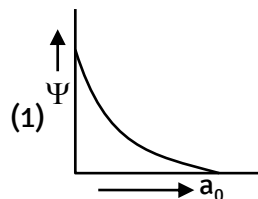
62. In P-atom find out the no. of pair of electrons for $\ell = 1$ and $m = 0$:

- (1) 3 (2) 1 (3) 2 (4) 0

63. The number of electrons in the M-shell of the element with atomic number 24 is:

- (1) 24 (2) 12
(3) 8 (4) 13

64. Which of the following graphs correspond to one node?



65. When on metal sheet fall λ_1 light will eject electron with v_1 velocity and with λ_2 light eject electron of v_2 velocity, what is value of $v_2^2 - v_1^2$?

- (1) $\frac{2hc}{m} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$ (2) $\frac{hc}{m} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$
(3) $\frac{2hc}{m} \left(\frac{1}{\lambda_1} - \frac{1}{\lambda_2} \right)$ (4) $\frac{m}{2hc} \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$

66. The uncertainties in the velocities of two particles A and B are 0.05 and 0.02 ms^{-1} respectively. The mass of B is five times to that of the mass of A. What is the ratio of uncertainties in their positions $\left(\frac{\Delta x_A}{\Delta x_B} \right)$?

- (1) 2 (2) 0.25
(3) 4 (4) 1

67. numbers n and l are:

- (a) $n = 4, l = 1$ (b) $n = 4, l = 0$
(c) $n = 3, l = 2$ (d) $n = 3, l = 1$

Following can be placed in order of increasing energy as:

- (1) (c) < (d) < (b) < (a) (2) (d) < (b) < (c) < (a)
(3) (b) < (d) < (a) < (c) (4) (a) < (c) < (b) < (d)

Exercise 3

1. Match the column :

	Column-I		Column-II
(i)	Cathode rays	(a)	Helium nuclei
(ii)	Dumb-bell	(b)	Electromagnetic radiation
(iii)	Alpha particles	(c)	p-orbital
(iv)	X-rays	(d)	Electrons

- (1) i-(a), ii-(c), iii-(d), iv-(b)
 (2) i-(c), ii-(b), iii-(a), iv-(d)
 (3) i-(d), ii-(c), iii-(a), iv-(b)
 (4) i-(d), ii-(a), iii-(b), iv-(c)

2. Frequency = ν , Time period = T, Energy of n^{th} orbit = E_n , radius of n^{th} orbit = r_n , Atomic number = Z, orbit number = n

	Column-I		Column-II
(i)	$\sqrt{\nu}$	(a)	n^3
(ii)	T	(b)	Z
(iii)	E_n	(c)	$\frac{1}{n^2}$
(iv)	r_n	(d)	$\frac{1}{Z}$

- (1) i-(d), ii-(c), iii-(b), iv-(a)
 (2) i-(b), ii-(a), iii-(c), iv-(d)
 (3) i-(c), ii-(b), iii-(a), iv-(d)
 (4) i-(a), ii-(d), iii-(b), iv-(c)

3. Match the following :

	Column-I		Column-II
(A)	Energy of ground State of He^+	(i)	+6.04 eV
(B)	Energy of ground State of He^+	(ii)	-27.2 eV
(C)	Kinetic energy of II excited State of He^+	(iii)	$8.72 \times 10^{-18} \text{ J}$
(D)	Ionisation potential of He^+	(iv)	-54.4 eV

- (1) A-(i), B-(ii), C-(iii), D-(iv)
 (2) A-(iv), B-(iii), C-(ii), D-(i)
 (3) A-(iv), B-(ii), C-(i), D-(iii)
 (4) A-(ii), B-(iii), C-(i), D-(iv)

4. Match the type of series given in column I with the wavelength range given in column II.

	Column-I		Column-II
(A)	Lyman	(1)	Ultraviolet
(B)	Paschen	(2)	Near infrared
(C)	Balmer	(3)	Far infrared
(D)	Pfund	(4)	Visible

- (1) A-(4), B-(3), C-(1), D-(2)
 (2) A-(4), B-(2), C-(1), D-(3)
 (3) A-(3), B-(1), C-(2), D-(4)
 (4) A-(1), B-(2), C-(4), D-(3)

5. Match the equation in column I with the name/type in column II

	Column-I		Column-II
(A)	$\Delta x, \Delta p \geq \frac{h}{4\pi}$	(1)	de-Broglie equation
(B)	$mvr = \frac{nh}{2\pi}$	(2)	Heisenberg uncertainty rule
(C)	$r_1 = \frac{h^2}{4\pi^2 m K e^2}$	(3)	Bohr's radius of H-atom
(D)	$\lambda = \frac{h}{\sqrt{2m(KE)}}$	(4)	Angular momentum is quantized

- (1) A-(2), B-(4), C-(3), D-(1)
 (2) A-(4), B-(1), C-(2), D-(3)
 (3) A-(3), B-(2), C-(1), D-(4)
 (4) A-(1), B-(4), C-(3), D-(2)

6. Match the Column. Select an appropriate code from the given options.

	Column I		Column II
(I)	$n = 1, l = 0$	(a)	4d
(II)	$n = 2, l = 0$	(b)	5p
(III)	$n = 4, l = 2$	(c)	1s
(IV)	$n = 5, l = 1$	(d)	2s

- (1) I-(b), II-(c), III-(a), IV-(d)
 (2) I-(a), II-(b), III-(c), IV-(d)
 (3) I-(c), II-(d), III-(a), IV-(b)
 (4) I-(c), II-(b), III-(a), IV-(d)

7. **Assertion:** All isotopes of a given element show the same type of chemical behaviour.
Reason: The chemical properties of an atom are controlled by the number of electrons in the atom.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is False.
 (4) If both Assertion & Reason are false.
8. **Assertion:** Thomson's atomic model is known as 'raisin pudding' model.
Reason: The atom is visualized as a pudding of positive charge with electrons (raisins) embedded in it.
 (1) Both A and R are correct; R is the correct explanation of A.
 (2) Both A and R are correct; R is not the correct explanation of A.
 (3) A is correct; R is incorrect.
 (4) A is incorrect; R is correct.
9. **Assertion:** Most of the α -particles striking gold foil in Rutherford's experiment were deflected.
Reason: The size of nucleus is very small as compared to the size of atom.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is False.
 (4) If both Assertion & Reason are false.
10. **Assertion:** Bohr model fails in case of multielectron species.
Reason: It does not mention electron-electron interaction.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is False.
 (4) If both Assertion & Reason are false.
11. **Assertion:** The radius of the first orbit of hydrogen atom is $0.529\text{\AA}$.
Reason: Radius for each circular orbit (r_n) = $0.529\text{\AA} (n^2 / Z)$, where $n=1,2,3$ and Z = atomic number.
 (1) Both A and R are correct; R is the correct explanation of A.
 (2) Both A and R are correct; R is not the correct explanation of A.
 (3) A is correct; R is incorrect.
 (4) A is incorrect; R is correct.
12. **Assertion:** The transition of electrons $n_3 \rightarrow n_2$ in H atom will emit greater energy than $n_4 \rightarrow n_3$.
Reason: n_3 and n_2 are closer to nucleus than n_4 .
 (1) Both A and R are correct; R is the correct explanation of A.
 (2) Both A and R are correct; R is not the correct explanation of A.
 (3) A is correct; R is incorrect.
 (4) A is incorrect; R is correct.
13. **Statement I:** Emitted radiation will fall in visible range when an electron jumps from $n = 4$ to $n = 2$ in H-atom.
Statement II: Balmer series radiations belong to visible range for hydrogen atom
 (1) Both Statement I and Statement II are false
 (2) Statement I is true but Statement II is false
 (3) Statement I is false but Statement II is true
 (4) Both Statement I and Statement II are true

14. **Assertion:** The orbital angular momentum of d-orbitals is $\sqrt{6} \frac{h}{2\pi}$.

Reason: Angular momentum of electron in orbit is $mvr = \frac{nh}{2\pi}$.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

15. **Assertion:** For $n = 3$, ℓ may be 0, 1 and 2 and m may be 0; 0 ± 1 ; 0 ± 1 and ± 2 .

Reason: For each value of n , there are 0 to $(n - 1)$ possible values of ℓ ; and for each value of ℓ , there are 0 to $\pm \ell$ values of m .

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

16. **Assertion:** 2p orbitals do not have any spherical nodes.

Reason: The number of spherical nodes in p-orbitals is given by $(n - 2)$, where n is principle quantum number.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

17. **Assertion:** Energies of the orbitals in hydrogen or hydrogen like species depend only on the quantum number ' n '.

Reason: Energies of the orbitals in multielectron atoms depend on quantum numbers ' n ' and ' ℓ '.

- (1) Both A and R are correct; R is the correct explanation of A.
- (2) Both A and R are correct; R is not the correct explanation of A.
- (3) A is correct; R is incorrect.
- (4) A is incorrect; R is correct.

18. **Assertion:** Half-filled and fully-filled degenerate orbitals are more stable.

Reason: Extra stability is due to the symmetrical distribution of electrons and exchange energy.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

19. **Assertion:** An orbital cannot have more than two electrons and they must have opposite spins.

Reason: No two electrons in an atom can have same set of all the four quantum numbers as per Pauli's exclusion principle.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false.

20. **Assertion:** The 19th electron in potassium atom enters into 4s-orbital and not the 3d-orbital.

Reason: $(n + \ell)$ rule is followed for determining the orbital of the lowest energy state.

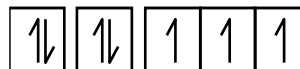
- (1) If both Assertion and Reason are true and the Reason is correct explanation of the Assertion.
- (2) If both Assertion and Reason are true but the Reason is not correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false.
- (4) If Assertion & Reason are false.

21. **Assertion:** Two electrons in an atom can have the same values of four quantum numbers.

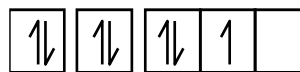
Reason: Two electrons in an atom can be present in the same shell, sub-shell and orbital and have the same spin.

- (1) Both A and R are correct; R is the correct explanation of A.
- (2) Both A and R are correct; R is not the correct explanation of A.
- (3) A is correct; R is incorrect.
- (4) Both A and R is incorrect

22. **Statement I:** The electronic configuration of nitrogen atom is represented as:



not as



Statement II: The configuration of ground state of an atom is the one which has the greatest multiplicity.

- (1) Both Statement I and Statement II are false
 - (2) Statement I is true but Statement II is false
 - (3) Statement I is false but Statement II is true
 - (4) Both Statement I and Statement II are true
23. **Assertion:** Energy of the orbital increases with increase of principal quantum number.

Reason: Energy is required in shifting away the negatively charged electron from the positively charged nucleus.

- (1) Both A and R are correct; R is the correct explanation of A.
- (2) Both A and R are correct; R is not the correct explanation of A.
- (3) A is correct; R is incorrect.
- (4) A is incorrect; R is correct.

Exercise 4 (Previous Year's Questions)

- A 0.66 kg ball is moving with a speed of 100 m/s. The associated wavelength will be ($h = 6.6 \times 10^{-34}$ J s) [AIPMT-2010]

(1) 6.6×10^{-34} m (2) 1.0×10^{-35} m
(3) 1.0×10^{-32} m (4) 6.6×10^{-32} m
- The energies E_1 and E_2 of two radiations are 25 eV and 50 eV respectively. The relation between their wavelengths i.e. λ_1 and λ_2 will be : [AIPMT-PRE-2011]

(1) $\lambda_1 = \lambda_2$ (2) $\lambda_1 = 2\lambda_2$
(3) λ_1 and $4\lambda_2$ (4) λ_1 and $\frac{1}{2}\lambda_2$
- According to the Bohr theory, which of the following transitions in the hydrogen atom will give rise to the least energetic photon? [AIPMT-MAIN-2011]

(1) $n = 5$ to $n = 3$ (2) $n = 6$ to $n = 1$
(3) $n = 5$ to $n = 4$ (4) $n = 6$ to $n = 5$
- The total number of atomic orbitals in fourth energy level of an atom is : [AIPMT-PRE-2011]

(1) 8 (2) 16
(3) 32 (4) 4
- If $n = 6$, the correct sequence for filling of electrons will be : [AIPMT-PRE-2011]

(1) $ns \rightarrow (n-2)f \rightarrow (n-1)d \rightarrow np$
(2) $ns \rightarrow (n-1)d \rightarrow (n-2)f \rightarrow np$
(3) $ns \rightarrow (n-2)f \rightarrow np \rightarrow (n-1)d$
(4) $ns \rightarrow np \rightarrow (n-1)d \rightarrow (n-2)f$
- Maximum number of electrons in a subshell with $\ell = 3$ and $n = 4$ is : [AIPMT-PRE-2012]

(1) 10 (2) 12
(3) 14 (4) 16
- The orbital angular momentum of a p electron is given as: [AIPMT-MAIN-2012]

(1) $\sqrt{\frac{3}{2}} \frac{h}{\pi}$ (2) $\sqrt{6} \cdot \frac{h}{2\pi}$
(3) $\frac{h}{\sqrt{2}\pi}$ (4) $\sqrt{3} \frac{h}{2\pi}$
- The correct set of four quantum numbers for the valence electrons of rubidium atom ($Z = 37$) is? [AIPMT-PRE-2012]

(1) 5, 0, 0, + 1/2 (2) 5, 1, 0, + 1/2
(3) 5, 1, 1, + 1/2 (4) 6, 0, 0 + 1/2
- The value of Planck's constant is 6.63×10^{-34} J s. The speed of light is 3×10^{17} nm s⁻¹. Which value is closest to the wavelength in nanometer of a quantum of light with a frequency of 6×10^{15} s⁻¹? [NEET UG-2013]

(1) 75 (2) 10
(3) 25 (4) 50
- Based on equation $E = -2.178 \times 10^{-18} \text{ J} \left(\frac{Z^2}{n^2} \right)$ certain conclusions are written. Which of them is not correct? [NEET UG-2013]

(1) For $n = 1$, the electron has more negative energy than it does for $n = 6$ which means that the electron is more loosely bound in the smallest allowed orbit.
(2) The negative sign in equation simply means that the energy of electron bound to the nucleus is lower than it would be if the electrons were at the infinite distance from the nucleus
(3) Larger the value of n , the larger is the orbit radius
(4) Equation can be used to calculate the change in energy when the electron changes orbit
- What is the maximum number of electrons that can be associated with the following set of quantum numbers? $n = 3$; $l = 1$ and $m = -1$. [NEET UG-2013]

(1) 2 (2) 10
(3) 6 (4) 4
- Magnetic moment 2.83 BM is given by which of the following ions? (At. nos. Ti = 22, Cr = 24, Mn = 25, Ni = 28): [AIPMT - 2014]

(1) Ti^{3+} (2) Ni^{2+}
(3) Cr^{3+} (4) Mn^{2+}

13. What is the maximum number of orbitals that can be identified with the following quantum numbers? $n = 3$, $\ell = 1$, $m_\ell = 0$

[AIPMT-2014]

- (1) 1 (2) 2
(3) 3 (4) 4

14. Calculate the energy in joule corresponding to light of wavelength 45 nm: (Planck's constant $h = 6.63 \times 10^{-34}$ J s ; Speed of light $c = 3 \times 10^8$ m s⁻¹)

[AIPMT-2014]

- (1) 6.67×10^{15} (2) 6.67×10^{11}
(3) 4.42×10^{-15} (4) 4.42×10^{-18}

15. The angular momentum of electron in 'd' orbital is equal to :-

[AIPMT-2015]

- (1) $\frac{\sqrt{2}h}{2\pi}$ (2) $\frac{2\sqrt{3}h}{2\pi}$
(3) $\frac{\sqrt{3}h}{2\pi}$ (4) $\frac{\sqrt{6}h}{2\pi}$

16. The number of d-electrons in Fe^{2+} ($Z = 26$) is not equal to the number of electrons in which one of the following?

[AIPMT-2015]

- (1) p-electrons in Cl ($Z = 17$)
(2) d-electrons in Fe ($Z = 26$)
(3) p-electrons in Ne ($Z = 10$)
(4) s-electrons in Mg ($Z = 12$)

17. Which is the correct order of increasing the energy of the listed orbitals in the atom of titanium? (At. no. $Z = 22$)

[RE-AIPMT-2015]

- (1) 3s 3p 3d 4s (2) 3s 3p 4s 3d
(3) 3s 4s 3p 3d (4) 4s 3s 3p 3d

18. Magnetic moment 2.84 B.M. is given by:

[AIPMT-2015]

- (1) Ti^{3+} (2) Cr^{2+}
(3) Co^{2+} (4) Ni^{2+}

19. Which of the following pairs of ions are isoelectronic and isostructural?

[AIPMT-2015]

- (1) ClO_3^- , CO_3^{2-} (2) SO_3^{2-} , NO_3^-
(3) ClO_3^- , SO_3^{2-} (4) CO_3^{2-} , SO_3^{2-}

20. Which one is the wrong statement?

[NEET-2016]

- (a) The uncertainty principle is $\Delta E \times \Delta t \geq \frac{h}{4\pi}$.
(b) Half-filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.
(c) The energy of 2s orbital is less than the energy of 2p orbital in case of hydrogen like atoms.
(d) de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$, where m = mass of the particle, v = group velocity of the particle.
(1) a (2) d
(3) c (4) b

21. Which one is the wrong statement?

[NEET-2017]

- (1) de-Broglie's wavelength is given by $\lambda = \frac{h}{mv}$; where m = mass of the particle, v = group velocity of the particle.
(2) The uncertainty principle is $\Delta E \times \Delta t \geq \frac{h}{4\pi}$
(3) Half-filled and fully filled orbitals have greater stability due to greater exchange energy, greater symmetry and more balanced arrangement.
(4) The energy of 2s orbital is less than the energy of 2p orbital in case of Hydrogen like atoms.

22. Match the metal ions given in Column I with the spin magnetic moments of the ions given in column II and assign the correct code :

[NEET-2018]

	Column-I		Column-II
(a)	Co^{3+}	(i)	$\sqrt{8}$ B.M.
(b)	Cr^{3+}	(ii)	$\sqrt{35}$ B.M.
(c)	Fe^{3+}	(iii)	$\sqrt{3}$ B.M.
(d)	Ni^{2+}	(iv)	$\sqrt{24}$ B.M.
		(v)	$\sqrt{15}$ B.M.

- (1) a-iv, b-i, c-ii, d-iii (2) a-i, b-ii, c-iii, d-iv
(3) a-iv, b-v, c-ii, d-i (4) a-iii, b-v, c-i, d-ii

23. Which one is a wrong statement? [NEET-2018]
 (1) The electronic configuration of N atom is:

$$\begin{array}{ccccc} 1s^2 & 2s^2 & 2p_x^1 & 2p_y^1 & 2p_z^1 \\ \boxed{\uparrow\downarrow} & \boxed{\uparrow\downarrow} & \boxed{\uparrow} & \boxed{\uparrow} & \boxed{\downarrow} \end{array}$$

 (2) An orbital is designated by three quantum numbers while an electron in an atom is designated by four quantum numbers.
 (3) Total orbital angular momentum of electron in 's' orbital is equal to zero.
 (4) The value of m for d_{z^2} is zero.
24. Which of the following series transitions in the spectrum of hydrogen atom falls in visible region? [NEET-2019]
 (1) Paschen series (2) Brackett series
 (3) Lyman series (4) Balmer series
25. 4d, 5p, 5f and 6p orbitals are arranged in the order of decreasing energy. The correct option is: [NEET-2019]
 (1) $6p > 5f > 4d > 5p$ (2) $5f > 6p > 4d > 5p$
 (3) $5f > 6p > 5p > 4d$ (4) $6p > 5f > 5p > 4d$
26. The number of protons, neutrons and electrons in, ${}_{71}^{175}\text{Lu}$ respectively, are: [NEET-2020]
 (1) 104, 71 and 71 (2) 71, 71 and 104
 (3) 175, 104 and 71 (4) 71, 104 and 71
27. The calculated spin only magnetic moment of Cr^{2+} ion is: [NEET-2020]
 (1) 4.90 BM (2) 5.92 BM
 (3) 2.84 BM (4) 3.87 BM
28. The number of angular nodes and radial nodes in 3s orbital are: [NEET 2020 Exam (Covid Re-exam)]
 (1) 0 and 2, respectively
 (2) 1 and 0, respectively
 (3) 3 and 0, respectively
 (4) 0 and 1, respectively
29. From the following pairs of ions which one is not an iso-electronic pair? [NEET-2021]
 (1) $\text{Fe}^{2+}, \text{Mn}^{2+}$ (2) $\text{O}^{2-}, \text{F}^{-}$
 (3) $\text{Na}^{+}, \text{Mg}^{2+}$ (4) $\text{Mn}^{2+}, \text{Fe}^{3+}$
30. A particular station of All India Radio, New Delhi, broadcasts on a frequency of 1368 kHz (kilohertz). The wavelength of the electromagnetic radiation emitted by the transmitter is: [NEET-2021]
 [speed of light, $c = 3.0 \times 10^8 \text{ ms}^{-1}$]
 (1) 21.92 cm (2) 219.3 m
 (3) 219.2 m (4) 2192 m
31. If radius of second Bohr orbit of the He^{+} ion is 105.8 pm, what is the radius of third Bohr orbit of Li^{+2} ion? [NEET-2022]
 (1) 15.87 pm (2) 1.587 pm
 (3) 158 Å (4) 158.7 pm
32. Identify the incorrect statement from the following. [NEET-2022]
 (1) All the five 4d orbitals have shapes similar to the respective 3d orbitals
 (2) In an atom, all the five 3d orbitals are equal in energy free state.
 (3) The shapes of d_{xy} , d_{yz} and d_{zx} , orbitals are similar to each other; and $d_{x^2-y^2}$ and d_{z^2} are similar to each other.
 (4) All the five 5d orbitals are different in size when compared to the respective 4d orbitals
33. Select the correct statements from the following: [NEET-2023]
 (A) Atoms of all elements are composed of two fundamental particles.
 (B) The mass of the electron is $9.10939 \times 10^{-31} \text{ kg}$.
 (C) All the isotopes of a given element show same chemical properties.
 (D) Protons and electrons are collectively, known as nucleons.
 (E) Dalton's atomic theory regarded the atom as an ultimate particle of matter.
 Choose the correct answer from the options given below:
 (1) C, D and E only (2) A and E only
 (3) B, C and E only (4) A, B and C only

34. The relation between n_m , (n_m = the number of permissible values of magnetic quantum number (m)) for a given value of azimuthal quantum number (ℓ), is: [NEET-2023]

(1) $\ell = 2n_m + 1$ (2) $n_m = 2\ell^2 + 1$
 (3) $n_m = \ell + 2$ (4) $\ell = \frac{n_m - 1}{2}$

35. The energy of an electron in the ground state ($n = 1$) for He^+ ion is $-xJ$, then that for an electron in $n = 2$ state for Be^{3+} ion in J is:

[NEET-2024]

(1) $-4x$ (2) $-\frac{4}{9}x$ (3) $-x$ (4) $-\frac{x}{9}$

36. 'Spin only' magnetic moment is same for which of the following ions? [NEET-2024]

- A) Ti^{3+} B) Cr^{2+}
 C) Mn^{2+} D) Fe^{2+}
 E) Sc^{3+}

Choose the most appropriate answer from the options given below:

- (1) B and C only (2) A and D only
 (3) B and D only (4) A and E only

37. Math List I With List II. [NEET-2024]

	List I (Quantum Number)		List II (Information provided)
A.	m_l	I.	Shape of orbital
B.	m_s	II.	Size of orbital
C.	ℓ	III.	Orientation of orbital
D.	n	IV.	Orientation of spin of electron

Choose the correct answer from the options given below.

- (1) A-III, B-IV, C-II, D-I (2) A-II, B-I, C-IV, D-III
 (3) A-I, B-IV, C-I, D-II (4) A-III, B-IV, C-I, D-II

38. The quantum numbers of four electrons are given below: [RE-NEET-2024]

(I) $n = 4; \ell = 2, m = -2; s = -\frac{1}{2}$

(II) $n = 3; \ell = 2, m = 1; s = +\frac{1}{2}$

(III) $n = 4; \ell = 1, m = 0; s = +\frac{1}{2}$

(IV) $n = 3; \ell = 1, m = -1; s = +\frac{1}{2}$

The correct decreasing order of energy of these electrons is:

- (1) IV > II > III > I
 (2) I > III > II > IV
 (3) III > I > II > IV
 (4) I > II > III > IV

39. Given below are two statements:

[RE-NEET-2024]

Statement I: The Balmer spectral line for H atom with lowest energy is located at –

$$\frac{5}{36}R_H\text{cm}^{-1}$$

(R_H = Rydberg constant)

Statement II: When the temperature of black body increases, the maxima of the curve (intensity vs wavelength) shifts to shorter wavelength.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Statement I is true but Statement II is false.
 (2) Statement I is false but Statement II is true.
 (3) Both Statement I and Statement II are true.
 (4) Both Statement I and Statement II are false.

ANSWER KEY

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	
Ans.	2	4	1	3	2	1	1	3	1	2	1	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	4	3	2	3	4	4	3	1	1	2	1	4	1	4	2	4	4	2	4
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	1	4	2	2	1	2	1	4	4	3	4	1	3	4	2	4	1	1	4
Que.	41	42	43	44	45	46	47	48	49											
Ans.	4	3	2	1	1	1	3	1	3											

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	2	2	1	1	4	4	1	1	1	2	1	2	4	2	1	4	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36				
Ans.	3	1	1	3	3	1	2	3	2	2	2	3	4	1	4	3				

Exercise 1.4

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	2	3	2	3	4	4	2	3	3	3	3	3	2	3	2	4	2	4	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	2	2	3	4	1	4	2	2	4	2	2	1	1	1	3	4	4	2	1
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	4	4	1	1	2	1	4	1	2	2	2	4	4	4	4	3	3	4	2	2

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	4	3	2	2	2	3	1	3	2	4	1	3	2	3	2	1	4	3	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	3	2	3	2	3	2	1	3	3	3	1	3	3	3	1	3	1	2	1	4
Que.	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
Ans.	2	2	1	1	4	3	3	1	4	1	2	2	3	2	3	1	2	3	2	4
Que.	61	62	63	64	65	66	67													
Ans.	4	2	4	2	1	1	2													

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	3	4	1	3	1	1	4	1	1	1	4	2	1	1	2	1	1	1
Que.	21	22	23																	
Ans.	4	4	1																	

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	2	1	3	3	1	4	1	1	2	1	4	4	1	2	4	3	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	
Ans.	4	3	1	4	3	4	1	1	1	2	4	3	3	4	3	3	4	2	3	

Chapter 3

REDOX REACTIONS

Chapter Summary

- Redox Reactions
- Concepts of Oxidation and Reduction
- The Rules to Derive Oxidation Number or Oxidation State
- Application of Oxidation Number
- Balancing of Redox Reaction
- Types of Redox Reactions
- Titration
 - (i) KMnO_4 Vs Oxalic Acid
 - (ii) KMnO_4 Vs Mohr Salt

Redox Reactions (Oxidation-Reduction)

Many Chemical reactions involve transfer of electrons from one chemical substance to another. These electron-transfer reactions are termed as **Oxidation-Reduction** or **Redox reactions**.

or

Those reactions which involve oxidation and reduction both simultaneously are known as oxidation and reduction or Redox Reaction.

or

Those reactions which involve increase in oxidation number and decrease in oxidation number simultaneously are known as Oxidation & Reduction or Redox Reactions.

Concepts of Oxidation and Reduction

There are four concepts for oxidation and reduction reactions.

- | | |
|-----------------------|-------------------------------|
| (A) Classical concept | (B) Modern concept |
| (C) Valency concept | (D) Oxidation number concept. |

(A) Classical Concept for Oxidation and Reduction:

According to this concept, oxidation and reduction can be explained as:

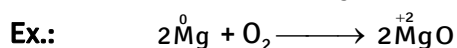
Oxidation		Reduction	
(1)	Addition of oxygen $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ (Oxidation of Magnesium)	(1)	Removal of oxygen $\text{CuO} + \text{C} \rightarrow \text{Cu} + \text{CO}$ (Reduction of cupric oxide)
(2)	Removal of Hydrogen $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$ (Oxidation of Hydrogen sulphide)	(2)	Addition of Hydrogen $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$ (Reduction of Nitrogen)
(3)	Addition of Electronegative element $\text{SnCl}_2 + \text{Cl}_2 \longrightarrow \text{SnCl}_4$ (Oxidation of stannous chloride)	(3)	Removal of Electronegative element $2\text{FeCl}_3 + \text{H}_2 \longrightarrow 2\text{FeCl}_2 + 2\text{HCl}$ (Reduction of ferric chloride)
(4)	Removal of Electropositive element $2\text{K}_2\text{MnO}_4 + \text{Cl}_2 \longrightarrow 2\text{KCl} + 2\text{KMnO}_4$ (Oxidation of Potassium manganate)	(4)	Addition of Electropositive element $\text{HgCl}_2(\text{aq}) + \text{Hg}(\text{s}) \rightarrow \text{Hg}_2\text{Cl}_2(\text{s})$ (addition of mercury to mercuric chloride)

(B) Modern Concept or Electronic Concept:

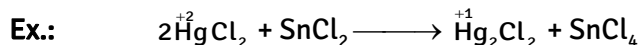
Oxidation	Reduction
According to this concept the process which involves the loss of one or more electrons from an atom or an ion or molecule is called oxidation (de-electronation) $\text{M} \longrightarrow \text{M}^{+n} + n\text{e}^-$ $\text{Mg} \longrightarrow \text{Mg}^{+2} + 2\text{e}^-$ $\text{M}^{+n_1} \longrightarrow \text{M}^{+n_2} + (n_2 - n_1)\text{e}^-$ $\text{Fe}^{+2} \longrightarrow \text{Fe}^{+3} + \text{e}^-$ (ferrous) (ferric) $\text{A}^{-n} \longrightarrow \text{A} + n\text{e}^-$ $2\text{Cl}^- \longrightarrow \text{Cl}_2 + 2\text{e}^-$ $\text{A}^{-n_1} \longrightarrow \text{A}^{-n_2} + (n_1 - n_2)\text{e}^-$ $[\text{Fe}(\text{CN})_6]^{4-} \longrightarrow [\text{Fe}(\text{CN})_6]^{3-} + \text{e}^-$	According to this concept, the process which involves gain of one or more electrons by an atom or ion or molecule is called reduction (electronation). $\text{M}^{+n} + n\text{e}^- \longrightarrow \text{M}$ $\text{Cu}^{+2} + 2\text{e}^- \longrightarrow \text{Cu}$ $\text{M}^{+n_2} + (n_2 - n_1)\text{e}^- \longrightarrow \text{M}^{+n_1}$ $\text{Sn}^{+4} + 2\text{e}^- \longrightarrow \text{Sn}^{+2}$ (Stannic ion) (Stannous ion) $\text{A} + n\text{e}^- \longrightarrow \text{A}^{-n}$ $\text{Cl} + \text{e}^- \longrightarrow \text{Cl}^-$ $\text{A}^{-n_1} + (n_2 - n_1)\text{e}^- \longrightarrow \text{A}^{-n_2}$ $[\text{Fe}(\text{CN})_6]^{3-} + \text{e}^- \longrightarrow [\text{Fe}(\text{CN})_6]^{4-}$

(C) Valency Concept of Oxidation and Reduction:

- (i) **Oxidation:** According to this concept, it is the process in which increase in (+) ve valency or decrease in (-) ve valency of a substance takes place is called oxidation.



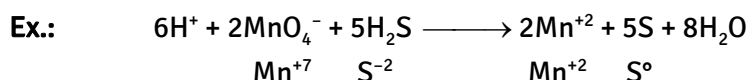
(ii) **Reduction:** According to this concept, the process in which (+) ve valency decreases or (-) ve valency increases is called reduction.



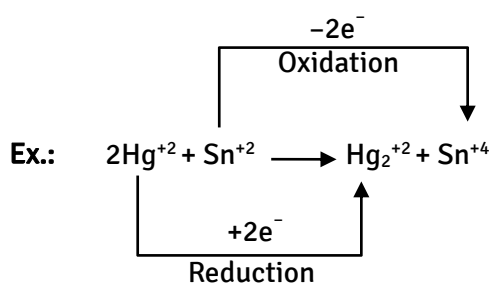
(D) **Oxidation Number Concept:**

(i) **Oxidation:** According to this concept, if the oxidation no. of an element in a reaction increases it is called oxidation.

(ii) **Reduction:** According to this concept, if the oxidation no. of an element in a reaction decreases it is called reduction.



Redox reaction involves two half reactions, one involving loss of electron called oxidation half reaction and the other involving gain of electrons called reduction half reaction.



Oxidation Number:

Definition:

It represents the number of electron(s) gained or lost by an atom when it changes to compound from a free state.

- (a) If electron(s) are gained by an atom in the formation of compound, oxidation number is given (-)ve sign.
- (b) If electron are lost by an atom in the formation of compound, oxidation number is given (+) ve sign.
- (c) It represents the real charge in case of ionic compounds and represents the imaginary charge in case of covalent compounds.

Oxidation State:

Definition:

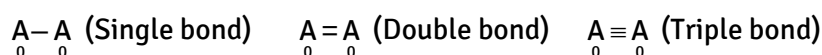
Oxidation state of an atom is defined as oxidation number per atom for all practical purposes. Oxidation state is often expressed as oxidation number.

- Oxidation number may be fractional/zero/whole number.

The Rules to Derive Oxidation Number or Oxidation State

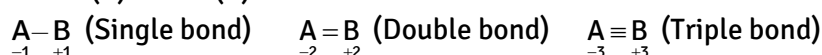
(i) **For Covalent Bond:**

(a) Covalent bond between same atoms



(b) Covalent bond between different atoms

- If E.N. (A) > E.N. (B)

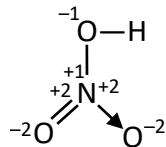


(ii) For Coordinate Bond:

(a) When coordinate bond is formed in between low electronegative element and high electronegative element then the e^- donor element shows +2 oxidation number whereas e^- acceptor element shows -2, oxidation number in this type of bond or bonded compounds.

Electronegativity of acceptor > Electronegativity of donor.

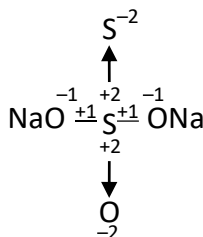
For example, in HNO_3 .



Here 'N' is a less electronegative element than O therefore, oxidation number of N = +2 and oxidation number of O = -2 (due to single dative bond).

(b) When coordinate bond is formed between the two same electronegative elements then the e^- donor element shows +2 oxidation number where e^- acceptor element shows -2 oxidation number. In this type of bond or bonded compound. Electronegativity of acceptor = electronegativity of donor.

For example: In $\text{Na}_2\text{S}_2\text{O}_3$



Here average oxidation number of 'S' in $\text{Na}_2\text{S}_2\text{O}_3$ is +2.

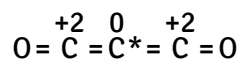
(c) When coordinate bond is formed in between high electronegative element to low electronegative element then there is no change in oxidation number will be shown by both the elements, which is

bonded by coordinate bond. e.g. HNC . $\text{H}-\overset{+1}{\text{N}}^{\overset{-1}{-}} \equiv \overset{0}{\text{C}}^{\overset{+2}{+}}$

(iii) Paradox of Fractional Oxidation Number:

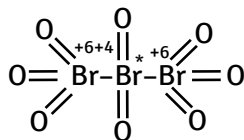
Fractional oxidation number is the average oxidation state of the element under examination and the structural parameters reveal that the element for whom fractional oxidation number is realised is actually present in different oxidation states. Structure of the species C_3O_2 , Br_3O_8 and $\text{S}_4\text{O}_6^{2-}$ reveal the following bonding situations.

(a) The element marked with asterisk in each species is exhibiting the different oxidation state (oxidation number) from rest of the atoms of the same element in each of the species. This reveals that in C_3O_2 , two carbon atoms are present in +2 oxidation state each whereas the third one is present in zero oxidation state and the average is 4/3. However, the realistic picture is +2 for two terminal carbons and zero for the middle carbon.



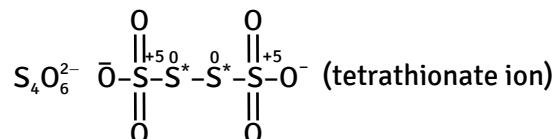
Structure of C_3O_2
(Carbon suboxide)

(b) Likewise, in Br_3O_8 , each of the two terminal bromine atoms are present in +6 oxidation state and the middle bromine is present in +4 oxidation state. Once again, the average, that is different from reality is $16/3$.



Structure of Br_3O_8 (tribromooctaoxide)

(c) In the same fashion, in the species average oxidation number of 'S', is 2.5, whereas the reality being +5, 0, 0 and +5 oxidation states respectively for each sulphur.



(d) Oxidation number of oxygen in superoxides of alkali metals (KO_2 , CsO_2 , RbO_2) is $-1/2$.

e.g. In Fe_3O_4 , Fe shows its oxidation number as $8/3$ as it is a mixed oxide and can be written as $\text{Fe}^{\text{II}}\text{O} \cdot \text{Fe}_2^{\text{III}}\text{O}_3$.

(e) Metals can exist in mixed oxidation states in non-stoichiometric compounds.

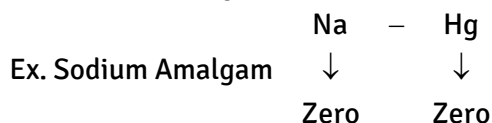
For example: Oxidation state of Fe in $\text{Fe}_{0.94}\text{O}$ is +2 and +3

(iv) Zero Oxidation Number or Oxidation State:

(a) The oxidation state of an element in its free state is zero.

Example oxidation state of Na, Cu, I, Cl, O, O_2 , P_4 , S_8 , N_2 , $\text{C}_{\text{graphite}}$, $\text{C}_{\text{diamond}}$, $\text{S}_{\text{rhombic}}$ etc. are zero.

(b) In case of alloy, oxidation number of each metal will be zero.



(c) Oxidation state of metal in metal carbonyl $[\text{M}(\text{CO})_x]$ is zero.

(v) Oxidation Number or Oxidation State for Compound or Polyatomic Ion:

(a) Algebraic sum of oxidation state of all the atoms in a neutral molecule is zero.

(b) Algebraic sum of oxidation state of all the atoms in a complex ion is equal to the charge present on it.

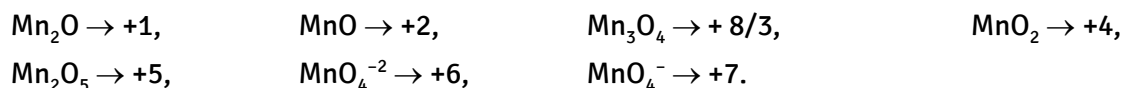
(c) Generally, oxidation state of Oxygen is -2 but in H_2O_2 it is -1 & in OF_2 it is $+2$.

(d) Generally, oxidation state of Hydrogen is $+1$ but in metallic hydrides it is -1 .

(e) Generally, oxidation state of Halogen atoms is -1 but in inter halogen compounds it changes.

(f) Generally, oxidation state of alkali metals is $+1$ and that of alkaline earth metals is $+2$.

(g) Oxidation state of transition elements varies from compound to compound. Mn has oxidation state from $+1$ to $+7$.



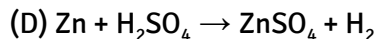
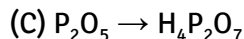
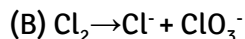
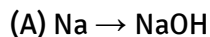
(h) In complex compounds, oxidation state of some neutral molecules (ligands) is zero.

Ex.: Neutral ligand CO , NH_3 , H_2O , C_2H_2 , C_2H_4

Negative ligand CN^- , SCN^- , $\text{C}_2\text{O}_4^{2-}$

Positive ligand CH_3^+ , NO^+ , C_2H_5^+

Ex.:1 State which of the following reactions neither oxidation nor reduction -



Ans. (C)

Sol. In the reaction $\text{P}_2\text{O}_5 \rightarrow \text{H}_4\text{P}_2\text{O}_7$

The O.N. of P in P_2O_5 is

$$2x + 5(-2) = 0 \text{ or } x = +5$$

The O.N. of P in $\text{H}_4\text{P}_2\text{O}_7$ is

$$4(+1) + 2(x) + 7(-2) = 0$$

$$2x = 10 \text{ or } x = +5$$

Since there is no change in O.N. of P, hence the above reaction is neither oxidation nor reduction.

Ex.:2 The oxidation no. of Cl in NOClO_4 is -

(A) +11

(B) +9

(C) +7

(D) +5

Ans. (C)

Sol. The compound may be written as $\text{NO}^+ \text{ClO}_4^-$. For ClO_4^- , Let Ox. No. of Cl = a

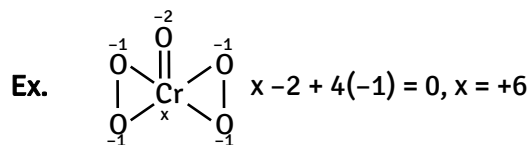
$$a + 4 \times (-2) = -1$$

$$a = +7$$

Hence, the oxidation no. of Cl in NOClO_4 is +7.

Points to Ponder

- Oxidation number of Cr in CrO_5



- Difference between valency and oxidation number:

	Valency	Oxidation Number
1.	It is the combining capacity of the element. No + or - sign is attached to it.	Oxidation number is the charge (real or imaginary) present on the atom of the element when it is in combination. It may have + or - sign.
2.	Valency of an element usually fixed.	Oxidation number of an element may have different values. It depends on the nature of the compound in which it is present.
3.	Valency is always a whole number.	Oxidation number of the element may be a whole number or fractional.
4.	Valency of the element is never zero except of noble gases.	Oxidation number of the element may be zero.
	e.g. Valency of N in HNO_3 is 4.	e.g. Oxidation number of N in HNO_3 is +5.

Exercise 1.1

- According to classical concept, oxidation involves:
 - (1) Addition of oxygen
 - (2) Addition of electronegative element
 - (3) Removal of either hydrogen or some electro positive element
 - (4) All of these
- A redox reaction is:
 - (1) Proton transfer reaction
 - (2) Electron transfer reaction
 - (3) Ion combination reaction
 - (4) A reaction in a solution
- According to modern concept, oxidation is:
 - (1) Electronation
 - (2) De-electronation
 - (3) Addition of oxygen
 - (4) Addition of electronegative element
- Oxidation takes place with:
 - (1) Gain of electrons
 - (2) Loss of electrons
 - (3) Increase in the valency of negative part
 - (4) Decrease in the valency of positive part
- The oxidation number of nitrogen in NH_2OH is:
 - (1) +1
 - (2) -1
 - (3) -3
 - (4) -2
- Oxidation number of sulphur in $\text{Na}_2\text{S}_2\text{O}_3$ would be:
 - (1) +2
 - (2) +4
 - (3) -2
 - (4) 0
- Oxidation number of C in CH_2Cl_2 is:
 - (1) +2
 - (2) +4
 - (3) -4
 - (4) 0
- The oxidation number of P is +3 in:
 - (1) H_3PO_3
 - (2) H_3PO_4
 - (3) HPO_3
 - (4) $\text{H}_4\text{P}_2\text{O}_7$
- The oxidation number of carbon in $\text{C}_{12}\text{H}_{22}\text{O}_{11}$ is:
 - (1) 0
 - (2) -6
 - (3) +6
 - (4) +2
- The oxidation number of chlorine in HOCl is:
 - (1) -1
 - (2) 0
 - (3) +1
 - (4) 2
- Oxidation Number of hydrogen in KH , MgH_2 and NaH respectively would be:
 - (1) -1, -1 and -1
 - (2) +1, +1, and +1
 - (3) +2, +1 and -2
 - (4) -2, -3 and -1
- The oxidation number of C in CH_4 , CH_3Cl , CH_2Cl_2 , CHCl_3 and CCl_4 are respectively:
 - (1) +4, +2, 0, -2, -4
 - (2) +2, +4, 0, -4, -2
 - (3) -4, -2, 0, +2, +4
 - (4) -2, -4, 0, +4, +2
- 1/3 oxidation state of nitrogen will be obtained in case of:
 - (1) Ammonia (NH_3)
 - (2) Hydrazoic acid (N_3H)
 - (3) Nitric oxide (NO)
 - (4) Nitrous oxide (N_2O)
- Which of the following halogens always shows only one oxidation state in compound:
 - (1) Cl
 - (2) F
 - (3) Br
 - (4) I
- Oxygen has the oxidation state of +2 in:
 - (1) SO_2
 - (2) CO_2
 - (3) H_2O_2
 - (4) OF_2
- The oxidation state of iodine in H_4IO_6^- is:
 - (1) +7
 - (2) -1
 - (3) +5
 - (4) +1
- Oxidation number of Fe in $\text{Fe}_{0.94}\text{O}$ is:
 - (1) 200
 - (2) 200/94
 - (3) 94/200
 - (4) none of these
- The charge on cobalt in $[\text{Co}(\text{CN})_6]^{3-}$ is:
 - (1) -6
 - (2) -3
 - (3) +3
 - (4) +6
- The oxidation number of Pt in $\text{K}[\text{PtNH}_3\text{Cl}_5]$ will be:
 - (1) +4
 - (2) -1
 - (3) +2
 - (4) +3
- Oxidation state of Cr in $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ is:
 - (1) +1
 - (2) 0
 - (3) +3
 - (4) +2
- The compound in which oxidation state of metal is zero:
 - (1) $\text{Fe}_2(\text{CO})_9$
 - (2) $\text{Ni}(\text{CO})_4$
 - (3) $\text{Fe}_3(\text{CO})_9$
 - (4) All the above
- Oxidation number of sodium in sodium amalgam is:
 - (1) +1
 - (2) 0
 - (3) -1
 - (4) +2
- In which of the following reactions, the underlined element has decreased its oxidation number during the reaction?
 - (1) $\underline{\text{Fe}} + \text{CuSO}_4 \rightarrow \text{Cu} + \text{FeSO}_4$
 - (2) $\underline{\text{H}_2} + \text{Cl}_2 \rightarrow 2\text{HCl}$
 - (3) $\underline{\text{C}} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$
 - (4) $\underline{\text{MnO}_2} + 4\text{HCl} \rightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$

24. In the reaction $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$ the oxidation state of S is:
 (1) Increased (2) Decreased
 (3) Remains same (4) Can't determine
25. Oxidation number of oxygen in hydrogen peroxide is:
 (1) -1 (2) +1 (3) 0 (4) -2
26. The oxidation number of phosphorus in $\text{Ba}(\text{H}_2\text{PO}_2)_2$ is:
 (1) +3 (2) +2 (3) +1 (4) -1
27. Oxidation number of C in HNC is:
 (1) +2 (2) -3 (3) +3 (4) zero
28. Oxidation number of carbon in carbon suboxide (C_3O_2) is:
 (1) $+\frac{2}{3}$ (2) $+\frac{4}{3}$ (3) +4 (4) $-\frac{4}{3}$
29. Oxidation state of hydrogen in CaH_2 is:
 (1) +1 (2) -1 (3) +2 (4) 0
30. Two oxidation states for chlorine are found in the compound:
 (1) CaOCl_2 (2) KCl (3) KClO_3 (4) Cl_2O_7
31. Compound $\text{YBa}_2\text{Cu}_3\text{O}_7$ is a super conductor. The O.N. of the copper in the compound will be: [O. No. of Y = +3]
 (1) $+\frac{7}{3}$ (2) zero (3) +2 (4) +1
32. Which of the following is not an example of redox reaction?
 (1) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7 \rightarrow \text{N}_2 + 4\text{H}_2\text{O} + \text{Cr}_2\text{O}_3$
 (2) $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
 (3) $2\text{K} + \text{F}_2 \rightarrow 2\text{KF}$
 (4) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{HCl}$
33. The reaction $\text{Co(s)} + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Co}^{2+}(\text{aq}) + \text{Cu(s)}$ is:
 (1) Oxidation (2) Reduction
 (3) Redox (4) Hydrolysis
34. The conversion of sugar $\text{C}_{12}\text{H}_{22}\text{O}_{11} \rightarrow \text{CO}_2$ is:
 (1) Oxidation
 (2) Reduction
 (3) Neither oxidation nor reduction
 (4) Both oxidation and reduction
35. Which one of the following is a redox reaction?
 (1) $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$
 (2) $2\text{NaCl} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + 2\text{HCl}$
 (3) $\text{HCl} + \text{AgNO}_3 \rightarrow \text{AgCl} + \text{HNO}_3$
 (4) $\text{NaOH} + \text{HCl} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
36. The oxidation state of sulphur in Caro's acid (H_2SO_5) and Marshal's acid ($\text{H}_2\text{S}_2\text{O}_8$) are:
 (1) S^{6+} , S^{6+} (2) S^{6+} , S^{7+}
 (3) S^{6+} , S^{5+} (4) S^{6+} , S^{6-}
37. Oxidation number of oxygen in KO_3 , KO_2 and K_2O is:
 (1) $-\frac{1}{3}$, $-\frac{1}{2}$, -2 (2) $-\frac{1}{3}$, $-\frac{1}{2}$, $-\frac{1}{2}$
 (3) -1, -1, -1 (4) -2, -2, -2
38. The oxidation state of +3 for phosphorus is in:
 (1) Hypophosphorous acid (H_3PO_2)
 (2) Meta-phosphoric acid (HPO_3)
 (3) Ortho-phosphoric acid (H_3PO_4)
 (4) Phosphorous acid (H_3PO_3)
39. The charge on cobalt in $[\text{Co}(\text{CN})_6]^{-3}$ is:
 (1) -6 (2) -3 (3) +3 (4) +6
40. Which of the following reactions do not involve oxidation-reduction?
 (1) $2\text{Rb} + 2\text{H}_2\text{O} \rightarrow 2\text{RbOH} + \text{H}_2$
 (2) $2\text{CuI}_2 \rightarrow 2\text{CuI} + \text{I}_2$
 (3) $\text{NH}_4\text{Cl} + \text{NaOH} \rightarrow \text{NaCl} + \text{NH}_3 + \text{H}_2\text{O}$
 (4) $3\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
41. Oxidation state of carbon atoms in diamond and graphite are:
 (1) +2, +4 (2) +4, +2
 (3) -4, 4 (4) zero, zero
42. Oxidation number of 'S' in $(\text{CH}_3)_2\text{SO}$ is:
 (1) +1 (2) 0 (3) +2 (4) +3
43. In which of the following compounds, an element exhibits two different oxidation states:
 (a) NH_4NO_3 (b) NH_4NO_2
 (c) N_3H (d) Br_3O_8
 (1) a,b (2) a,b,c
 (3) a,c,d (4) All

Application of Oxidation Number

(1) To Compare the Strength of Oxyacid and Base:

Strength of acid $\propto$ Oxidation Number of central element

$$\text{Strength of base} \propto \frac{1}{\text{Oxidation Number of central element}}$$

Example: (a) Order of acidic strength in HClO , HClO_2 , HClO_3 , HClO_4 will be.

Solution:

Oxidation Number of chlorine

HClO (Hypo Chlorous acid)

+1

HClO₂ (Chlorous acid)

+3

HClO_3 (Chloric acid)

+5

HClO₄ (Perchloric acid)

+7

So, the order of acidic strength will be $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HClO}$

(b) Order of acidic strength in $\text{H}_2\text{N}_2\text{O}_2$, HNO_2 , HNO_3 will be.

+1

+3

+5

$$\text{H}_2\text{N}_2\text{O}_2$$
 HNO_2 HNO_3

Hyponitrous acid

Nitrous acid

Nitric acid

(weakest acid)

(strongest acid)

- **Exception:** $\text{H}_3\text{PO}_4 < \text{H}_3\text{PO}_3 < \text{H}_3\text{PO}_2$ (Acidic strength, explained through resonance).

(2) To Determine the Oxidising and Reducing Nature:

(a) Oxidising Agents or Oxidants (O.A.):

The substances which accept electrons in a chemical reaction i.e., electron acceptors are oxidising agent.

- (i) Oxidising agents are Lewis acids.
- (ii) Substances which can oxidise others and get reduce itself.
- (iii) Substances which show the decrement in oxidation number.

(b) Reducing Agents or Reductant (R.A.):

The substance which donates electrons in a chemical reaction are reducing agent i.e., electron donors are reducing agents.

- (i) Reducing agents are Lewis bases.
- (ii) Substances which can reduce other and get oxidised itself.
- (ii) Substances which show the increment in oxidation number.

(c) Maximum oxidation number of an element is equal to its group number in the periodic table.

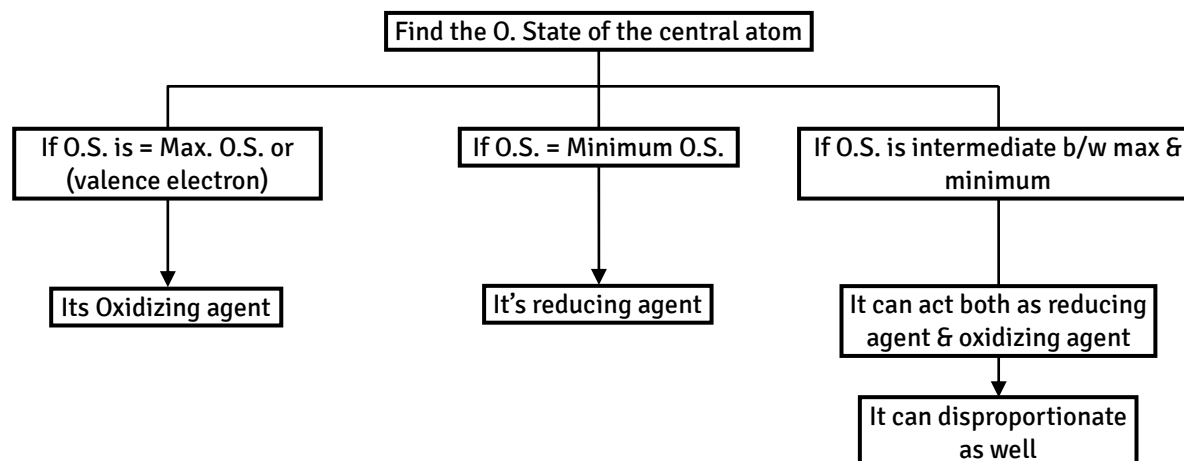
(d) Minimum oxidation number of an element is equal to (group number -8)

- 1 Group elements always shows +1 as oxidation number
- 2 Group elements always shows +2 as oxidation number
- 3 Group elements always shows +3 oxidation number
- 4 Group shows -4 to +4 oxidation number
- 5 Group shows -3 to +5 oxidation number
- 6 Group shows -2 to +6 oxidation number
- 7 Group shows -1 to +7 oxidation number (except F)

Inert gases always shows zero oxidation number (except Xe, Kr)

Sr. No.	Reaction	Oxidant (Getting Reduced)	Reductant (Getting Oxidised)
1.	$C + O_2 \longrightarrow CO_2$	O [0 $\longrightarrow$ -2]	C [0 $\longrightarrow$ +4]
2.	$PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$	O [-1 $\longrightarrow$ -2]	S [-2 $\longrightarrow$ +6]
3.	$Sn + 2F_2 \longrightarrow SnF_4$	F [0 $\longrightarrow$ -1]	Sn [0 $\longrightarrow$ +4]
4.	$SO_2 + 2H_2O + Cl_2 \longrightarrow 2HCl + H_2SO_4$	Cl [0 $\longrightarrow$ -1]	S [+4 $\longrightarrow$ +6]
5.	$I_2 + 10HNO_3 \longrightarrow 2HIO_3 + 10NO_2 + 4H_2O$	N [+5 $\longrightarrow$ +4]	I [0 $\longrightarrow$ +5]
6.	$CuO + H_2 \longrightarrow Cu + H_2O$	Cu [+2 $\longrightarrow$ 0]	H [0 $\longrightarrow$ +1]
7.	$2KMnO_4 + 3H_2SO_4 + 5H_2S \longrightarrow K_2SO_4 + 2MnSO_4 + 8H_2O + 5S$	Mn [+7 $\longrightarrow$ +2]	S [-2 $\longrightarrow$ 0]
8.	$H_2O_2 + Ag_2O \longrightarrow 2Ag + H_2O + O_2$ (Oxidation of H_2O_2)	Ag [+1 $\longrightarrow$ 0]	O [-1 $\longrightarrow$ 0]
9.	$H_2SO_4 + 2HI \longrightarrow SO_2 + I_2 + 2H_2O$	S [+6 $\longrightarrow$ +4]	I [-1 $\longrightarrow$ 0]

Flow chart to identify: Whether a particular substance is an oxidising or reducing agent.

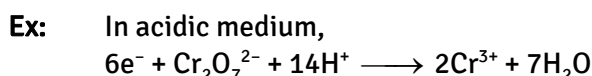


- If central element of any compound is in maximum oxidation state, then it will act as only oxidant.
Ex: $HClO_4$, H_2SO_4 , CO_2 , $K_2Cr_2O_7$, K_2MnO_4
- If central element of compound is in minimum oxidation state, then it will act as only reductant.
Ex: NH_3 , PH_3 , CH_4 , H_2S
- If oxidation state of central element in any compound is intermediate then compound can act as reductant as well as oxidant both.
Ex: H_2O_2 , CO , HNO_2 , H_2SO_3

(3) Equivalent Weight of Oxidising/Reducing Agents in Redox Reaction:

The equivalent weight of an oxidising agent is the weight which accepts one mole electron in a chemical reaction.

(a) Equivalent weight of an oxidant (get reduced) = $\frac{\text{Molecular weight}}{\text{Number of electrons gained by one mole}}$



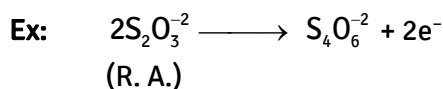
Here atom which undergoes reduction is Cr in which oxidation state is decreasing from +6 to +3

$$\begin{aligned}\text{Equivalent weight of } K_2Cr_2O_7 &= \frac{\text{Molecular weight of } K_2Cr_2O_7}{3 \times 2} = \frac{\text{Molecular weight}}{6} \\ &= \frac{294.21}{6} = 49.03\end{aligned}$$

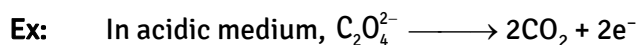
Note: [6 in denominator indicates that 6 electrons are gained by 1 mole $Cr_2O_7^{2-}$ as it is clear from the given balanced equation]

- (b) The equivalent weight of a reducing agent is the weight which donates one mole electron in a chemical reaction.

$$\text{Equivalent weight of a reductant (get oxidised)} = \frac{\text{Molecular weight}}{\text{Number of electrons lost by one mole}}$$

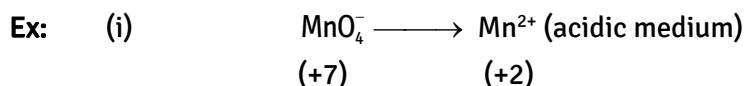


$$\text{equivalent weight of } S_2O_3^{2-} = \frac{2M}{2} = M$$

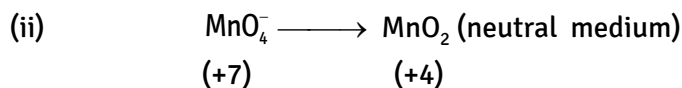


$$\text{Here, Total electrons lost} = 2 \text{ So, equivalent weight} = \frac{\text{Molecular weight}}{2}$$

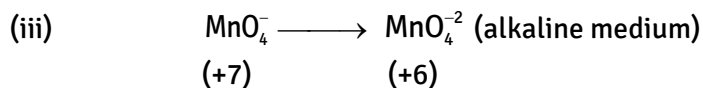
- (c) In different condition a compound may have different equivalent weight. Because, it depends upon the number of electrons gained or lost by that compound in that reaction.



$$\text{Here 5 electrons are taken so equivalent weight} = \frac{\text{Molecular weight of } KMnO_4}{5} = \frac{158}{5} = 31.6$$



$$\text{Here, only 3 electrons are gained, so equivalent weight} = \frac{\text{Molecular weight of } KMnO_4}{3} = \frac{158}{3} = 52.7$$



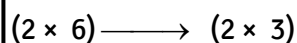
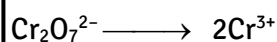
$$\text{Here, only one electron is gained, so equivalent weight} = \frac{\text{Molecular weight of } KMnO_4}{1} = 158$$

Note:-

It is important to note that $KMnO_4$ acts as an oxidant in every medium although with different strength which follows the order –

acidic medium > neutral medium > alkaline medium

while, $K_2Cr_2O_7$ acts as an oxidant only in acidic medium as follows



It is clear that $KMnO_4$ is better oxidant than $K_2Cr_2O_7$.

Some Oxidizing Agents/Reducing Agents with eq. wt.

Species	Changed to	Reaction	Electrons exchanged or change in O.N.	Eq. Wt.
MnO_4^- (O.A.)	Mn^{+2} in acidic medium	$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	5	$E = \frac{M}{5}$
MnO_4^- (O.A.)	MnO_2 in neutral medium	$\text{MnO}_4^- + 3\text{e}^- + 2\text{H}_2\text{O} \longrightarrow \text{MnO}_2 + 4\text{OH}^-$	3	$E = \frac{M}{3}$
MnO_4^- (O.A.)	MnO_4^{2-} in basic medium	$\text{MnO}_4^- + \text{e}^- \longrightarrow \text{MnO}_4^{2-}$	1	$E = \frac{M}{1}$
$\text{Cr}_2\text{O}_7^{2-}$ (O.A.)	Cr^{3+} in acidic medium	$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	6	$E = \frac{M}{6}$
MnO_2 (O.A.)	Mn^{2+} in acidic medium	$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \longrightarrow \text{Mn}^{2+} + 2\text{H}_2\text{O}$	2	$E = \frac{M}{2}$
Cl_2 (O.A.) In bleaching powder	Cl^-	$\text{Cl}_2 + 2\text{e}^- \longrightarrow 2\text{Cl}^-$	2	$E = \frac{M}{2}$
CuSO_4 (O.A.) In iodometric titration	Cu^+	$\text{Cu}^{2+} + \text{e}^- \longrightarrow \text{Cu}^+$	1	$E = \frac{M}{1}$
$\text{S}_2\text{O}_3^{2-}$ (R.A.)	$\text{S}_4\text{O}_6^{2-}$	$2\text{S}_2\text{O}_3^{2-} \longrightarrow \text{S}_4\text{O}_6^{2-} + 2\text{e}^-$	2 (for two molecules)	$E = \frac{2M}{2} = M$
H_2O_2 (O.A.)	H_2O	$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \longrightarrow 2\text{H}_2\text{O}$	2	$E = \frac{M}{2}$
H_2O_2 (R.A.)	O_2	$\text{H}_2\text{O}_2 \longrightarrow \text{O}_2 + 2\text{H}^+ + 2\text{e}^-$ (O.N. of oxygen in H_2O_2 is -1 per atom)	2	$E = \frac{M}{2}$
Fe^{2+} (R.A.)	Fe^{3+}	$\text{Fe}^{2+} \longrightarrow \text{Fe}^{3+} + \text{e}^-$	1	$E = \frac{M}{1}$

Law of Equivalence:

The law states that one equivalent of an element combine with one equivalent of the other, and in a chemical reaction, equivalent and mill equivalent of reactants react in equal amount to give same number of equivalent or milli equivalents of products separately.

According: (i) $aA + bB \longrightarrow mM + nN$
 $\text{m.eq of A} = \text{m.eq of B} = \text{m.eq of M} = \text{m.eq of N}$

(ii) In a compound M_xN_y
 $\text{m.eq of M}_x\text{N}_y = \text{m.eq of M} = \text{m.eq of N}$

For Acid-Base (Neutralization Reaction) or Redox Reaction:

$N_1V_1 = N_2V_2$ is always true.

But $M_1V_1 = M_2V_2$ (may or may not be true)

But $M_1 \times n_1 \times V_1 = M_2 \times n_2 \times V_2$ (always true where n terms represent n-factor)

(4) To Determine the Molecular Formula of Compound:

Ex: Suppose that there are three atoms A, B, C and their oxidation number are +6, -1, -2, respectively. Then the molecular formula of compound may be.

Sol: Since, the charge on a free compound is zero. So
 $+6 + (-1 \times 4) + (-2)$ or $+6 + (-1 \times 2) + (-2 \times 2)$
 $+6 + (-6)$ $+6 + (-2) + (-4) = -6$
 So molecular formula, AB_4C or AB_2C_2 .

Ex.3: Which of the following acts as both oxidant and reductant -
 (A) HNO_3 (B) HNO_2
 (C) Both HNO_3 & HNO_2 (D) Neither HNO_3 nor HNO_2

Ans. (B)

Sol. O.N. of N in HNO_2 is +3

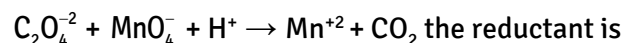
Max. O.N. of N is +5

Min. O.N. of N is -3

Thus O.N. of N in HNO_2 can show an increase or decrease as the case may be. That is why HNO_2 acts as oxidant and reductant both.

O.N. of N in HNO_3 is +5, Hence it can act only as an oxidant.

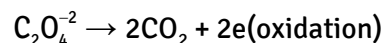
Ex.4: In the reaction



(A) $C_2O_4^{2-}$ (B) H^+ (C) MnO_4^- (D) None of these

Ans. (A)

Sol. In the above reaction $C_2O_4^{2-}$ acts as a reductant because it is oxidised to CO_2 as:



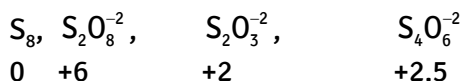
$C_2O_4^{2-}$ reduces MnO_4^- to Mn^{+2} ion in solution.

Ex.5: The order of increasing O.N. of S in S_8 , $S_2O_8^{2-}$, $S_2O_3^{2-}$, $S_4O_6^{2-}$ is given below -

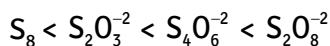
(A) $S_8 < S_2O_8^{2-} < S_2O_3^{2-} < S_4O_6^{2-}$ (B) $S_2O_8^{2-} < S_2O_3^{2-} < S_4O_6^{2-} < S_8$
 (C) $S_2O_8^{2-} < S_8 < S_4O_6^{2-} < S_2O_3^{2-}$ (D) $S_8 < S_2O_3^{2-} < S_4O_6^{2-} < S_2O_8^{2-}$

Ans. (D)

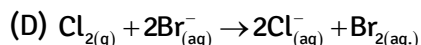
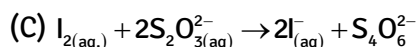
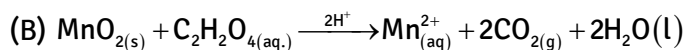
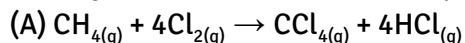
Sol. The O.Ns. of S are shown below along with the compounds



Hence the order of increasing O.N. of S is



Ex.6: Identify the oxidized and reduced species in the following reactions:



Sol. Reduced Oxidised

(A) Cl_2	CH ₄
(B) MnO_2	$C_2H_2O_4$
(C) I_2	$S_2O_3^{2-}$
(D) Cl_2	Br^-

Ex.7: Equivalent weight of Fe_2O_3 in terms of its mol. weight in the change $\text{Fe}_3\text{O}_4 \rightarrow \text{Fe}_2\text{O}_3$ is:

- (A) M (B) M/2 (C) M/3 (D) 3M/2

Ans. (D)

Sol. $2[\text{Fe}^{+8/3}]_3 \rightarrow 3[\text{Fe}^{3+}]_2 + 2e$

$$\therefore E_{\text{Fe}_2\text{O}_3} = \frac{M}{2/3}$$

Points to Ponder

- When metals participate in chemical reaction, they are always oxidised. Thus, metals always behave like reducing agents.

Exercise 1.2

- Find the correct increasing order of acidic strength in the following:
 - (1) $\text{NO} < \text{N}_2\text{O}_3 < \text{NO}_2 < \text{N}_2\text{O}_5$
 - (2) $\text{NO} < \text{N}_2\text{O}_3 < \text{N}_2\text{O}_5 < \text{NO}_2$
 - (3) $\text{NO} < \text{NO}_2 < \text{N}_2\text{O}_3 < \text{N}_2\text{O}_5$
 - (4) $\text{N}_2\text{O}_3 < \text{NO} < \text{N}_2\text{O}_5 < \text{NO}_2$
- A, B and C are three elements forming a compound in oxidation states of +2, +5 and -2 respectively. What could be the compound?
 - (1) $\text{A}_2(\text{BC})_2$
 - (2) $\text{A}_2(\text{BC}_4)_3$
 - (3) $\text{A}_3(\text{BC}_4)_2$
 - (4) ABC
- In acidic medium, reaction $\text{MnO}_4^- \rightleftharpoons \text{Mn}^{2+}$ is an example of -
 - (1) Oxidation by three electrons
 - (2) Reduction by three electrons
 - (3) Oxidation by five electrons
 - (4) Reduction by five electrons
- In a reaction, $\text{H}_2\text{O} + \text{C} \rightarrow \text{CO} + \text{H}_2$
 - (1) H_2O is the reducing agent
 - (2) H_2O is the oxidising agent
 - (3) carbon is the oxidising agent
 - (4) oxidation-reduction does not occur
- Identify oxidising & reducing agent $\text{PbS} + 4\text{H}_2\text{O}_2 \rightarrow \text{PbSO}_4 + 4\text{H}_2\text{O}$
 - (1) PbS, H_2O_2
 - (2) H_2O_2 , PbS
 - (3) H_2O_2 both
 - (4) PbS both
- In a reaction of H_2O (steam) + C (glowing) $\rightarrow \text{CO} + \text{H}_2$
 - (1) H_2O is the reducing agent
 - (2) H_2O is the oxidising agent
 - (3) Carbon is the oxidising agent
 - (4) Oxidation-reduction does not occur
- Reaction
 - (A) $\text{S}^{2-} + 4\text{H}_2\text{O}_2 \rightarrow \text{SO}_4^{2-} + 4\text{H}_2\text{O}$
 - (B) $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 The true statement regarding the above reactions is:
 - (1) H_2O_2 acts as reductant in both the reactions.
 - (2) H_2O_2 acts as oxidant in reaction (A) and reductant in reaction (B).
 - (3) H_2O_2 acts as an oxidant in both the reactions.
 - (4) H_2O_2 acts as reductant in reaction (A) and oxidant in reaction (B).
- If an element is in its lowest oxidation state, under proper conditions it can act as:
 - (1) Reducing agent
 - (2) An oxidising agent
 - (3) Oxidising as well as reducing agent
 - (4) Neither oxidising nor reducing agent
- Which compound cannot be used as reducing agent?
 - (1) CO_2
 - (2) HNO_2
 - (3) H_3PO_3
 - (4) H_2SO_3
- Which one can act as both oxidising & reducing agent?
 - (1) HNO_2
 - (2) H_2O_2
 - (3) H_2SO_3
 - (4) all

11. Which compound cannot be used as oxidising agent?
 (1) O_3 (2) HNO_3
 (3) $KMnO_4$ (4) NH_3
12. Oxidation number of Mn can be-
 (1) +2 to +6 (2) +2, +3
 (3) +2 to +7 (4) +2, +8
13. Oxidation number of nitrogen can be-
 (1) From +5 to -3 (2) From -5 to -3
 (3) From -5 to +3 (4) From +10 to +6
14. Fluorine does not show positive oxidation state due to the -
 (1) Absence of s-orbitals
 (2) Absence of p-orbitals
 (3) Absence of d-orbitals
 (4) Highest electro negativity
15. Oxidation number of iodine varies from -
 (1) -1 to +1 (2) -1 to +7
 (3) +3 to +5 (4) -1 to +5
16. In which SO_2 acts as oxidant, while reacting with:
 (1) Acidified $KMnO_4$ (2) Acidified $K_2Cr_2O_7$
 (3) H_2S (4) Acidified C_2H_5OH
17. Which one of the compound cannot decolourise an acidified solution of $KMnO_4$?
 (1) SO_2 (2) $FeCl_3$
 (3) H_2O_2 (4) $FeSO_4$
18. In which of the following reactions H_2O_2 acts as a reducing agent?
 (i) $H_2O_2 + 2H^+ + 2e^- \longrightarrow 2H_2O$
 (ii) $H_2O_2 - 2e^- \longrightarrow O_2 + 2H^+$
 (iii) $H_2O_2 + 2e^- \longrightarrow 2OH^-$
 (iv) $H_2O_2 + 2OH^- - 2e^- \longrightarrow O_2 + 2H_2O$
 (1) i, ii (2) iii, iv
 (3) i, iii (4) ii, iv
19. Which of the following equation depicts reducing nature of H_2O_2 ?
 (1) $2[Fe(CN)_6]^{4-} + 2H^+ + H_2O_2 \longrightarrow 2[Fe(CN)_6]^{3-} + 2H_2O$
 (2) $I_2 + H_2O_2 + 2OH^- \longrightarrow 2I^- + 2H_2O + O_2$
 (3) $Mn^{2+} + H_2O_2 \longrightarrow Mn^{4+} + 2OH^-$
 (4) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$
20. In the reaction
 $MnO_4^- + SO_3^{2-} + H^+ \longrightarrow SO_4^{2-} + Mn^{2+} + H_2O$
 (1) MnO_4^- and H^+ both are reduced
 (2) MnO_4^- is reduced and H^+ is oxidised
 (3) MnO_4^- is reduced and SO_3^{2-} is oxidised
 (4) MnO_4^- is oxidised and SO_3^{2-} is Reduced
21. Thiosulphate reacts differently with iodine and bromine in the reactions given below:
 $2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$
 $S_2O_3^{2-} + 4Br_2 + 5H_2O \longrightarrow 2SO_4^{2-} + 8Br^- + 10H^+$
 Which of the following statements in correct?
 (1) Bromine is a stronger oxidant than iodine.
 (2) Bromine is a weaker oxidant than iodine.
 (3) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reactions.
 (4) Bromine undergoes oxidation and iodine undergoes reduction in these reactions.
22. Molecular weight of $KMnO_4$ in acidic medium and neutral medium will be respectively -
 (1) $7 \times$ equivalent weight and $2 \times$ equivalent weight
 (2) $5 \times$ equivalent weight and $3 \times$ equivalent weight
 (3) $4 \times$ equivalent weight and $5 \times$ equivalent weight
 (4) $2 \times$ equivalent weight and $4 \times$ equivalent weight
23. The eq. wt. of iodine in,
 $I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$ is:
 (1) M_w (2) $M_w/2$
 (3) $M_w/4$ (4) $M_w/6$
24. In the following unbalanced redox reaction,
 $Cu_3P + Cr_2O_7^{2-} \longrightarrow Cu^{2+} + H_3PO_4 + Cr^{3+}$
 Equivalent weight of H_3PO_4 is -
 (1) $M/3$ (2) $M/6$ (3) $M/7$ (4) $M/8$
25. Equivalent weight of oxidising agent will be-
 $2H_2 + O_2 \longrightarrow 2H_2O$
 (1) 1 (2) 32 (3) 2 (4) 8

26. The Molecular weight of KBrO_3 is M . What is its equivalent weight if the reaction is -
 $\text{BrO}_3^- \rightarrow \text{Br}^-$ (acidic medium)
 (1) M (2) $M/4$
 (3) $M/6$ (4) $6M$
27. In the reaction, $\text{VO} + \text{Fe}_2\text{O}_3 \rightarrow \text{FeO} + \text{V}_2\text{O}_5$, the eq. wt. of V_2O_5 is equal to its:
 (1) Mol. wt. (2) Mol. wt./8
 (3) Mol. wt./6 (4) None of these
28. The equivalent weight of $\text{Na}_2\text{S}_2\text{O}_3$ (Mol. wt = M) in the reaction,
 $2\text{Na}_2\text{S}_2\text{O}_3 + \text{I}_2 \rightarrow \text{Na}_2\text{S}_4\text{O}_6 + 2\text{NaI}$ is -
 (1) $M/4$ (2) $M/3$ (3) $M/2$ (4) M
29. Equivalent weight of FeC_2O_4 in the change, $\text{FeC}_2\text{O}_4 \rightarrow \text{Fe}^{3+} + \text{CO}_2$ is:
 (1) $M/3$ (2) $M/6$ (3) $M/2$ (4) $M/1$
30. What is the equivalent weight of NH_3 in the given reaction?
 $3\text{CuO} + 2\text{NH}_3 \rightarrow 3\text{Cu} + \text{N}_2 + 3\text{H}_2\text{O}$
 (1) 17 (2) $\frac{17}{4}$
 (3) $\frac{17}{2}$ (4) $\frac{17}{3}$
31. What is the equivalent weight of P_4 in the following reaction?
 $\text{P}_4 + \text{NaOH} \rightarrow \text{NaH}_2\text{PO}_2 + \text{PH}_3$
 (1) $\frac{31}{4}$ (2) $\frac{31}{3}$
 (3) $\frac{31}{2}$ (4) $31 \times 4/3$
32. In acidic medium, equivalent weight of $\text{K}_2\text{Cr}_2\text{O}_7$ (Mol. wt. = M) is-
 (1) $M/3$ (2) $M/4$
 (3) $M/6$ (4) $M/2$
33. Equivalent weight of H_3PO_2 when it disproportionates into PH_3 and H_3PO_3 is: (mol. wt. of $\text{H}_3\text{PO}_2 = M$)
 (1) M (2) $\frac{3M}{4}$
 (3) $\frac{M}{2}$ (4) $\frac{M}{4}$
34. Equivalent weight of N_2 in the change $\text{N}_2 \rightarrow \text{NH}_3$ is:
 (1) $\frac{28}{6}$ (2) 28
 (3) $\frac{28}{2}$ (4) $\frac{28}{3}$
35. For the reaction, $\text{N}_2 + 3\text{H}_2 \rightarrow 2\text{NH}_3$ if E_1 and E_2 are equivalent masses of NH_3 and N_2 respectively, then $E_1 - E_2$ is:
 (1) 1 (2) 2 (3) 3 (4) 4
36. The number of mole of oxalate ions oxidised by one mole of MnO_4^- in acidic medium is:
 (1) $1/5$ (2) $2/5$ (3) $5/2$ (4) 5
37. 5.6 g KOH completely neutralises 0.1 mole of H_3PO_x . Thus, x is:
 (1) 2 (2) 3 (3) 4 (4) 1
38. The number of mole of ferrous oxalate oxidized by one mole of KMnO_4 is:
 (1) $1/5$ (2) $3/5$ (3) $2/3$ (4) $5/3$
39. How many milligrams of iron (Fe^{2+}) are equal to 1 mL of 0.1055 N $\text{K}_2\text{Cr}_2\text{O}_7$ equivalent?
 (1) 5.9 mg (2) 0.59 mg
 (3) 59 mg (4) 5.9×10^{-3} mg
40. n-factor of H_3PO_2 during its disproportionation is:
 $3\text{H}_3\text{PO}_2 \rightarrow \text{PH}_3 + 2\text{H}_3\text{PO}_3$
 (1) 1 (2) 2 (3) $\frac{4}{3}$ (4) $\frac{3}{4}$
41. To find the n-factor in the following chemical changes.
 (i) $\text{KMnO}_4 \xrightarrow{\text{H}^+} \text{Mn}^{+2}$
 (ii) $\text{KMnO}_4 \xrightarrow{\text{H}_2\text{O}} \text{Mn}^{+4}$
 (iii) $\text{KMnO}_4 \xrightarrow{\text{OH}^- \text{ (concentrated basic medium)}} \text{Mn}^{+6}$
 (iv) $\text{K}_2\text{Cr}_2\text{O}_7 \xrightarrow{\text{H}^+} \text{Cr}^{3+}$
 (v) $\text{C}_2\text{O}_4^{2-} \rightarrow \text{CO}_2$
 (vi) $\text{FeSO}_4 \rightarrow \text{Fe}_2\text{O}_3$
 (vii) $\text{Fe}_2\text{O}_3 \rightarrow \text{FeSO}_4$

Balancing of Redox Reaction

- (a) Oxidation number change method.
(b) Ion electron method.

(a) Oxidation Number Change Method:

This method was given by Jonson. In a balanced redox reaction, total increase in oxidation number must be equal to total decreases in oxidation number. This equivalence provides the basis for balancing redox reactions.

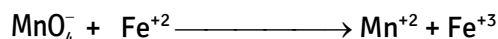
The general procedure involves the following steps: -

- Select the atom in oxidising agent whose oxidation number decreases and indicate the gain of electrons.
- Select the atom in reducing agent whose oxidation number increases and write the loss of electrons.
- Now cross multiply i.e. multiply oxidising agent by the number of loss of electrons and reducing agent by number of gain of electrons.
- Balance the number of atoms on both sides whose oxidation numbers change in the reaction.
- In order to balance oxygen atoms, add H_2O molecules to the side deficient in oxygen. Then balance the number of H atoms by adding H^+ ions to the side deficient in H atoms.

Ex: Balance the following reaction by the oxidation number method -



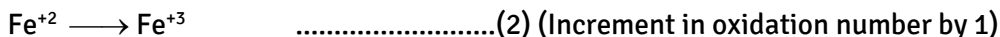
Sol: Write the oxidation number of all the atoms.



change in oxidation number has occurred in Mn and Fe.



.....(1) (Decrement in oxidation number by 5)

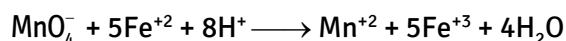


.....(2) (Increment in oxidation number by 1)

To make increase and decrease equal, eq. (2) is multiplied by 5.



To balance oxygen, $4\text{H}_2\text{O}$ are added to R.H.S. and to balance hydrogen, 8H^+ are added to L.H.S.

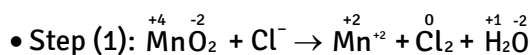


This is the balanced equation.

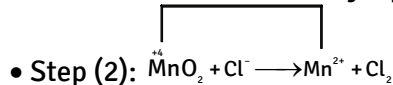
Ex: Balance the following chemical reaction by oxidation number method and write their skeleton equation:

Chloride ions reduce manganese dioxide to manganese (II) ions in acidic medium and get itself oxidized to chlorine gas.

Sol: (i) The skeleton equation for the given process is $\text{MnO}_2 + \text{Cl}^- \longrightarrow \text{Mn}^{2+} + \text{Cl}_2$

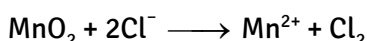


O.N. decreases by 2 per Mn

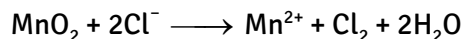


O.N. increases by 1 per Cl

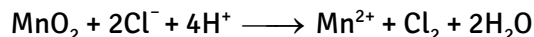
- Step (3): Equalize the increase/decrease in O.N. by multiplying MnO_2 by 1 and Cl^- by 2



- Step (4): Balance other atoms except H and O. Here they are all balanced.
- Step (5): Balance O atoms by adding H_2O molecules to the side falling short of O atoms.



- Step (6): Balance H atoms by adding H^+ ions to the side falling short of H atoms.

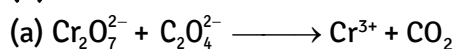


(b) Ion-Electron Method:

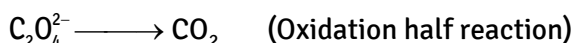
This method was given by Jette and La mav. The following steps are followed while balancing redox reaction (equations) by this method.

- Write the equation in ionic form.
- Split the redox equation into two half reactions, one representing oxidation and the other representing reduction.
- Balance these half reactions separately and then add by multiplying with suitable coefficients so that the electrons are cancelled.
- Add both half reaction and then balance the atoms other than 'O' and 'H'.
- Then balance oxygen atoms by adding H_2O molecules to the side deficient in oxygen. The number of H_2O molecules added is equal to the deficiency of oxygen atoms.
- Balancing of H
 - In acidic medium: Add same number of H^+ ions to the required side as the number of H-atoms are required.
 - If basic medium: Add same number of H_2O molecules as required H-atoms are added to the required side and simultaneously the same number of OH^- ions are added to opposite side.
- Balance the charge by electrons to the side which is rich in +ve charges. i.e. deficiency in electrons.
Number of electrons added is equal to the deficiency.
- Multiply the half equations with suitable coefficients to equalize the number of electrons.
- Add these half equations to get an equation which is balanced with respect to charge and atoms.

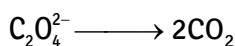
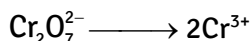
Ex: (A) Acidic Medium:



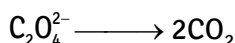
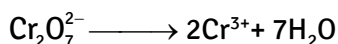
(b) Write both the half reaction.



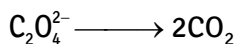
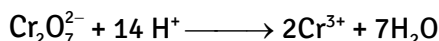
(c) Atoms other than H and O are balanced.



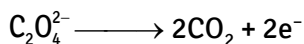
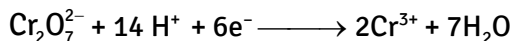
(d) Balance O-atoms by the addition of H_2O to another side



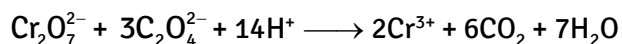
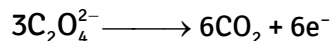
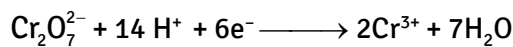
(e) Balance H-atoms by the addition of H^+ to another side



(f) Now, balance the charge by the addition of electron (e^-).

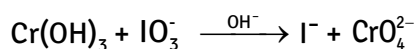


- (g) Multiply equations by a constant to get the same number of electrons on both sides. In the above case, second equation is multiplied by 3 and then added to first equation.

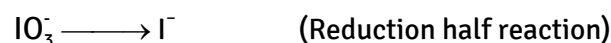


Ex.: (B) Alkaline Medium:

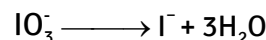
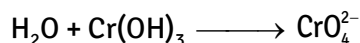
- (a) Consider the reaction



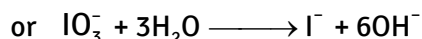
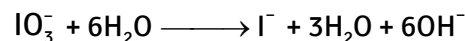
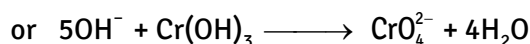
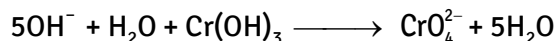
- (b) Separate the two half reactions.



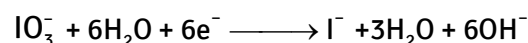
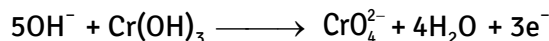
- (c) Balance O-atoms by adding H_2O .



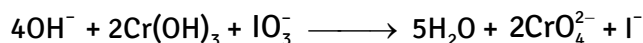
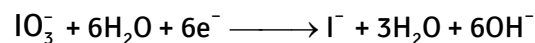
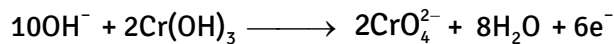
- (d) Balance H-atoms by adding H_2O to side having deficiency and an equal number of OH^- ions to the other side.



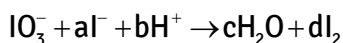
- (e) Balance the charges by electrons



- (f) Multiply first equation by 2 and add to second to give



Ex.8: In the balanced chemical reaction,



a, b, c and d respectively correspond to:

(A) 5, 6, 3, 3

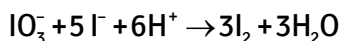
(B) 5, 3, 6, 3

(C) 3, 5, 3, 6

(D) 5, 6, 5, 5

Ans. (A)

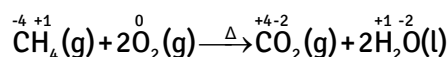
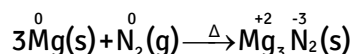
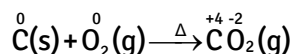
Sol. The balanced equation will be:



Types of Redox Reactions

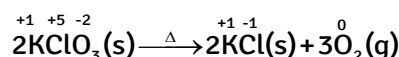
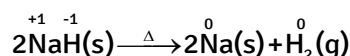
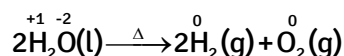
1. Combination Redox Reactions:

- A combination reaction may be denoted in the manner: $A + B \longrightarrow C$
- Either A or B or both A and B must be in the elemental form.
- All combustion reactions, as well as other reactions involving elements other than dioxygen, are combination redox reactions.

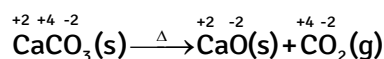


2. Decomposition Redox Reactions:

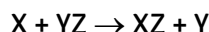
- A decomposition reaction leads to the breakdown of a compound into two or more components at least one of which must be in the elemental state.



- This may also be noted here that all decomposition reactions are not redox reactions. For example, decomposition of calcium carbonate is not a redox reaction because there is no change in oxidation number of elements.



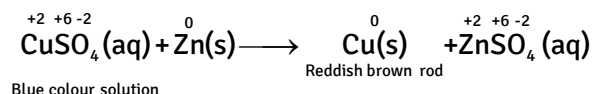
3. Displacement Reactions: In a displacement reaction, an ion (or an atom) in a compound is replaced by other (ion or atom) of another element. It may be denoted as:



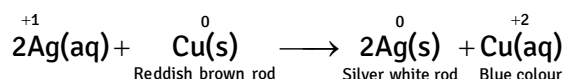
Displacement reactions fit into two categories: metal displacement and non-metal displacement.

(a) Metal Displacement:

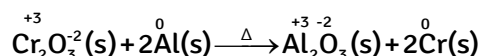
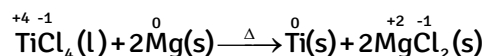
- A metal in a compound can be displaced by another metal in the uncombined state.



While $ZnSO_4(aq) + Cu(s) \longrightarrow$ no reaction.



So reactivity order $Zn > Cu > Ag$

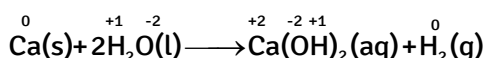
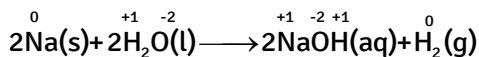


- In each case, the reducing metal is a better reducing agent than the one that is being reduced which evidently shows more capability to lose electrons as compared to the one that is reduced.

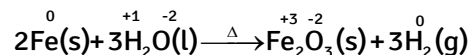
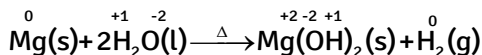
(b) Non-Metal Displacement:

- The non-metal displacement redox reactions include hydrogen displacement and a rarely occurring reaction involving oxygen displacement.

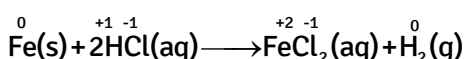
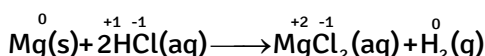
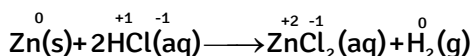
- All alkali metals and some alkaline earth metals (Ca, Sr, and Ba) Which are very good reductants, will displace hydrogen from cold water.



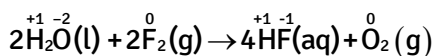
- Less active metals such as magnesium and iron react with steam to produce dihydrogen gas:



- Many metals, including those which do not react with cold water as well as with steam are capable of displacing hydrogen from acids. Cadmium and tin are the examples of such metals. A few examples for the displacement of hydrogen from acids are:

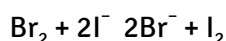
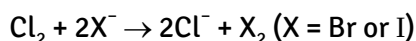
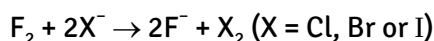


- Above reactions can be used to prepare dihydrogen gas in the laboratory. Very less active metals such as silver (Ag), and gold (Au) do not react even with hydrochloric acid.
- The power of halogens as oxidising agents decreases as we move down from fluorine to iodine in group 17 of the periodic table. This implies that fluorine is so reactive that it can replace chloride, bromide and iodide ions in solution. In fact, fluorine is so reactive that it attacks water and displaces the oxygen of water:



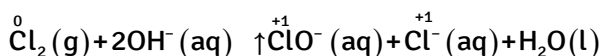
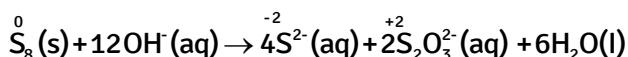
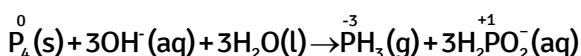
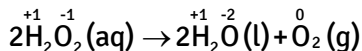
It is for this reason that the displacement reactions of chlorine, bromine and iodine using fluorine are not generally carried out in aqueous solution.

In general, a halogen oxidises halides ions of higher atomic number:



4. Disproportionation Reactions:

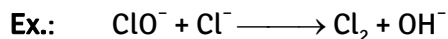
- In a disproportionation reaction, an element in one oxidation state is simultaneously oxidised and reduced.
- One of the reacting substances in a disproportionation reaction always contains an element that can exist in at least three oxidation states.



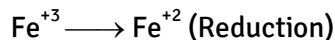
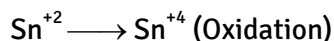
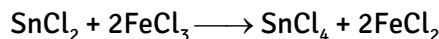
- While $2\text{F}_2(\text{g}) + 2\text{OH}^-(\text{aq}) \longrightarrow 2\text{F}^-(\text{aq}) + \text{OF}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- Fluorine being the most electronegative element, it cannot exhibit any positive oxidation state. This means among halogens; fluorine does not show a disproportionation tendency.

5. Comproportionation Reaction:

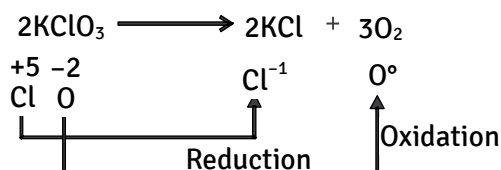
Reverse of disproportionation reaction known as comproportionation reaction.

**6. Intermolecular Redox Reaction:**

Oxidation and reduction take place when molecules of two different substances react with each other. Called intermolecular redox reaction.

**7. Intra Molecular Redox Reaction:**

During the chemical reaction, if oxidation and reduction take place in single compound then reaction is called intramolecular redox reaction.

**Points to Ponder****Titration****(i) KMnO_4 Vs Oxalic Acid:**

Potassium permanganate is a strong oxidising agent and in the presence of sulfuric acid it acts as a powerful oxidising agent. In acidic medium the oxidising ability of KMnO_4 is represented by the following equation. In acidic solution,



Solution containing MnO_4^- ions are purple in colour and the solution containing Mn^{2+} ions are colourless and hence permanganate solution is decolourised when added to a solution of a reducing agent. The moment there is an excess of potassium permanganate present the solution becomes purple. Thus, KMnO_4 serves as self-indicator in acidic solution.

Potassium permanganate is standardized against pure oxalic acid. It involves a redox reaction. Oxalic acid is oxidised to carbon dioxide by KMnO_4 , which itself gets reduced to MnSO_4 . Oxalic acid reacts with potassium permanganate in the following way.

The chemical reaction at room temperature is given below.



The overall reaction takes place in the process is:



The ionic equation involved in the process is given below.

Reduction Half reaction: - $[\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}] \times 2$

Oxidation Half reaction: - $[\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{CO}_2 + 2\text{e}^-] \times 5$

Overall Ionic reaction: - $2\text{MnO}_4^- + 16\text{H}^+ + 5\text{C}_2\text{O}_4^{2-} \rightarrow 2\text{Mn}^{2+} + 10\text{CO}_2 + 8\text{H}_2\text{O}$

This titration cannot be carried out in the presence of acids like nitric acid or hydrochloric acid because itself is an oxidising agent. So hydrochloric acid chemically reacts with KMnO_4 solution forming chlorine which is also an oxidising agent.

(ii) KMnO_4 Vs Mohr Salt:

Potassium permanganate is strong oxidant in the presence of sulphuric acid. Mohr salt is a double salt forming a single crystalline structure having the formula $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$. The chemical name for Mohr's salt is ferrous ammonium sulphate.

In this Mohr salt acts as a reducing agent and potassium permanganate acts as an oxidising agent. So, the reaction between Mohr's and potassium permanganate is a redox reaction. In this redox reaction, ferrous ion from Mohr's salt gets oxidised and pink coloured of manganese present in potassium permanganate, which is in the +7-oxidation state gets reduced to colourless Mn^{+2} state.

The chemical reaction and the molecular chemical equation is given below.

Reduction half reaction - $2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 3\text{H}_2\text{O} + 5[\text{O}]$

Oxidation half reaction -

$[2\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O} + \text{H}_2\text{SO}_4 + [\text{O}] \rightarrow \text{Fe}_2(\text{SO}_4)_3 + 2(\text{NH}_4)_2\text{SO}_4 + 13\text{H}_2\text{O}] \times 5$

Overall reaction -

$2\text{KMnO}_4 + 10\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O} + 8\text{H}_2\text{SO}_4 \rightarrow \text{K}_2\text{SO}_4 + 2\text{MnSO}_4 + 5\text{Fe}_2(\text{SO}_4)_3 + 10(\text{NH}_4)_2\text{SO}_4 + 68\text{H}_2\text{O}$

The ionic equation involved in the process is given below.

Oxidation half reaction - $[\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} + \text{e}^-] \times 5$

Reduction half reaction - $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$

Overall ionic equation - $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$

This titration is based upon oxidation-reduction titrations. When ferrous ammonium sulphate solution is titrated against potassium permanganate in the presence of acidic medium by sulphuric acid. Acidic medium is necessary in order to prevent precipitation of manganese oxide.

Here KMnO_4 acts as a self-indicator and this titration is called permanganate titration.

- All combustion reaction are example of redox reaction.
- On decomposition of a compound if product is present in elemental state, it is a redox reaction.
- Double displacement reaction is not a redox reaction.
- Single displacement reaction is an example of redox reaction.
- All neutralisation reaction are non-redox reactions.

Exercise 1.3

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14. In the equation,
 $\text{NO}_2^- + \text{H}_2\text{O} \longrightarrow \text{NO}_3^- + 2\text{H}^+ + n\text{e}^-$,
 n stands for:
 (1) 1 (2) 2 (3) 3 (4) 4
15. The reaction is balanced if,
 $5\text{H}_2\text{O}_2 + \text{XClO}_2 + 2\text{OH}^- \longrightarrow \text{XCl}^- + \text{YO}_2 + 6\text{H}_2\text{O}$:
 (1) X = 5, Y = 2 (2) X = 2, Y = 5
 (3) X = 4, Y = 10 (4) X = 5, Y = 5
16. Which are correct about:
 (i) $\text{Pb}_3\text{O}_4 + 8\text{HCl} \longrightarrow 3\text{PbCl}_2 + \text{Cl}_2 + 4\text{H}_2\text{O}$
 (ii) $\text{Pb}_3\text{O}_4 + 4\text{HNO}_3 \longrightarrow \text{Pb}(\text{NO}_3)_2 + \text{PbO}_2 + 2\text{HCl}$
 (a) Pb_3O_4 is a mixed oxide having bases PbO and PbO_2 in 2:1 ratio
 (b) Reaction (i) is acid-base reaction of PbO and HCl along with redox change between PbO_2 and HCl
 (c) Reaction (ii) is simply acid-base reaction between PbO and HNO_3 .
 (d) PbO_2 does not show redox change with HNO_3 .
 (1) a, b, c (2) a, b
 (3) a, d (4) All
17. In the reaction, $x\text{BrO}_3^- + y\text{Cr}^{3+} + z\text{H}_2\text{O} \longrightarrow \text{Br}_2 + \text{HCrO}_4^- + \text{H}^+$, the coefficients x, y and z are:
 (1) 6, 8, 22 (2) 6, 10, 22
 (3) 6, 10, 11 (4) 6, 10, 20
18. Which one is not correct for the reaction:
 $(\text{CN})_2 + \text{H}_2\text{O} \longrightarrow \text{HCN} + \text{HOCN}$?
 (1) A disproportionation reaction
 (2) Ox. no. of C in $(\text{CN})_2$, HCN and HOCN are +3, +2 and +4 respectively
 (3) Ox. no. of N in $(\text{CN})_2$, HCN and HOCN are +3, +2 and +4 respectively
 (4) None of the above
19. Which of the following is a disproportionation reaction: -
 (1) $\text{Cu}_2\text{O} + 2\text{H}^+ \longrightarrow \text{Cu} + \text{Cu}^{2+} + \text{H}_2\text{O}$
 (2) $2\text{CrO}_4^{2-} + 2\text{H}^+ \longrightarrow \text{Cr}_2\text{O}_7^{2-} + \text{H}_2\text{O}$
 (3) $\text{CaCO}_3 + 2\text{H}^+ \longrightarrow \text{Ca}^{+2} + \text{H}_2\text{O} + \text{CO}_2$
 (4) $\text{Cr}_2\text{O}_7^{2-} + 2\text{OH}^- \longrightarrow 2\text{CrO}_4^{2-} + \text{H}_2\text{O}$
20. The decomposition of KClO_3 to KCl and O_2 on heating is an example of:
 (1) Intermolecular redox change
 (2) Intramolecular redox change
 (3) Disproportionation or auto redox change
 (4) None
21. Which of the following examples does not represent disproportionation?
 (1) $\text{MnO}_2 + 4\text{HCl} \longrightarrow \text{MnCl}_2 + \text{Cl}_2 + 2\text{H}_2\text{O}$
 (2) $2\text{H}_2\text{O}_2 \longrightarrow 2\text{H}_2\text{O} + \text{O}_2$
 (3) $4\text{KClO}_3 \longrightarrow 3\text{KClO}_4 + \text{KCl}$
 (4) $3\text{Cl}_2 + 6\text{NaOH} \longrightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2$

Exercise 2

- Amongst the following, identify the species with an atom in +6 oxidation state:
 (1) MnO_4^- (2) $[\text{Cr}(\text{CN})_6]^{3-}$
 (3) $[\text{NiF}_6]^{2-}$ (4) CrO_2Cl_2
- The oxidation number of arsenic atom in H_3AsO_4 is:
 (1) -1 (2) -3 (3) +3 (4) +5
- In substance $\text{Mg}(\text{HXO}_3)$, the oxidation number of X is:
 (1) 0 (2) +2 (3) +3 (4) +4
- The oxidation number of iron in potassium ferricyanide $\text{K}_3\text{Fe}(\text{CN})_6$ is:
 (1) Two (2) Six
 (3) Three (4) Four
- Oxidation number of P in KH_2PO_3 is:
 (1) -1 (2) -3 (3) +5 (4) +3
- Maximum oxidation state is present in:
 (1) CrO_2Cl_2 and MnO_4^-
 (2) MnO_2
 (3) $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{Co}(\text{CN})_6]^{3-}$
 (4) MnO
- The sum of oxidation states of sulphur in $\text{H}_2\text{S}_2\text{O}_8$ is:
 (1) +2 (2) +6 (3) +7 (4) +12
- Iodine shows the highest oxidation state in the compound:
 (1) KI (2) KI_3 (3) IF_5 (4) KIO_4
- Select the compound in which the oxidation number of oxygen is -1:
 (1) H_2O (2) O_2F_2
 (3) Na_2O (4) BaO_2
- In which of the following compounds iron has lowest oxidation state?
 (1) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (2) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (3) $\text{Fe}(\text{CO})_5$
 (4) $\text{Fe}_{0.94}\text{O}$
- In which of the following compounds of Cr, the oxidation number of Cr is not +6?
 (1) CrO_3 (2) CrO_2Cl_2
 (3) Cr_2O_3 (4) $\text{K}_2\text{Cr}_2\text{O}_7$
- Oxidation state of Cr in $\text{K}_3[\text{Cr}(\text{C}_2\text{O}_4)_3]$ is:
 (1) +1 (2) 0 (3) +3 (4) +2
- Oxidation state of P is minimum in:
 (1) $\text{H}_4\text{P}_2\text{O}_7$ (2) H_3PO_2
 (3) H_3PO_4 (4) H_3PO_3
- Which statement is wrong:
 (1) Oxidation number of oxygen is +1 in peroxides
 (2) Oxidation number of oxygen is +2 in oxygen difluoride
 (3) Oxidation number of oxygen is $-\frac{1}{2}$ in superoxides
 (4) Oxidation number of oxygen is -2 in most of its compound
- Oxidation state of cobalt in $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{SO}_4$ is -
 (1) 0 (2) +4 (3) -2 (4) +3
- The oxidation state of the most electronegative element in the products of the reaction between BaO_2 and H_2SO_4 are:
 (1) 0 and -1 (2) -1 and -2
 (3) -2 and 0 (4) -2 and +1
- Which of the following shows nitrogen with its increasing order of oxidation number?
 (1) $\text{NO} < \text{N}_2\text{O} < \text{NO}_2 < \text{NO}_3^- < \text{NH}_4^+$
 (2) $\text{NH}_4^+ < \text{N}_2\text{O} < \text{NO}_2 < \text{NO}_3^- < \text{NO}$
 (3) $\text{NH}_4^+ < \text{N}_2\text{O} < \text{NO} < \text{NO}_2 < \text{NO}_3^-$
 (4) $\text{NH}_4^+ < \text{NO} < \text{N}_2\text{O} < \text{NO}_2 < \text{NO}_3^-$

18. A sulphur containing species that cannot be a reducing agent is:
 (1) SO_2 (2) SO_3^{2-}
 (3) H_2SO_4 (4) $\text{S}_2\text{O}_3^{2-}$
19. In the reaction,
 $\text{C} + 4\text{HNO}_3 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O} + 4\text{NO}_2$, HNO_3
 acts as:
 (1) An oxidising agent
 (2) An acid
 (3) A reducing agent
 (4) None of them
20. When H_2 reacts with Na, it acts as:
 (1) oxidizing agent (2) reducing agent
 (3) both of these (4) None
21. Which of the following reagent can act as reducing agent with SO_2 ?
 (1) F_2 (2) KMnO_4
 (3) HNO_3 (4) H_2S
22. In which of the following reaction H_2O_2 acts as reducing agent?
 (1) $2\text{FeCl}_2 + 2\text{HCl} + \text{H}_2\text{O}_2 \rightarrow 2\text{FeCl}_3 + 2\text{H}_2\text{O}$
 (2) $\text{Cl}_2 + \text{H}_2\text{O}_2 \rightarrow 2\text{HCl} + \text{O}_2$
 (3) $2\text{HI} + \text{H}_2\text{O}_2 \rightarrow 2\text{H}_2\text{O} + \text{I}_2$
 (4) $\text{H}_2\text{SO}_3 + \text{H}_2\text{O}_2 \rightarrow \text{H}_2\text{SO}_4 + \text{H}_2\text{O}$
23. HNO_2 acts as an oxidant with which one of the following reagents?
 (1) KMnO_4 (2) H_2S
 (3) $\text{K}_2\text{Cr}_2\text{O}_7$ (4) Br_2
24. In the following change,
 $3\text{Fe} + 4\text{H}_2\text{O} \rightarrow \text{Fe}_3\text{O}_4 + 4\text{H}_2$.
 If the atomic weight of iron is 56, then its equivalent weight will be:
 (1) 42 (2) 21 (3) 63 (4) 84
25. The n-factor of FeS_2 during its oxidation as:
 $\text{FeS}_2 \rightarrow \text{Fe}_2\text{O}_3 + \text{SO}_2$
 (1) 10 (2) 11
 (3) 2 (4) 8
26. In the reaction,
 $8\text{Al} + 3\text{Fe}_3\text{O}_4 \rightarrow 4\text{Al}_2\text{O}_3 + 9\text{Fe}$,
 the number of electrons transferred from reductant to oxidant is -
 (1) 8 (2) 4 (3) 16 (4) 24
27. In weak alkaline medium KMnO_4 converted into:
 (1) MnO_2 (2) K_2MnO_4
 (3) $\text{Mn}(\text{OH})_2$ (4) MnO
28. When KMnO_4 reacts with H_2O_2 in acidic medium then which of following is formed?
 (1) Mn^{+2} (2) Mn^{+7}
 (3) MnO_2 (4) Mn_2O_3
29. What mass of MnO_2 is reduced by 35 ml of 0.16 N oxalic acid in acidic solution? The skeleton equation is:
 $\text{MnO}_2 + \text{H}^+ + \text{H}_2\text{C}_2\text{O}_4 \rightarrow \text{CO}_2 + \text{H}_2\text{O} + \text{Mn}^{2+}$
 (1) 8.7 gm (2) 0.24 gm
 (3) 0.84 gm (4) 43.5 gm
30. 10 ml of 0.3 M oxalic acid titrate against KMnO_4 in weak basic medium find moles of KMnO_4 required
 (1) 2 m mol (2) 2 mol
 (3) 12 m mol (4) 6 m mol
31. Why is HCl not used to make the medium acidic in redox titrations of KMnO_4 in acidic medium?
 (1) Both HCl and KMnO_4 act as oxidising agents.
 (2) KMnO_4 also oxidises HCl into Cl_2
 (3) KMnO_4 is a weaker oxidising agent than HCl
 (4) KMnO_4 acts as a reducing agent in the presence of HCl
32. How many mole of FeSO_4 , $\text{H}_2\text{C}_2\text{O}_4$ and FeC_2O_4 are oxidised separately by one mole of KMnO_4 in acidic medium respectively:
 (1) 5; 5/2; 5/3 (2) 2/5; 2; 5
 (3) 5/3; 3/5; 2/5 (4) 3/5; 2; 5

33. 100 ml of 0.2 N oxalic acid titrate against 0.1 N KMnO_4 in acidic medium find volume of KMnO_4 required
 (1) 200 ml (2) 100 ml
 (3) 50 ml (4) None of these
34. 20 ml of 0.3 M oxalic acid titrate against 0.2 M KMnO_4 in alkaline medium find volume of KMnO_4 required
 (1) 60 ml (2) 30 ml
 (3) 12 ml (4) None of these
35. 1.26 g crystalline oxalic acid titrate against 0.2 M KMnO_4 in acidic medium find volume of KMnO_4 required
 (1) 20 ml (2) 40 ml
 (3) 10 ml (4) None of these
36. 6×10^{-3} mol $\text{K}_2\text{Cr}_2\text{O}_7$ reacts completely with 9×10^{-3} mol X^{n+} to give XO_3^- and Cr^{3+} . The value of 'n' is:
 (1) 1 (2) 2 (3) 3 (4) 5
37. 0.2 g of a sample of H_2O_2 required 10 mL of N KMnO_4 in a titration in the presence of H_2SO_4 . Purity of H_2O_2 is:
 (1) 25% (2) 85% (3) 65% (4) 95%
38. When KMnO_4 is titrated against $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ in acidic medium, the equivalent mass of KMnO_4 is -
 (1) $\frac{\text{Molecular mass}}{10}$
 (2) $\frac{\text{Molecular mass}}{5}$
 (3) $\frac{\text{Molecular mass}}{2}$
 (4) Molecular mass
39. An element A in a compound ABD has oxidation number A^n . It is oxidised by $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium. In the experiment 1.68×10^{-3} mole of $\text{K}_2\text{Cr}_2\text{O}_7$ were used for 3.26×10^{-3} mole of ABD. The new oxidation number of A after oxidation is:
 (1) 3 (2) 3-n
 (3) n-3 (4) +n
40. When ferrous oxalate is titrated against $\text{K}_2\text{Cr}_2\text{O}_7$, meq. of Fe^{2+} , $\text{C}_2\text{O}_4^{2-}$ and $\text{Cr}_2\text{O}_7^{2-}$ in the redox reaction are X, Y and Z respectively, then:
 (1) $X = Y$ (2) $X + Y = Z$
 (3) $X + 2Y = Z$ (4) $2X + 6Y = 6Z$
41. A compound of Xe and F is found to have 53.5% Xe. What is the oxidation number of Xe in this compound?
 (Xe atomic mass = 131.29, F=19)
 (1) -4 (2) 0
 (3) +4 (4) +6
42. The value of n in the following equation is:
 $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + n\text{Fe}^{2+} \rightarrow 2\text{Cr}^{3+} + n\text{Fe}^{3+} + 7\text{H}_2\text{O}$
 (1) 2 (2) 3
 (3) 7 (4) 6
43. In the balanced equation:
 $[\text{Zn} + \text{H}^+ + \text{NO}_3^- \rightarrow \text{NH}_4^+ + \text{Zn}^{+2} + \text{H}_2\text{O}]$
 coefficient of NH_4^+ is:
 (1) 4 (2) 3
 (3) 2 (4) 1

44. The number of electrons required to balance the following equation are:
 $\text{NO}_3^- + 4\text{H}^+ \rightarrow 2\text{H}_2\text{O} + \text{NO}$
(1) 2 on right side (2) 3 on left side
(3) 3 on right side (4) 5 on left side
45. In the following reaction the value of 'X' is -
 $\text{H}_2\text{O} + \text{SO}_3^{2-} \rightarrow \text{SO}_4^{2-} + 2\text{H}^+ + \text{X}$
(1) 4e^- (2) 3e^-
(3) 2e^- (4) 1e^-
46. The reaction,
 $2\text{K}_2\text{MnO}_4 + \text{Cl}_2 \rightarrow 2\text{KMnO}_4 + 2\text{KCl}$
is an example of -
(1) Redox
(2) Reduction
(3) Neutralization
(4) Disproportionation
47. In the reaction:
 $\text{MnO}_4^- + \text{SO}_3^{2-} + \text{H}^+ \rightarrow \text{SO}_4^{2-} + \text{Mn}^{2+} + \text{H}_2\text{O}$
(1) MnO_4^- and H^+ both are reduced.
(2) MnO_4^- is reduced and H^+ is oxidised.
(3) MnO_4^- is reduced and SO_3^{2-} is oxidised
(4) MnO_4^- is oxidised and SO_3^{2-} is reduced.
48. Which of the following reaction involves neither oxidation nor reduction:
(1) $\text{CrO}_4^{2-} \rightarrow \text{Cr}_2\text{O}_7^{2-}$
(2) $\text{Cr} \rightarrow \text{CrCl}_3$
(3) $\text{Na} \rightarrow \text{Na}^+$
(4) $2\text{S}_2\text{O}_3^{2-} \rightarrow \text{S}_4\text{O}_6^{2-}$
49. In which of the following reactions the underlined substance is oxidised?
(1) $3\text{Mg} + \underline{\text{N}_2} \rightarrow \text{Mg}_3\text{N}_2$
(2) $2\text{KI} + \underline{\text{Br}_2} \rightarrow 2\text{KBr} + \text{I}_2$
(3) $\underline{\text{CuO}} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$
(4) $\underline{\text{CO}} + \text{Cl}_2 \rightarrow \text{COCl}_2$
50. Select the example of disproportionation reaction:
(1) $\text{BaCl}_2 + \text{H}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{HCl}$
(2) $\text{NH}_4\text{NO}_3 \rightarrow \text{N}_2\text{O} + 2\text{H}_2\text{O}$
(3) $4\text{H}_3\text{PO}_3 \rightarrow \text{PH}_3 + 3\text{H}_3\text{PO}_4$
(4) $\text{AgCl} + 2\text{NH}_3 \rightarrow \text{Ag}(\text{NH}_3)_2\text{Cl}$

Exercise 3

1. Which of the following is correctly match:

	Compound		Oxidation number
(A)	AlF_3	(P)	+2.5
(B)	P_4O_{10}	(Q)	+7
(C)	HClO_4	(R)	+5
(D)	$\text{S}_4\text{O}_6^{2-}$	(S)	+3

- (1) (A)-P, (B)-Q, (C)-R, (D)-S
 (2) (A)-S, (B)-R, (C)-Q, (D)-P
 (3) (A)-S, (B)-R, (C)-P, (D)-Q
 (4) (A)-P, (B)-Q, (C)-S, (D)-R

2. Match the column I with column II and mark the appropriate choice.

Column I (Compound)	Column II (Oxidation state of Fe)
(A) $\text{K}_3[\text{Fe}(\text{OH})_6]$	(i) +8/3
(B) $\text{K}_2[\text{FeO}_4]$	(ii) +2
(C) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$	(iii) +3
(D) Fe_3O_4	(iv) +6

- (1) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (i), C $\rightarrow$ (ii), (D) $\rightarrow$ (iv)
 (2) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (iv), C $\rightarrow$ (ii), (D) $\rightarrow$ (i)
 (3) (A) $\rightarrow$ (i), (B) $\rightarrow$ (iii), C $\rightarrow$ (ii), (D) $\rightarrow$ (iv)
 (4) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (ii), C $\rightarrow$ (i), (D) $\rightarrow$ (iii)

3. Match the half-reactions (in column I) with change in oxidation number (in column II)

Column-I	Column-II
(A) $\text{Cl}^- \rightarrow \text{ClO}_4^-$	(1) 2
(B) $\text{Cr}^{3+} \rightarrow \text{CrO}_5$	(2) 8
(C) $\text{H}_2\text{O}_2 \rightarrow \text{O}_2$	(3) 0
(D) $\text{CrO}_2^{2+} \rightarrow \text{CrO}_4^{2-}$	(4) 3

- (1) A-(2), B-(4), C-(1), D-(3)
 (2) A-(4), B-(2), C-(1), D-(3)
 (3) A-(3), B-(1), C-(4), D-(2)
 (4) A-(2), B-(1), C-(3), D-(4)

4. Match the reactions given in Column I with Equivalent wt. given in Column II.

Column I	Column II
(A) $\text{MnO}_4^- \rightarrow \text{Mn}^{+2}$	(p) M/6
(B) $\text{Cr}_2\text{O}_7^{2-} \rightarrow \text{Cr}^{+3}$	(q) M/5
(C) $\text{I}^- \rightarrow \text{I}_2$	(r) M/3
(D) $\text{FeC}_2\text{O}_4 \rightarrow \text{Fe}^{+3} + \text{CO}_2$	(s) M/1

- (1) A-(p), B-(r), C-(s), D-(q)
 (2) A-(s), B-(r), C-(p), D-(q)
 (3) A-(q), B-(p), C-(s), D-(r)
 (4) A-(r), B-(q), C-(p), D-(s)

5. Match incorrect equivalent mass of reactant in following column?
 (M = molar mass of reactant):-

Column-I	Column-II
(1) $\text{FeS}_2 \rightarrow \text{Fe}^{3+} + \text{SO}_3$	$\frac{M}{15}$
(2) $\text{P}_2\text{H}_4 \rightarrow \text{P}_4\text{H}_2 + \text{PH}_3$	$\frac{5M}{6}$
(3) $2\text{Mn}^{+7} \rightarrow \text{Mn}^{2+} + \text{Mn}^{+6}$	$\frac{M}{3}$
(4) $\text{Ba}(\text{MnO}_4)_2 \xrightarrow{\text{H}^+} \text{Mn}^{2+}$	$\frac{M}{5}$

6. Match Column I with Column II

Column-I	Column-II
(A) $\text{V}_2\text{O}_5(\text{s}) + 5\text{Ca}(\text{s}) \rightarrow 2\text{V}(\text{s}) + 5\text{CaO}(\text{s})$	(p) Disproportionation reaction
(B) $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \xrightarrow{\Delta} \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$	(q) Decomposition reaction
(C) $\text{P}_4(\text{s}) + 3\text{OH}(\text{aq}) + 3\text{H}_2\text{O}(\text{l}) \rightarrow \text{PH}_3(\text{g}) + 3\text{H}_2\text{PO}_2^-(\text{aq})$	(r) Combustion reaction
(D) $2\text{KClO}_3(\text{s}) \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$	(s) Displacement reaction

- (1) A-(s), B-(q), C-(r), D-(p)
 (2) A-(s), B-(r), C-(p), D-(q)
 (3) A-(r), B-(s), C-(q), D-(p)
 (4) A-(r), B-(s), C-(p), D-(q)

7. **Assertion:** Oxidation state of Hydrogen is +1 in H_2O while -1 in CaH_2 .
Reason: CaH_2 is a metal hydride and for metal hydrides, hydrogen is assigned the oxidation number of -1.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is false.
 (4) If both Assertion & Reason are false
8. **Assertion:** Oxidation number of carbon in CH_2O is zero.
Reason: CH_2O (formaldehyde) is a organic compound.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is false.
 (4) If both Assertion & Reason are false
9. **Assertion:** Oxidation number of Cr in CrO_5 is +6.
Reason: In (CrO_5) , four oxygen atoms are involved in peroxide linkage.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is false.
 (4) If both Assertion & Reason are false

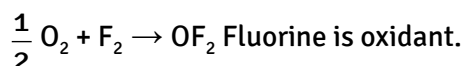
10. **Assertion:** $\text{H}_2\text{S} + \text{Cl}_2 \rightarrow 2\text{HCl} + \text{S}$
 In the above reaction, Cl has been oxidised to Cl^- while S^{2-} has been reduced to S
Reason: In a reaction the element whose oxidation number decreases is reduced and the element whose oxidation number increases is oxidised.
 (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
 (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
 (3) If Assertion is True but the Reason is false.
 (4) If both Assertion & Reason are false
11. **Assertion:** H_2SO_4 cannot act as reducing agent.
Reason: Sulphur cannot increase its oxidation number beyond +6.
 (1) Both Statement I and Statement II are false
 (2) Statement I is true but Statement II is false
 (3) Statement I is false but Statement II is true
 (4) Both Statement I and Statement II are true
12. **Assertion:** In the reaction,
 $2\text{Cu}_2\text{O}_{(s)} + \text{Cu}_2\text{S}_{(s)} \rightarrow 6\text{Cu}_{(s)} + \text{SO}_{2(g)}$
 copper acts as a reductant and sulphur acts as an oxidant.
Reason: The given reaction is not a redox reaction.
 (1) If both assertion and reason are true and reason is the correct explanation of assertion.
 (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (3) If assertion is true but reason is false.
 (4) If both assertion and reason are false.

13. Assertion: Nitrous acid (HNO_2) may act as an oxidising agent as well as a reducing agent.

Reason: The oxidation number of Nitrogen remains same in all the compounds.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are false

14. Statement-I : In the reaction,



Statement-II: Fluorine cannot show positive oxidation state.

- (1) Both Statement I and Statement II are false
- (2) Statement I is true but Statement II is false
- (3) Statement I is false but Statement II is true
- (4) Both Statement I and Statement II are true

15. Assertion: $[\text{Fe}(\text{CN})_6]^{4-} \rightarrow \text{Fe}^{3+} + \text{CO}_2 + \text{NO}_3^-$, the equivalent weight of reactant is 3.47 .

Reason: Equivalent weight of reactant

$$= \frac{\text{Mol.wt.}}{61}$$

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is false.
- (4) If both Assertion & Reason are false

16. Assertion: Equivalent weight of KMnO_4 in acidic medium is $M/5$ (M = molecular weight) while in faintly basic medium, it is equal to $M/3$.

Reason: In acidic medium, 1 mol of MnO_4^- gains 5 mole electrons while in faintly basic medium it gains 3 mole electrons.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is false.
- (4) If both Assertion & Reason are false

17. Assertion: MnO_4^- is always reduced to Mn^{+2} .

Reason: Decrease in oxidation number or gaining of electron means oxidation.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is false.
- (4) If both Assertion & Reason are false

18. Assertion: Decomposition of potassium chlorate is an example of redox reaction.

Reason: There is no change in the oxidation number of potassium in decomposition of potassium chlorate.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

19. **Assertion:** ClO_4^- does not show disproportionation reaction.

Reason: In ClO_4^- , chlorine is present in its highest oxidation state.

- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

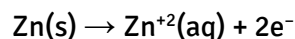
20. **Assertion:** All halogens under go disproportionation reaction in alkaline medium.

Reason: All halogens exhibit variable oxidation states.

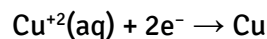
- (1) If both assertion and reason are true and reason is the correct explanation of assertion.
- (2) If both assertion and reason are true but reason is not the correct explanation of assertion.
- (3) If assertion is true but reason is false.
- (4) If both assertion and reason are false.

21. **Assertion:**

$\text{Zn(s)} + \text{Cu}^{+2}(\text{aq}) \rightarrow \text{Zn}^{+2}(\text{aq}) + \text{Cu(s)}$ can be split into following half reactions



(Oxidation half reaction)



(Reduction half reaction)

Reason: Every redox reaction can be split into two reactions, one representing loss of electrons and the other representing gain of electrons.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is false.
- (4) If both Assertion & Reason are false

Exercise 4 (Previous Year's Questions)

- Correct decreasing order of oxidising strength is: [AIPMT-2010]
 - (1) MnO_4^- , VO_2^+ , $\text{Cr}_2\text{O}_7^{2-}$
 - (2) $\text{Cr}_2\text{O}_7^{2-}$, MnO_4^- , VO_2^+
 - (3) $\text{Cr}_2\text{O}_7^{2-}$, VO_2^+ , MnO_4^-
 - (4) MnO_4^- , $\text{Cr}_2\text{O}_7^{2-}$, VO_2^+
- How much amount of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is required for liberation of 2.54 g I_2 when titrated with KI? [AIPMT-2011]
 - (1) 2.5 g
 - (2) 4.99 g
 - (3) 2.4 g
 - (4) 1.2 g
- When Cl_2 gas reacts with hot and concentrated sodium hydroxide solution, the oxidation number of chlorine changes from: [AIPMT-2012]
 - (1) Zero to -1 and zero to +3
 - (2) Zero to +1 and zero to -3
 - (3) Zero to +1 and zero to -5
 - (4) Zero to -1 and zero to +5
- In which of the following compounds, nitrogen exhibits highest oxidation state? [AIPMT-2012]
 - (1) N_3H
 - (2) NH_2OH
 - (3) N_2H_4
 - (4) NH_3
- The reaction of aqueous KMnO_4 with H_2O_2 in acidic conditions gives: [AIPMT-2014]
 - (1) Mn^{+4} and O_2
 - (2) Mn^{2+} and O_2
 - (3) Mn^{2+} and O^3
 - (4) Mn^{4+} and MnO_2
- (a) $\text{H}_2\text{O}_2 + \text{O}_3 \rightarrow \text{H}_2\text{O} + 2\text{O}_2$
 - (b) $\text{H}_2\text{O}_2 + \text{Ag}_2\text{O} \rightarrow 2\text{Ag} + \text{H}_2\text{O} + \text{O}_2$
 Role of hydrogen peroxide in the above reactions is respectively: [AIPMT-2014]
 - (1) Oxidizing in (a) and reducing in (b)
 - (2) Reducing in (a) and oxidizing in (b)
 - (3) Reducing in (a) and (b)
 - (4) Oxidizing in (a) and (b)
- Assuming complete ionization, same moles of which of the following compounds will require the least amount of acidified KMnO_4 for complete oxidation? [RE-AIPMT-2015]
 - (1) FeC_2O_4
 - (2) $\text{Fe}(\text{NO}_2)_2$
 - (3) FeSO_4
 - (4) FeSO_3
- Lowest oxidation state of phosphorous is in: [NEET-2017]
 - (1) H_3PO_2
 - (2) H_3PO_4
 - (3) $\text{H}_4\text{P}_2\text{O}_7$
 - (4) H_3PO_3
- For the redox reaction,

$$\text{MnO}_4^- + \text{C}_2\text{O}_4^{2-} + \text{H}^+ \rightarrow \text{Mn}^{2+} + \text{CO}_2 + \text{H}_2\text{O}$$
 the correct coefficients of the reactants for the balanced reaction are: [NEET-2018]
 - (1) $\text{MnO}_4^- \rightarrow 2$, $\text{C}_2\text{O}_4^{2-} \rightarrow 16$, $\text{H}^+ \rightarrow 5$
 - (2) $\text{MnO}_4^- \rightarrow 2$, $\text{C}_2\text{O}_4^{2-} \rightarrow 5$, $\text{H}^+ \rightarrow 16$
 - (3) $\text{MnO}_4^- \rightarrow 16$, $\text{C}_2\text{O}_4^{2-} \rightarrow 5$, $\text{H}^+ \rightarrow 2$
 - (4) $\text{MnO}_4^- \rightarrow 5$, $\text{C}_2\text{O}_4^{2-} \rightarrow 16$, $\text{H}^+ \rightarrow 2$
- Which of the following reactions are disproportionation reaction? [NEET-2019]
 - (a) $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}^0$
 - (b) $2\text{KMnO}_4 \xrightarrow{\Delta} \text{K}_2\text{MnO}_4 + \text{MnO}_2 + \text{O}_2$
 - (c) $3\text{MnO}_4^{2-} + 4\text{H}^+ \rightarrow 2\text{MnO}_4^- + \text{MnO}_2 + 2\text{H}_2\text{O}$
 - (d) $2\text{MnO}_4^- + 3\text{Mn}^{2+} + 2\text{H}_2\text{O} \rightarrow 5\text{MnO}_2 + 4\text{H}^+$
 Select the correct option from the following-
 - (1) (a), (c) and (d)
 - (2) (a) and (d) only
 - (3) (a) and (c) only
 - (4) (a), (b) and (c)
- What is the change in oxidation number of carbon in the following reaction? [NEET-2020]

$$\text{CH}_4(\text{g}) + 4\text{Cl}_2(\text{g}) \rightarrow \text{CCl}_4(\text{l}) + 4\text{HCl}(\text{g})$$
 - (1) 0 to +4
 - (2) -4 to +4
 - (3) 0 to -4
 - (4) +4 to +4

12. The oxidation number of the underlined atom in the following species:
Identify the incorrect option.

[NEET 2020 Exam (Covid Re-exam)]

- (1) $\text{Cu}_2\underline{\text{O}}$ is -1
(2) $\underline{\text{Cl}}\text{O}_3^-$ is +5
(3) $\text{K}_2\underline{\text{Cr}}_2\text{O}_7$ is +6
(4) $\text{H}\underline{\text{Au}}\text{Cl}_4$ is +3

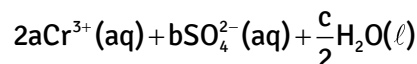
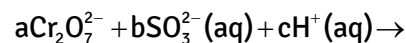
13. Which of the following reactions is the metal displacement reaction? Choose the right option. [NEET-2021]

- (1) $2\text{Pb}(\text{NO}_3)_2 \rightarrow 2\text{PbO} + 4\text{NO}_2 + \text{O}_2 \uparrow$
(2) $2\text{KClO}_3 \xrightarrow{\Delta} 2\text{KCl} + 3\text{O}_2$
(3) $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} \text{Al}_2\text{O}_3 + 2\text{Cr}$
(4) $\text{Fe} + 2\text{HCl} \rightarrow \text{FeCl}_2 + \text{H}_2 \uparrow$

14. In the neutral or faintly alkaline medium, KMnO_4 oxidises iodide into iodate. The change in oxidation state of manganese in this reaction is form: [NEET-2022]

- (1) +6 to +4 (2) +7 to +3
(3) +6 to +5 (4) +7 to +4

15. On balancing the given redox reaction,



the coefficients a, b and c are found to be, respectively- [NEET 2023]

- (1) 3, 8, 1 (2) 1, 8, 3
(3) 8, 1, 3 (4) 1, 3, 8

16. Which reaction is NOT a redox reaction?

[NEET 2024]

- (1) $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$,
(2) $\text{BaCl}_2 + \text{Na}_2\text{SO}_4 \rightarrow \text{BaSO}_4 + 2\text{NaCl}$
(3) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
(4) $2\text{KClO}_3 + \text{I}_2 \rightarrow 2\text{KIO}_3 + \text{Cl}_2$

17. The oxidation states not show by Mn in given reaction is: [RE-NEET 2024]



(A) +6 (B) +2 (C) +4 (D) +7 (E) +3
Choose the most appropriate answer from the options given below:

- (1) D and E only (2) B and D only
(3) A and B only (4) B and E only

ANSWER KEY

Exercise 1.1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	2	2	2	2	1	4	1	1	3	1	3	2	2	4	1	2	3	1	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	4	2	4	1	1	3	1	2	2	1	1	4	3	1	1	1	1	4	3	3
Que.	41	42	43																	
Ans.	4	2	4																	

Exercise 1.2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	1	3	4	2	2	2	2	1	1	4	4	3	1	4	2	3	2	4	2	3
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	1	2	2	4	4	3	3	4	1	4	4	3	2	1	1	3	1	4	1	3
Que.	41																			
Ans.	(i) 5 (ii) 3 (iii) 1 (iv) 6 (v) 2 (vi) 1 (vii) 2																			

Exercise 1.3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	2	3	2	3	1	3	1	2	2	3	1	2	2	4	2	3	1	2
Que.	21																			
Ans.	1																			

Exercise 2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	4	4	3	3	4	1	4	4	4	3	3	3	2	1	4	2	3	3	1	1
Que.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Ans.	4	2	2	2	2	4	1	1	2	1	2	1	1	1	1	1	2	2	2	2
Que.	41	42	43	44	45	46	47	48	49	50										
Ans.	4	4	4	2	3	1	3	1	4	3										

Exercise 3

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	1	3	4	2	1	2	1	4	4	4	3	4	1	1	4	2	1	4
Que.	21																			
Ans.	1																			

Exercise 4 (Previous Year's Questions)

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17			
Ans.	4	2	4	1	2	3	3	1	2	3	2	1	3	4	4	2	4			